CANONICAL SUBGROUPS AND PERFECTOID SHIMURA VARIETIES

AYAN NATH

Abstract. We develop a group-theoretic analogue of the theory of canonical subgroups which works uniformly for integral canonical models of Shimura varieties defined over $\mathbf{Z}_p$, with bounds on its overconvergence radii in terms of group-theoretic Hasse invariants. This is achieved using prismatic Dieudonné theory. We apply this to prove that all minimally compactified Shimura varieties at infinite level, including those of exceptional type, are perfectoid spaces for all sufficiently large primes.

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1. Introduction

Since Scholze's breakthrough [Sch15] establishing that Shimura varieties of Hodge type at infinite level are perfectoid spaces, there has been a concerted effort to extend this result to all Shimura varieties, as conjectured in [Sch16, Conjecture 2.1]. Subsequent works have enlarged the scope of this theorem, encompassing abelian-type [She17] and pre-abelian type [HJ23] Shimura varieties essentially by reducing it to the Hodge-type case. More recently, He [He26] proved pointwise perfectoidness at infinite level for arbitrary Shimura varieties using Sen theory. However, the global perfectoidness of exceptional Shimura varieties has remained out of reach due to the absence of integral canonical models, as well as there being no reasonable way to relate these Shimura varieties with abelian varieties. The recent breakthrough of Bakker, Shankar, and Tsimerman [BST25a] establishing the existence of integral canonical models for sufficiently large primes is what allows us to approach this problem, coupled with the prismatic Dieudonné theory formulated as in Gardner–Madapusi [GM24]. The aim of this paper is to leverage these tools to establish the global perfectoidness of arbitrary Shimura varieties at infinite level, including those of exceptional type for sufficiently large primes.

Theorem A (Theorem 5.6.6)**.** *Let (G,X) be a Shimura datum with reflex field E. For all sufficiently large primes p, the following holds. Fix a complete algebraically closed field $C/\mathbf{Q}_p$ and an embedding $E \hookrightarrow C$. Fix any open compact subgroup $K^p \subset G(\mathbf{A}_f^p)$. For any open compact subgroup $K_p \subset G(\mathbf{Q}_p)$, let $\mathcal{S}^*_{K^pK_p}$ denote the adic space over $\operatorname{Spa} C$ associated with the base change of the minimal compactification $\operatorname{Sh}^*_{K^pK_p}(G,X)$ along $E \hookrightarrow C$. Then there is a perfectoid space $\mathcal{S}^*_{K^p}$ such that*

$$\mathcal{S}^*_{K^p} \sim \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathcal{S}^*_{K^pK_p}$$

as diamonds over $\operatorname{Spd} C$. Moreover, there is a canonical $G(\mathbf{Q}_p)$-equivariant Hodge–Tate period map

$$\pi_{\mathrm{HT}} \colon \mathcal{S}^*_{K^p} \to \mathscr{F}\!\ell_{G,\mu}$$

of adic spaces over $\operatorname{Spa} C$ which is functorial in the tame level. Finally, the boundary of $\mathcal{S}^*_{K^p}$ is Zariski-closed.

Remark 1.2. The restriction to sufficiently large primes p in Theorem A stems solely from our reliance on the existence of integral canonical models for exceptional Shimura varieties, which is currently established by Bakker, Shankar, and Tsimerman [BST25a] only when p is ineffectively sufficiently large. Consequently, once integral canonical models are known to exist for small primes p, the identical argument will apply without change to establish the perfectoidness at infinite level for all primes p.

Scholze's method relies on the theory of canonical subgroups of abelian varieties. To transpose this machinery to general Shimura varieties, we replace Barsotti–Tate groups with group-theoretic analogues. For Shimura data of Hodge type, this is usually done by considering 'p-divisible groups with G-structure'. Such an interpretation is not known for arbitrary Shimura data, even in the general abelian-type case. The theory of (n-truncated) $(\mathcal{G},\mu)$-apertures developed by Gardner and Madapusi [GM24] serves as a group-theoretic replacement for (n-truncated) Barsotti–Tate groups functioning uniformly even when G is an exceptional reductive group. Let $\mathcal{G}$ be a reductive group scheme over $\mathbf{Z}_p$ and $\mu\colon \mathbf{G}_m \to \mathcal{G}$ a miniscule cocharacter. In brief, over a p-complete ring R, an n-truncated $(\mathcal{G},\mu)$-aperture is the data of a $\mathcal{G}$-torsor $\mathcal{E}_1$ over the mod-p^n syntomification $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ that is "bounded by μ".

When reduced modulo p, any such aperture naturally induces an F-zip with $\mathcal{G}$-structure, also called a $\mathcal{G}$-zip, over R/pR in the sense of Pink, Wedhorn, and Ziegler [PWZ15]. In the classical setting of de Rham cohomology of a smooth projective variety in characteristic p, an F-zip encodes the decreasing Hodge filtration and the increasing conjugate filtration together with the semilinear Cartier isomorphism of their associated gradeds. In the group-theoretic framework, a $\mathcal{G}$-zip consists of a $\mathcal{G}$-torsor equipped with two parabolic reductions: a reduction to the negative parabolic $\mathcal{P}_\mu^-$ (the de Rham realization, or Hodge filtration) and a reduction to the positive parabolic $\mathcal{P}_\mu^+$ (the conjugate realization, or conjugate filtration) together with a semilinear isomorphism of their associated Levi reductions (Cartier isomorphism). The work of Goldring and Koskivirta [GK19] explains how to attach Hasse invariants to any $\mathcal{G}$-zip, which they call group-theoretic Hasse invariants, measuring failure of ordinariness.

Our key result establishes a uniform *integral* theory of canonical subgroups for these group-theoretic analogues of Barsotti–Tate groups:

Theorem C (Canonical subgroup for 1-truncated apertures, Theorem 4.1.3)**.** *Let $\mathcal{G}/\mathbf{Z}_p$ be a reductive group scheme and $\mu\colon \mathbf{G}_m \to \mathcal{G}$ a minuscule cocharacter. Let $\mathcal{E}_1 \to R^{\mathrm{syn}} \otimes \mathbf{F}_p$ be a $\mathcal{G}$-torsor representing a 1-truncated $(\mathcal{G}, \mu)$-aperture over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R. Let $\varepsilon \geqslant 0$ be the p-adic valuation of its group-theoretic Hasse invariant. If $\varepsilon < \frac{p}{p+1}$, then there exists a unique reduction $\mathcal{E}_1^- \subset \mathcal{E}_1$ of the structure group to $\mathcal{P}_\mu^+$ that agrees with the reduction of structure induced by the conjugate filtration modulo $p^{1-\varepsilon}$.*

By iterating this construction, we obtain analogues of higher-level canonical subgroups as well (Theorem 4.2.8). Specializing to the case of $\mathcal{G} = \mathrm{GL}_h$, we deduce a parallel statement for Barsotti–Tate groups. Remarkably, even in this classical setting, we obtain stronger results:

Corollary D (Canonical subgroup for 1-truncated Barsotti–Tate groups, Theorem 4.3.1)**.** *Let G be a 1-truncated Barsotti–Tate group over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R with reduction G_0 modulo p. If the Hasse invariant of G has p-adic valuation $\varepsilon < \frac{p}{p+1}$, then G admits a unique finite locally-free subgroup scheme which agrees with the kernel of relative Frobenius $\mathrm{Ker}(G_0 \to G_0^{(p)})$ modulo $p^{1-\varepsilon}$.*

Corollary E (Level-n canonical subgroup for p-divisible groups, Theorem 4.3.3)**.** *Let $n \geqslant 1$ be an integer, and let G be a p-divisible group over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R with reduction G_0 modulo p. If the Hasse invariant of G has p-adic valuation $\varepsilon < \frac{p}{p^{n-1}(p+1)}$, then $G[p^n]$ admits a unique finite locally-free subgroup scheme which agrees with the kernel of n-th iterated relative Frobenius isogeny $\mathrm{Ker}(G_0 \to G_0^{(p^n)})$ modulo $p^{1-p^{n-1}\varepsilon}$.*

The theory of overconvergent canonical subgroups has a rich historical development. The case of elliptic curves was completely resolved by Katz [Kat73] and Lubin [Lub79]. For abelian varieties of general dimension, Abbes and Mokrane [AM04] resolved the level-1 case (that is, for p-torsion points) by coupling the description given by Bloch and Kato of p-adic vanishing cycles on smooth projective varieties having good reduction with the ramification theory developed by Abbes and Saito. Andreatta and Gasbarri [AG06] subsequently recovered the result of Abbes–Mokrane via alternative global methods involving abelian schemes. Tian [Tia10] then extended the result of Abbes–Mokrane to truncated Barsotti–Tate groups of level 1 by utilizing resolutions of such groups by abelian schemes and Bloch–Kato results on the associated vanishing cycles. For higher levels, Conrad [Con06] established overconvergence in general for p^n-torsion points of abelian schemes for all n, though without explicit valuation bounds. The case of Hilbert modular varieties and other classical Shimura varieties was studied in detail by Andreatta, Iovita, and Pilloni [AIP15]. Finally, a comprehensive theory of canonical subgroups of arbitrary level $n \geqslant 1$ for Barsotti–Tate groups was established by Fargues [Far11] using Harder–Narasimhan and Abbes–Saito ramification filtrations, while Rabinoff [Rab12] investigated arbitrary level canonical subgroups via completely different methods.

Compared to this extensive literature, our results differ fundamentally in both methodology and scope. A unifying characteristic of these classical approaches is their restriction to complete discrete valuation rings of mixed characteristic. In contrast, our linear-algebraic approach via prismatic theory operates uniformly over arbitrary flat p-adic formal schemes. Furthermore, as detailed in Section 4.3, our overconvergence valuation bounds ($\varepsilon < \frac{1}{p^{n-2}(p+1)}$ for level n) strictly improve upon the classical thresholds of Fargues [Far11] and Scholze [Sch15]. These bounds are strong in the sense that they recover sharp valuation bounds of Katz *et al* when specialized to elliptic curves. While Rabinoff [Rab12] obtains larger overconvergence radii, his definition differs conceptually as it does not a priori involve lifting the kernel of the relative Frobenius; similarly, Fargues characterizes canonical subgroups via Abbes–Saito ramification filtrations rather than relative Frobenius kernels.

1.1. **Overview of proofs.** Let (G, X) be a Shimura datum with reflex field E, and let $\mathcal{G}$ be a reductive model of G over $\mathbf{Z}_p$. Fix a place $v \mid p$ of E. The Shimura datum determines a minuscule cocharacter $\mu_v\colon \mathbf{G}_{\mathrm{m},\mathcal{O}_{E,v}} \to \mathcal{G}^c_{\mathcal{O}_{E,v}}$, where $\mathcal{G}^c$ is the *cuspidal quotient* of $\mathcal{G}$ (cf. Setup 3.1.1). For each integer $n \geqslant 1$, we can then attach the smooth p-adic formal Artin stack $\mathrm{BT}_n^{\mathcal{G}^c,-\mu_v}$ over $\operatorname{Spf} \mathcal{O}_{E,v}$ of n-truncated $(\mathcal{G}^c, -\mu_v)$-apertures constructed in [GM24]. Let $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v} := \varprojlim_n \mathrm{BT}_n^{\mathcal{G}^c,-\mu_v}$ denote their inverse limit. These stacks generalize the formal stacks of (truncated) polarized p-divisible groups of height $2g$ in the case of the Siegel Shimura datum, providing a group-theoretic framework for defining "p-divisible groups with $\mathcal{G}^c$-structure" without requiring one to explicitly work with p-divisible groups.

Following the construction of integral canonical models $\mathscr{S}_K$ over $\mathcal{O}_{E,(v)}$ for arbitrary Shimura varieties by Bakker, Shankar, and Tsimerman [BST25a], recent work of Madapusi and Youcis [MY26] provides an alternative characterization of these models via a Serre–Tate property: there is a canonical, formally étale *syntomic realization* map of p-adic formal stacks over $\operatorname{Spf} \mathcal{O}_{E,v}$,

$$\varpi \colon \widehat{\mathscr{S}_K} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v},$$

such that the induced $\mathcal{G}^c(\mathbf{Z}_p)$-local system on the generic fiber recovers the canonical étale local system $\mathbf{Et}_{K,p}$ (Theorem 3.1.5). In the Siegel case, this is precisely the map classifying the universal polarized p-divisible group over the p-adic completion of the integral canonical model of the Siegel modular variety. Once we have an analogue of a universal p-divisible group over integral canonical models of exceptional Shimura varieties, it is conceivable that a theory of canonical subgroups can be used to prove infinite-level perfectoidness, just as in Scholze's work [Sch15]. For technical simplicity, we prefer to develop a theory of canonical subgroups under the assumption that μ_v is defined over $\mathbf{Z}_p$, which corresponds to assuming that the local reflex field E_v is $\mathbf{Q}_p$ (as occurs when $E = \mathbf{Q}$). This is sufficient to establish infinite-level perfectoidness for integral canonical models whose local reflex field is $\mathbf{Q}_p$; the general case is then deduced via an embedding argument using the fact that a closed adic subspace of a perfectoid space is perfectoid.

Let us now explain the proof of Theorem C. For simplicity, consider the case where $\mathfrak{X} = \operatorname{Spf} R$ for a p-torsion-free perfectoid algebra R over $\mathbf{Z}_p^{\mathrm{cycl}}$. The general case is deduced by quasisyntomic descent. Let $R^\flat$ denote the tilt of R with pseudo-uniformizer $t = p^\flat$, and let φ be the absolute Frobenius on $R^\flat$. Then a 1-truncated $(\mathcal{G}, \mu)$-aperture over R is equivalent to the data of a $\mathcal{G}$-bundle $\mathcal{T}$ over $R^\flat$ equipped with a modification $m \colon \varphi^*\mathcal{T}[1/t] \xrightarrow{\sim} \mathcal{T}[1/t]$ that is "bounded by μ". When $\mathcal{G} = \mathrm{GL}_h$, this reduces to a 1-truncated Breuil–Kisin–Fargues module over R (cf. Example 2.2.17). In this setting, finite locally free subgroup schemes correspond to m-stable submodules. In our group-theoretic situation, this translates to an m-stable reduction of structure of $\mathcal{T}$ to a parabolic subgroup. For canonical subgroups, it suffices to find a reduction of structure to the positive parabolic $\mathcal{P}_\mu^+$ associated with μ such that it reduces modulo $p^{1-\varepsilon}$ to the reduction of structure induced by the conjugate filtration over $R^\flat/tR^\flat \simeq R/pR$. This latter condition encodes the fact that the subgroup lifts the Frobenius kernel. To conclude, we choose an arbitrary, not necessarily m-stable, lift of the conjugate filtration and repeatedly apply m to obtain a unique "fixed-point" m-stable reduction via a t-adic contraction mapping principle argument.

Theorem C, combined with the upcoming work of Lee and Madapusi [LM] defining and constructing stacks of p-isogenies of $(\mathcal{G}, \mu)$-apertures, is sufficient to prove Theorem A for Shimura varieties with local reflex field $\mathbf{Q}_p$ admitting integral canonical models, essentially following Scholze's method [Sch15]. The only additional ingredient is a relative perfectness result for the reduction modulo p of the syntomic realization map (Corollary 3.2.6). Interestingly, this yields a purely formal explanation for the existence of canonical Frobenius lifts over the μ-ordinary locus of Shimura varieties, which we record in Theorem 3.3.9.

For general Shimura varieties with reflex fields larger than $\mathbf{Q}$, we employ an embedding argument. This can be achieved by a routine Weil restriction trick in the pre-abelian type case; however, for exceptional Shimura varieties this does not apply directly because the bound on p for the existence of integral canonical models is ineffective. To overcome this obstacle, we prove a result (Theorem 3.8.1 in Section 3.8) on embedding an E_6 Shimura variety into an E_7 Shimura variety with a totally real reflex field. Any Shimura variety with a totally real reflex field can then be embedded into a bigger Shimura variety with reflex field $\mathbf{Q}$ via Weil restriction. Since the global closed embedding of Shimura data spreads out to integral canonical models for sufficiently large p, pulling back the infinite-level perfectoid property from the ambient space completes the proof of Theorem A.

1.2. **Structure of the paper.** In Section 2, we quickly review some basic facts about prismatic F-gauges, $(\mathcal{G}, \mu)$-apertures, and the theory of group-theoretic Hasse invariants via $\mathcal{G}$-zips. In Section 3, we recall the theory of integral canonical models, p-Hecke correspondences, minimal compactifications, study p-Hecke correspondences over the μ-ordinary locus, and prove a result on embeddings of exceptional Shimura varieties. In Section 4, we establish our main local result Theorem C and construct Frobenius lift quasi-isogenies. Finally, in Section 5, we apply our result on canonical subgroups to prove Theorem A by mirroring Scholze's method [Sch15].

1.3. **Notations and conventions.**

- Throughout this article, p denotes a prime number.
- (Cyclotomic extensions) We denote by $\mathbf{Z}_p^{\mathrm{cycl}}$ the p-adic completion of $\mathbf{Z}_p[\mu_{p^\infty}]$, and by $\mathbf{Q}_p^{\mathrm{cycl}} := \mathbf{Z}_p^{\mathrm{cycl}}[1/p]$ the cyclotomic extension of $\mathbf{Q}_p$.
- (Completions) For any commutative ring or module M and an ideal $I \subset M$, $\widehat{M} := \varprojlim_n M/I^n M$ denotes its classical I-adic completion. Whenever we refer to a "p-complete" ring, module, formal scheme, or stack, we always mean *derived* p-completeness.
- (Witt vectors and $\mathrm{A}_{\mathrm{inf}}$) For any commutative ring R, we write $W(R)$ for the ring of p-typical Witt vectors with coefficients in R, and $W_n(R) := W(R)/V^n W(R)$ for the ring of n-truncated Witt vectors of length $n \geqslant 1$, equipped with the canonical Frobenius φ and Verschiebung V operators. For a perfectoid ring R, let $R^\flat := \varprojlim_\varphi R/p$ denote its tilt, which is a perfect $\mathbf{F}_p$-algebra of characteristic p. We denote by $\mathrm{A}_{\mathrm{inf}}(R) := W(R^\flat)$ the associated infinitesimal period ring of Fontaine, equipped with its canonical Frobenius φ and Fontaine's map $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$. The kernel of $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$ is a principal ideal generated by a non-zero-divisor ξ, and we let $\widetilde{\xi} := \varphi^{-1}(\xi)$ denote a generator of $\varphi^{-1}(\operatorname{Ker}\theta)$.
- (Reductive groups) For a reductive group scheme $\mathcal{G}$ over a base ring S (such as $\mathbf{Z}_p$, $\mathbf{F}_p$, or an unramified ring of integers $\mathcal{O}$), we denote by $\mathbf{B} \subset \mathcal{G}$ a Borel subgroup scheme and by $\mathbf{T} \subset \mathbf{B}$ a maximal torus. Let $\mathbf{X}_*(\mathbf{T}) := \operatorname{Hom}_S(\mathbf{G}_m, \mathbf{T})$ and $\mathbf{X}^*(\mathbf{T}) := \operatorname{Hom}_S(\mathbf{T}, \mathbf{G}_m)$ denote the groups of cocharacters and characters of $\mathbf{T}$, respectively. For a cocharacter $\mu\colon \mathbf{G}_m \to \mathcal{G}$, we denote by $\mathcal{P}_\mu^+ \subset \mathcal{G}$ and $\mathcal{P}_\mu^- \subset \mathcal{G}$ the associated positive and negative parabolic subgroup schemes defined by the dynamic grading of μ and $-\mu$, respectively, with common Levi quotient $\mathcal{M}_\mu := \mathcal{P}_\mu^+ \cap \mathcal{P}_\mu^-$. When writing bounds or cocharacters such as $-\nu$, $-\mu$, or $-\lambda$, it is implicitly understood that we take their dominant representatives in the chosen Weyl chamber, i.e., $-w_0\nu$, $-w_0\mu$, and $-w_0\lambda$, where w_0 is the longest element of the Weyl group.
- (Adic spaces) For a p-adic field $K/\mathbf{Q}_p$, we write $\operatorname{Spa} K$ for $\operatorname{Spa}(K, \mathcal{O}_K)$. We use script font $\widehat{\mathscr{S}}, \mathfrak{X}, \ldots$ to denote p-adic formal schemes over $\mathbf{Z}_p$ or $\mathcal{O}_K$, and calligraphic font $\mathcal{S}, \mathcal{X}, \ldots$ to denote their generic fibers as analytic adic spaces over $\mathbf{Q}_p$ or $\operatorname{Spa} K$. For an adic space $\mathcal{X}/K$, we denote by $X^\diamond$ its diamond over $\operatorname{Spd} K$.

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2. Prismatic Dieudonné theory

2.1. **Prismatic F-gauges.** We briefly recall some basic facts about cohomological stacks of Bhatt–Lurie and Drinfeld.

Definition 2.1.1 (Quasisyntomic morphisms)**.** A map $R \to S$ of p-complete rings is called p**-quasisyntomic** (or simply **quasisyntomic**) if it is p-completely flat (meaning $S/^{\mathbf{L}}p$ is flat over $R/^{\mathbf{L}}p$), and if the cotangent complex $\mathbf{L}_{S/R}$ has p-complete Tor amplitude in $[-1,0]$ (meaning $\mathbf{L}_{S/R} \otimes^{\mathbf{L}} \mathbf{F}_p$ has Tor amplitude in $[-1,0]$ over $S/^{\mathbf{L}}p$). Furthermore, $R \to S$ is a **quasisyntomic cover** if it is quasisyntomic and $S/^{\mathbf{L}}p$ is faithfully flat over $R/^{\mathbf{L}}p$. These properties are stable under base change along arbitrary morphisms of p-complete rings.

Example 2.1.2. The prototypical example of a quasisyntomic cover is $\mathbf{Z}_p\langle T\rangle \to \mathbf{Z}_p\langle T^{1/p^\infty}\rangle$.

Definition 2.1.3 (Quasisyntomic formal schemes)**.** A p-complete bounded p^∞-torsion ring R is **quasisyntomic** if the structure map $\mathbf{Z}_p \to R$ is quasisyntomic. A p-adic formal scheme X is **quasisyntomic** if it admits an open cover by affine formal schemes $\operatorname{Spf} R$ where each R is a quasisyntomic ring.

Remark 2.1.4 (Relation to classical notions)**.** In the p-complete context, this definition naturally extends the classical algebraic geometry concepts of locally complete intersection (lci) rings and syntomic morphisms—which are traditionally characterized as flat and lci—to the non-Noetherian and non-finite-type realm.

Definition 2.1.5 (Quasiregular semiperfectoid rings)**.** A ring R is called **quasiregular semiperfectoid (qrsp)** if it is a quasisyntomic ring and there exists a perfectoid ring S equipped with a surjective ring homomorphism $S \twoheadrightarrow R$.

Example 2.1.6. Any p-complete bounded p^∞-torsion quotient of a perfectoid ring by a finite regular sequence is quasiregular semiperfectoid.

Proposition 2.1.7. *Let R be a quasisyntomic ring. There exists a quasisyntomic cover $R \to R'$ such that R' is quasiregular semiperfectoid. Moreover, every term in the associated Čech nerve $R'^{\bullet}$ is a quasiregular semiperfectoid ring.*

Proof. See [BMS19, Lemmas 4.27 and 4.29]. □

Example 2.1.8 (Perfectoid covers of smooth rings)**.** Let R be a p-completely smooth $\mathbf{Z}_p$-algebra. While R itself is typically not perfectoid, it always admits a quasisyntomic cover by a perfectoid ring. To see this, we can first localize R in the p-completely étale topology to assume that there exists a p-completely étale morphism $\mathbf{Z}_p\langle T_1, \dots, T_n\rangle \to R$. Letting $\mathbf{Z}_p^{\mathrm{cycl}}$ denote the p-adic completion of $\mathbf{Z}_p[\mu_{p^\infty}]$, we have the standard perfectoid cover $P = \mathbf{Z}_p^{\mathrm{cycl}}\langle T_1^{1/p^\infty}, \dots, T_n^{1/p^\infty}\rangle_p^\wedge$ of $\mathbf{Z}_p\langle T_1, \dots, T_n\rangle$ given by adjoining p^∞-th roots of the variables. Base changing along our étale map yields the p-complete tensor product $S = R\widehat{\otimes}_{\mathbf{Z}_p\langle T_1,\dots,T_n\rangle} P$. Because p-completely étale extensions of perfectoid rings are again perfectoid, S is a perfectoid ring. The induced map $R \to S$ is a quasisyntomic cover.

To any (derived) p-adic formal scheme $\mathfrak{X}$, one can canonically attach three p-adic formal stacks: the **prismatization** $\mathfrak{X}^{\mathbb{\Delta}}$, the **(Nygaard) filtered prismatization** $\mathfrak{X}^{\mathcal{N}}$, and the **syntomification** $\mathfrak{X}^{\mathrm{syn}}$. When $\mathfrak{X} = \operatorname{Spf} R$, we often denote these by $R^{\mathbb{\Delta}}, R^{\mathcal{N}}$, and R^{syn}. These stacks are not classical in general: they are derived stacks that are covered flat locally by the spectra of animated rings.

Recollection 2.1.9 (Absolute prismatization)**.** The prismatization $\mathfrak{X}^{\mathbb{\Delta}}$ classifies Cartier–Witt divisors on $\mathfrak{X}$. Specifically, given an object $(A, I, \operatorname{Spf} A/I \to \mathfrak{X})$ in the absolute prismatic site $\mathfrak{X}_{\mathbb{\Delta}}$, there is a natural morphism $\iota_{(A,I)} \colon \operatorname{Spf} A \to X^{\mathbb{\Delta}}$. As explained in [BL22, Construction 3.10], this map sends an A-algebra B (with B being p-nilpotent) to the Cartier–Witt divisor $I \otimes_A W(B) \to W(B)$, where $A \to W(B)$ is the unique δ-ring homomorphism lifting the structure map $A \to B$.

Recollection 2.1.10 (Nygaard filtration)**.** As demonstrated by Bhatt and Lurie, evaluating these stacks on quasiregular semiperfectoid (qrsp) rings yields objects governed by Nygaard-filtered absolute prismatic cohomology. For a qrsp ring R, its **absolute prismatic cohomology** is given by the initial prism $(\mathbb{\Delta}_R, I_R)$. The **Nygaard filtration** $\mathrm{Fil}^\bullet_{\mathcal{N}}\mathbb{\Delta}_R$ on $\mathbb{\Delta}_R$ is given by

$$\mathrm{Fil}^i_{\mathcal{N}}\mathbb{\Delta}_R = \{x \in \mathbb{\Delta}_R \;:\; \varphi(x) \in \mathrm{Fil}^i_{I_R}\mathbb{\Delta}_R\},$$

with $\mathrm{Fil}^\bullet_{I_R}\mathbb{\Delta}_R$ denoting the I_R-adic filtration. By construction, the Nygaard filtration satisfies quasisyntomic descent and preserves sifted colimits.

Recollection 2.1.11 (Filtered prismatization and syntomification)**.** The value of the filtered prismatization $\mathfrak{X}^{\mathcal{N}}$ is canonically identified with the formal Rees stack of the **Nygaard filtered absolute prismatic cohomology** $\mathrm{Fil}^{\bullet}_{\mathcal{N}}{\Delta}_R$ on test qrsp rings R. Under this correspondence, j_{dR} corresponds to stripping away the filtration, and j_{HT} is induced by the divided Frobenius map $\mathrm{Fil}^{\bullet}_{\mathcal{N}}{\Delta}_R \to \mathrm{Fil}^{\bullet}_{I_R}{\Delta}_R$. The stack $\mathfrak{X}^{\mathcal{N}}$ is a *filtered stack*, meaning it sits over the formal Artin stack $\mathbf{A}^1/\mathbf{G}_m$ which classifies pairs of a line bundle $\mathcal{L}$ and a cosection $\mathcal{L} \to C$ on p-complete test rings C. The fiber over the open substack $\mathrm{Spf}\,\mathbf{Z}_p \times B\mathbf{G}_m \subset \mathrm{Spf}\,\mathbf{Z}_p \times \mathbf{A}^1/\mathbf{G}_m$ is canonically identified with $\mathfrak{X}^{\Delta}$. This open substack is referred to as the **de Rham locus** of $X^{\mathcal{N}}$, and we denote its inclusion by $j_{\mathrm{dR}}\colon X^{\Delta} \to X^{\mathcal{N}}$. Additionally, there is a second open immersion $j_{\mathrm{HT}}\colon X^{\Delta} \to X^{\mathcal{N}}$ whose image is disjoint from the de Rham locus, known as the **Hodge–Tate locus**. The **syntomification** X^{syn} is constructed as the coequalizer of j_{dR} and j_{HT}. Consequently, it comes with a natural open immersion $j_{\Delta}\colon X^{\Delta} \to X^{\mathrm{syn}}$.

Recollection 2.1.12 (The de Rham point and its filtered version)**.** There exist canonical morphisms

$$x^{\mathcal{N}}_{\mathrm{dR}}\colon \mathbf{A}^1/\mathbf{G}_m \times \mathfrak{X} \to \mathfrak{X}^{\mathcal{N}} \quad \text{and} \quad x_{\mathrm{dR}}\colon \mathfrak{X} \to \mathfrak{X}^{\Delta},$$

where the latter is simply the restriction of the former to $B\mathbf{G}_m$. When evaluated on a test qrsp ring R, $x^{\mathcal{N}}_{\mathrm{dR}}$ translates to the morphism of formal Rees stacks induced by the canonical filtered map from $\mathrm{Fil}^{\bullet}_{\mathcal{N}}{\Delta}_R$ to R, endowing R with the trivial descending filtration. Using the canonical étale map $\mathfrak{X}^{\mathcal{N}} \to \mathfrak{X}^{\mathrm{syn}}$, we get maps $\mathbf{A}^1/\mathbf{G}_m \times \mathfrak{X} \to \mathfrak{X}^{\mathrm{syn}}$ and $\mathfrak{X} \to \mathfrak{X}^{\mathrm{syn}}$ which we also denote by $x^{\mathcal{N}}_{\mathrm{dR}}$ and x_{dR} respectively. These are called **(filtered) de Rham points**.

Example 2.1.13 (The case of perfectoid rings)**.** For a perfectoid ring R, its absolute prismatic cohomology is given by $\mathrm{A}_{\mathrm{inf}}(R)$ equipped with the Cartier divisor induced by Fontaine's map $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$. Thus, the prismatization R^{Δ} is canonically identified with the affine formal scheme $\mathrm{Spf}\,\mathrm{A}_{\mathrm{inf}}(R)$, and the de Rham point $x_{\mathrm{dR}}\colon \mathrm{Spf}\,R \to R^{\Delta}$ corresponds exactly to the closed immersion defined by θ. The Nygaard filtration is given by $\mathrm{Fil}^i_{\mathcal{N}}\mathrm{A}_{\mathrm{inf}}(R) = \widetilde{\xi}^i\mathrm{A}_{\mathrm{inf}}(R)$, where $\widetilde{\xi}$ is a generator of $\varphi^{-1}(\mathrm{Ker}\,\theta)$. The filtered prismatization $R^{\mathcal{N}}$ is the formal Rees stack associated to this filtration, which admits the explicit quotient presentation $R^{\mathcal{N}} \simeq [\mathrm{Spf}\,\mathrm{A}_{\mathrm{inf}}(R)\{u,v\}/(uv-\widetilde{\xi})/\mathbf{G}_m]$, where the action of $\mathbf{G}_m$ assigns weights 1 and -1 to u and v respectively. The syntomification R^{syn} is a stack whose category of vector bundles $\mathrm{Vect}(R^{\mathrm{syn}})$ is naturally equivalent to the category of Breuil–Kisin–Fargues modules over $\mathrm{A}_{\mathrm{inf}}(R)$, i.e., finite projective $\mathrm{A}_{\mathrm{inf}}(R)$-modules M equipped with an isomorphism $\varphi^*M[1/\widetilde{\xi}] \xrightarrow{\sim} M[1/\widetilde{\xi}]$.

Proposition 2.1.14. *For any p-completely étale map $R \to S$ of p-complete rings, the resulting stack morphisms*

$$S^{\mathcal{N}} \to R^{\mathcal{N}}, \qquad S^{\Delta} \to R^{\Delta}$$

are (p, I)-completely étale.

Proof. See, for instance, [GM24, Proposition 6.12.1]. □

A key feature of quasisyntomic covers is that taking prismatizations or syntomifications converts them into flat covers.

Theorem 2.1.15 (Quasisyntomic descent)**.** *Let $X \to Y$ be a quasisyntomic cover of bounded quasisyntomic p-adic formal schemes. Then the induced morphism of formal stacks*

$$X^{\Delta} \to Y^{\Delta}, \quad X^{\mathcal{N}} \to Y^{\mathcal{N}}, \quad \text{and} \quad X^{\mathrm{syn}} \to Y^{\mathrm{syn}}$$

are surjective in the flat topology.

Proof. See [Pen26, Proposition 2.19, Corollary 2.21]. □

Definition 2.1.16 (Vector bundles in prismatic F-gauges)**.** A **vector bundle in prismatic F-gauges** (or **vector bundle F-gauge**) on $\mathfrak{X}$ is defined to be a vector bundle on the syntomification $\mathfrak{X}^{\mathrm{syn}}$. The category of such objects is denoted by $\mathrm{Vect}\,\mathfrak{X}^{\mathrm{syn}}$.

Recollection 2.1.17 (Prismatic F-crystals)**.** Following [BS23], let us recall the definitions of *prismatic crystals* and *prismatic F-crystals* on $\mathfrak{X}$. These are modules over the absolute prismatic site $\mathfrak{X}_{\Delta}$ satisfying the usual crystal condition. Consider the structure sheaf $\mathcal{O}_{\Delta}$ and the generalized Cartier divisor $\mathcal{I}_{\Delta} \to \mathcal{O}_{\Delta}$, which on an object $(A, I, \operatorname{Spf} A/I \to \mathfrak{X})$ evaluate to A and $I \to A$, respectively. A **prismatic crystal** of vector bundles on $\mathfrak{X}$ is a locally free $\mathcal{O}_{\Delta}$-module $\mathcal{E}$ of finite rank satisfying the crystal condition. A **prismatic F-crystal** consists of a prismatic crystal $\mathcal{E}$ together with an isomorphism

$$\varphi_{\mathcal{E}} \colon \varphi^*\mathcal{E}[1/\mathcal{I}_{\Delta}] \xrightarrow{\sim} \mathcal{E}[1/\mathcal{I}_{\Delta}].$$

The category of prismatic F-crystals is denoted by $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta})$. More concretely, a prismatic crystal assigns a finite locally free A-module $\mathcal{E}(A, I)$ to each prism (A, I) over X, compatible with base change. A prismatic F-crystal structure then corresponds to a family of isomorphisms $\mathcal{E}(A, I)[1/I] \otimes_{A,\varphi} A \xrightarrow{\sim} \mathcal{E}(A, I)[1/I]$ that respect the crystal structure. Similarly, a **mod p^n prismatic F-crystal** is a vector bundle $\mathcal{E}$ on the ringed site $(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ together with an isomorphism $\varphi_{\mathcal{E}} \colon \varphi^*\mathcal{E}[1/\mathcal{I}_{\Delta}] \xrightarrow{\sim} \mathcal{E}[1/\mathcal{I}_{\Delta}]$. We denote the category of mod p^n prismatic F-crystals by $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$.

Recollection 2.1.18 (Prismatic F-gauges to prismatic F-crystals)**.** If a vector bundle on $\mathfrak{X}^{\Delta}$ is the pullback of a vector bundle F-gauge along $j_{\Delta} \colon X^{\Delta} \to \mathfrak{X}^{\mathrm{syn}}$, the gluing data defining $\mathfrak{X}^{\mathrm{syn}}$ automatically equips it with the structure of a prismatic F-crystal. This establishes a canonical functor $\mathrm{Vect}(\mathfrak{X}^{\mathrm{syn}}) \to \mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta})$. This functor is fully faithful when $\mathfrak{X}$ is flat and p-quasisyntomic by [GL25, Corollary 3.53]. Mod p^n versions of these objects can be defined analogously, and the fully faithful functor extends to the mod p^n setting: $\mathrm{Vect}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \to \mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$.

Recollection 2.1.19 (Tannakian formalism)**.** Let $\mathcal{G}$ be a smooth affine group scheme over $\mathbf{Z}_p$. Following [IKY24], a **prismatic $\mathcal{G}$-torsor with F-structure** on $\mathfrak{X}$ is a pair $(\mathcal{P}, \varphi_{\mathcal{P}})$, where $\mathcal{P}$ is a $\mathcal{G}$-torsor on the absolute prismatic site $\mathfrak{X}_{\Delta}$ and $\varphi_{\mathcal{P}} \colon (\varphi^*\mathcal{P})[1/\mathcal{I}_{\Delta}] \xrightarrow{\sim} \mathcal{P}[1/\mathcal{I}_{\Delta}]$ is an isomorphism of $\mathcal{G}$-torsors. The category of such objects is denoted by $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta})$.

A **$\mathcal{G}$-bundle in prismatic F-crystals** on $\mathfrak{X}$ is an exact $\mathbf{Z}_p$-linear $\otimes$-functor from the category $\mathrm{Rep}_{\mathbf{Z}_p}(\mathcal{G})$ of representations of $\mathcal{G}$ on finite free $\mathbf{Z}_p$-modules to the category $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta})$ of prismatic F-crystals on $\mathfrak{X}$. We denote the category of such $\mathcal{G}$-objects by $\mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta})$.

The mod p^n versions of these concepts are defined analogously, giving categories $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ and $\mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$. The category of prismatic F-crystals over $\mathfrak{X}$ possesses a well-behaved Tannakian formalism. Specifically, by [IKY24, Proposition 1.28], there are canonical equivalences of categories:

$$\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}) \xrightarrow{\sim} \mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}) \quad \text{and} \quad \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n) \xrightarrow{\sim} \mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n).$$

Recollection 2.1.20 (Étale realization)**.** Let $\mathfrak{X}$ be a flat quasisyntomic p-adic formal scheme. Let $\mathcal{G}$ be a smooth affine group scheme over $\mathbf{Z}_p$. Write $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ for the category of mod p^n prismatic $\mathcal{G}$-torsors with F-structure on $\mathfrak{X}$. We claim that, by Recollection 2.1.18 and the Tannakian formalism (Recollection 2.1.19), there is a natural functor

$$G \colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \to \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n).$$

Indeed, given any $\mathcal{G}$-torsor $\mathcal{P} \in \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathcal{N}} \otimes \mathbf{Z}/p^n\mathbf{Z})$, we obtain a natural correspondence

$$j^*_{\mathrm{HT}}\mathcal{P} \xleftarrow{a} \varphi^*\pi^*\mathcal{P} \xrightarrow{b} \varphi^* j^*_{\mathrm{dR}}\mathcal{P}$$

where b is defined by applying φ^* to the canonical map $\pi^*\mathcal{P} \to j^*_{\mathrm{dR}}\mathcal{P}$ (arising via functoriality of pullback along $j_{\mathrm{dR}} \colon \mathfrak{X}_{\Delta} \to \mathfrak{X}^{\mathcal{N}}$ as a map of stacks over $\mathfrak{X}_{\Delta}$), while a is defined by adjunction from the analogous map $\pi^*\mathcal{P} \to j^*_{\mathrm{HT}}\mathcal{P}$ as well as the observation that $j_{\mathrm{HT}} \circ \pi$ equals the Frobenius φ on $\mathfrak{X}_{\Delta}$. Passing to vector bundles via faithful representations of $\mathcal{G}$, a local calculation by descent to the qrsp case shows that both a and b become isomorphisms after inverting $\mathcal{I}_{\Delta}$. If $\mathcal{P} = j^*_{\mathcal{N}}\mathcal{Q}$ comes from a $\mathcal{G}$-torsor $\mathcal{Q} \in \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z})$, then we have canonical identifications $j^*_{\mathrm{HT}}\mathcal{P} \simeq j^*_{\mathrm{dR}}\mathcal{P}$, so inverting $\mathcal{I}_{\Delta}$ in the diagram above provides the desired object $G(\mathcal{Q}) \in \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$.

Using the functor G, we obtain the composition

$$\mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \xrightarrow{G} \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}/p^n) \xrightarrow{(-)[1/\mathcal{I}_{\mathbb{\Delta}}]} \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathcal{I}_{\mathbb{\Delta}}]/p^n)^{\varphi^*=1} \simeq \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(\mathfrak{X}^{\mathrm{ad}}),$$

where $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(\mathfrak{X}^{\mathrm{ad}})$ denotes the category of pro-étale $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$-local systems on the adic generic fiber $\mathfrak{X}^{\mathrm{ad}}$. This is the **étale realization** $T_{\text{ét}}\colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(\mathfrak{X}^{\mathrm{ad}})$. Taking the inverse limit over n, we obtain the étale realization functor for $\mathcal{G}$-torsors on $\mathfrak{X}^{\mathrm{syn}}$:

$$T_{\text{ét}}\colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}}) \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(\mathfrak{X}^{\mathrm{ad}}).$$

Remark 2.1.21 (Modification of syntomic torsors)**.** Defining a modification of $\mathcal{G}$-torsors over $\mathfrak{X}^{\mathrm{syn}}$ along '$p = 0$' requires some care.

(1) **Evaluating on prisms**: By Recollection 2.1.19, a $\mathcal{G}$-torsor over $\mathfrak{X}^{\mathrm{syn}}$ gives a $\mathcal{G}$-bundle in prismatic F-crystals. For any p-torsion free prism $(A, I) \in (\mathfrak{X})_{\mathbb{\Delta}}$, A is a genuine commutative ring where p is a non-zero divisor, making $A[1/p]$ well-defined. A *modification* of $\mathcal{G}$-torsors $\mathcal{P}_1 \dashrightarrow \mathcal{P}_2$ over $\mathfrak{X}^{\mathrm{syn}}$ can be defined as a compatible system of $\mathcal{G}$-torsor isomorphisms $\mathcal{P}_1(A, I)[1/p] \xrightarrow{\sim} \mathcal{P}_2(A, I)[1/p]$ across all p-torsion free prisms $(A, I) \in (\mathfrak{X})_{\mathbb{\Delta}}$.
(2) **Pushforwards**: Alternatively, one can also avoid inverting p entirely. Given a cocharacter λ of $\mathcal{G}$ defined over $\mathbf{Z}_p$ and a $\mathcal{G}$-torsor $\mathcal{P}_1$ over $\mathfrak{X}^{\mathrm{syn}}$, a modification of constant type λ can be defined if $\mathcal{P}_1$ admits a reduction of structure to the group scheme $\mathcal{H}_\lambda$ defined in Remark 2.2.10 (using $d = p$). Specifically, the modified $\mathcal{G}$-torsor is defined as the pushforward $\mathcal{P}_2 := \mathcal{P}_1^- \times^{\mathcal{H}_\lambda, c_\lambda} \mathcal{G}$ (cf. Remark 2.2.10). By construction, this modification is of constant type λ.
(3) **Vinberg monoids**: There is a third way to define modifications using Vinberg monoids which is essentially a rephrasing of the second approach. The Vinberg monoid should be viewed as a 'local model' for the space of modifications. This will be explained in Section 3.5.

These perspectives are compatible and give us a good notion of modification of syntomic torsors at least when $\mathfrak{X}$ is flat and quasisyntomic. Furthermore, a modification of $\mathcal{G}$-torsors over $\mathfrak{X}^{\mathrm{syn}}$ naturally induces an isomorphism between their associated pro-étale local systems over $\mathfrak{X}^{\mathrm{ad}}$. This follows essentially by construction of the étale realization functor.

2.2. $(\mathcal{G}, \mu)$**-apertures.** We recall the main results of [GM24]. For more details, see *loc. cit.* or [MY26, §3].

Setup 2.2.1. Let $\mathcal{G}$ be a smooth connected affine group scheme over $\mathbf{Z}_p$, $\mathcal{O}$ the ring of integers in a finite unramified extension of $\mathbf{Q}_p$, and $\mu\colon \mathbf{G}_{\mathrm{m},\mathcal{O}} \to \mathcal{G}_{\mathcal{O}}$ a 1-bounded cocharacter. This means that the weights on $\mathrm{Lie}\,\mathcal{G}_{\mathcal{O}}$ via the adjoint action are bounded above by 1. Let $\mathcal{P}_\mu^- \subset \mathcal{G}$ (resp. $\mathcal{P}_\mu^+ \subset \mathcal{G}$) be the smooth subgroup scheme whose Lie algebra is identified with the sum of the weight-i spaces in $\mathrm{Lie}\,\mathcal{G}_{\mathcal{O}}$ for $i \leqslant 0$ (resp. $i \geqslant 0$). In practice, $\mathcal{G}$ will be reductive in which case this is same as saying μ is miniscule and $\mathcal{P}_\mu^\pm$ is a pair of opposite parabolic subgroups associated with μ.

Definition 2.2.2 ($(\mathcal{G}, \mu)$-apertures)**.** For any p-complete (animated) $\mathcal{O}$-algebra R, an n**-truncated** $(\mathcal{G}, \mu)$**-aperture** over R is a $\mathcal{G}$-torsor $\mathcal{E}$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ satisfying the following condition: for every geometric point $R \to \kappa$, the restriction of $(x_{\mathrm{dR}}^{\mathcal{N}})^*\mathcal{E}$ over $B\mathbf{G}_m \times \mathrm{Spec}\,\kappa$ is isomorphic to that of $\mathcal{Q}_\mu$, where $\mathcal{Q}_\mu$ is the $\mathcal{G}$-torsor over $B\mathbf{G}_{\mathrm{m},\mathcal{O}}$ classified by $B\mu\colon B\mathbf{G}_{\mathrm{m},\mathcal{O}} \to B\mathcal{G}_{\mathcal{O}}$. The ∞-groupoid of n-truncated apertures over R is denoted $\mathrm{BT}_n^{\mathcal{G},\mu}(R)$, and the assignment $R \mapsto \mathrm{BT}_n^{\mathcal{G},\mu}(R)$ is a (derived) p-adic formal prestack. Also define

$$\mathrm{BT}_\infty^{\mathcal{G},\mu} := \varprojlim_n \mathrm{BT}_n^{\mathcal{G},\mu}.$$

Objects in $\mathrm{BT}_\infty^{\mathcal{G},\mu}(R)$ are called **(**∞**-truncated)** $(\mathcal{G}, \mu)$**-apertures** over R.

Theorem 2.2.3. *The p-adic formal prestack $\mathrm{BT}_n^{\mathcal{G},\mu}$ is represented by a 0-dimensional quasi-compact smooth p-adic formal Artin stack over $\mathcal{O}$ with affine diagonal. Furthermore, the natural map $\mathrm{BT}_{n+1}^{\mathcal{G},\mu} \to \mathrm{BT}_n^{\mathcal{G},\mu}$ is smooth and surjective.*

Proof. See [GM24, Theorem D]. □

Example 2.2.4 (The case of GL_n)**.** Consider the case when $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$ where $d \leqslant h$ and μ_d is the miniscule cocharacter defined over $\mathbf{Z}_p$ given by the diagonal matrix

$$\mu_d(z) = \mathrm{diag}(\underbrace{z, z, \ldots, z}_{h}, \underbrace{1, \ldots, 1}_{h-d}).$$

It is shown in [GM24, Theorem 11.2.7] that $\mathrm{BT}_n^{\mathrm{GL}_h, \mu_d}$ is canonically isomorphic to the moduli stack of n-truncated Barsotti-Tate groups of height h and dimension d. Hence, Theorem 2.2.3 recovers a classical result of Grothendieck [Ill85].

Definition 2.2.5 (Breuil–Kisin frames)**.** Let R be a p-complete ring. A **Breuil–Kisin frame** for R is a prism $\underline{A} = (A, I)$ such that A is p-complete and p-torsion free, $I \subset A$ is an ideal and $R = A/I$.

Assumption 2.2.6. From now on, the underlying ideal I for Breuil–Kisin frames $\underline{A} = (A, I)$ we consider is going to be principal. This is a harmless assumption because any prism admits a faithfully flat cover by an orientable prism.

Example 2.2.7 (Frames for perfectoid rings)**.** If R is a perfectoid ring, then $\underline{\Delta}_R := (\Delta_R, \mathrm{Fil}^1_{\mathcal{N}} \Delta_R)$ is a Breuil–Kisin frame for R. We call this the $\mathrm{A}_{\mathrm{inf}}$ **frame**.

Example 2.2.8 (Classical Breuil–Kisin frames)**.** Let K be a finite totally ramified extension of an unramified field $K_0/\mathbf{Q}_p$. Let π be a uniformizer of $\mathcal{O}_K$ with minimal polynomial $E(u) \in \mathcal{O}_{K_0}[u]$ over K_0. The classical Breuil–Kisin frame for $\mathcal{O}_K$ is the prism $\underline{\mathfrak{S}} = (\mathfrak{S}, (E(u)))$, where $\mathfrak{S} := \mathcal{O}_{K_0}[\![u]\!]$ is equipped with the Frobenius φ acting as the natural Frobenius on $\mathcal{O}_{K_0}$ and satisfying $\varphi(u) = u^p$. The structure map $\mathfrak{S} \to \mathcal{O}_K$ evaluates u at π.

Remark 2.2.9 (Dilatations)**.** Let R be a ring a $d \in R$ a non-zerodivisor. Let $\mathcal{G}$ be a flat affine group scheme over R, and let $H_N \hookrightarrow \mathcal{G}_{R/d^N R}$ be a closed subgroup scheme, $N \geqslant 1$. The dilatation of $\mathcal{G}$ along H_N is the flat affine R-group scheme $\mathcal{G}^{[H_N]}$ characterized by the following universal property: for every flat R-algebra A,

$$\mathcal{G}^{[H_N]}(A) = \{\, g \in \mathcal{G}(A) \mid g \bmod d^N \in H_N(A/d^N A) \,\}.$$

The structural group morphism $\mathcal{G}^{[H_N]} \longrightarrow \mathcal{G}$ is an isomorphism over $\operatorname{Spec} R[1/d]$. This is the Néron blow-up, or dilatation, of $\mathcal{G}$ along H_N; see [BLR90, §3.2].

Remark 2.2.10 (Modifications via pushforwards)**.** Let $\mathcal{G}$ be a reductive group scheme over R and $d \in R$ a non-zerodivisor. A **modification** of $\mathcal{G}$-torsors is a triple $(\mathcal{T}_1, \mathcal{T}_2, m)$, where $\mathcal{T}_1$ and $\mathcal{T}_2$ are $\mathcal{G}$-torsors on $\operatorname{Spec} R$ and $m : \mathcal{T}_1|_{d \neq 0} \xrightarrow{\sim} \mathcal{T}_2|_{d \neq 0}$ is an isomorphism of $\mathcal{G}$-torsors. After fixing $\mathcal{T}_1$, isomorphism classes of such pairs $(\mathcal{T}_2, m)$ are equivalently sections of the associated affine Grassmannian fibration $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G}}$, $\mathrm{Gr}_{\mathcal{G}} = L\mathcal{G}/L^+\mathcal{G}$, where $L\mathcal{G}(A) = \mathcal{G}(A[1/d])$ and $L^+\mathcal{G}(A) = \mathcal{G}(A)$. If λ is a dominant cocharacter for which the Schubert bound $\mathrm{Gr}_{\mathcal{G}, \leqslant \lambda} \subset \mathrm{Gr}_{\mathcal{G}}$ is defined, then the modification is bounded by λ precisely when the associated section factors through $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G}, \leqslant \lambda}$. The relative position may vary in the Schubert stratification of $\mathrm{Gr}_{\mathcal{G}, \leqslant \lambda}$, taking values $\lambda' \leqslant \lambda$.

There is, however, a canonical group scheme construction for the open stratum. Suppose that $\lambda : \mathbf{G}_{m,R} \to \mathcal{G}$ is defined over R. The image $\lambda(d)$ lies in $\mathcal{G}(R[1/d])$. Define a group scheme $\mathcal{H}_\lambda$ characterised by: for every flat R-algebra A,

$$\mathcal{H}_\lambda(A) = \{\, h \in \mathcal{G}(A) \mid \lambda(d) h \lambda(d)^{-1} \in \mathcal{G}(A) \,\}.$$

Then the inclusion $i : \mathcal{H}_\lambda \hookrightarrow \mathcal{G}$ and the conjugation morphism

$$c_\lambda = \operatorname{Int} \lambda(d) : \mathcal{H}_\lambda \longrightarrow \mathcal{G}$$

are homomorphisms of R-group schemes. If $\mathcal{P}$ is a right $\mathcal{H}_\lambda$-torsor, then

$$\mathcal{T}_1 = \mathcal{P} \times^{\mathcal{H}_\lambda, i} \mathcal{G}, \qquad \mathcal{T}_2 = \mathcal{P} \times^{\mathcal{H}_\lambda, c_\lambda} \mathcal{G}$$

are $\mathcal{G}$-torsors, and over $\operatorname{Spec} R[1/d]$ there is a canonical isomorphism

$$m_\lambda : \mathcal{T}_1|_{d\neq 0} \xrightarrow{\sim} \mathcal{T}_2|_{d\neq 0}, \qquad [p,g] \longmapsto [p, \lambda(d)g].$$

This modification is, étale locally on $\operatorname{Spec} R$, the standard modification of relative position λ.

Conversely, a modification whose associated section of $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G},\leqslant\lambda}$ factors through the open Schubert stratum $\mathrm{Gr}_{\mathcal{G},\lambda}$ is recovered from the $\mathcal{H}_\lambda$-torsor of isomorphisms with the standard modification. To conclude, the data of a modification $(\mathcal{T}_1, \mathcal{T}_2, m)$ of constant type λ is equivalent to the data of a reduction of structure of $\mathcal{T}_1$ to $\mathcal{H}_\lambda$.

Remark 2.2.11 (Relation with parabolic dilatations)**.** Assume $\mathcal{G}$ is a reductive group scheme over R and $\lambda\colon \mathbf{G}_{m,R} \to \mathcal{G}$ is a cocharacter defined over R. Let $\mathcal{P}_\lambda^+ \subset \mathcal{G}$ (resp. $\mathcal{U}_\lambda^-$) be the parabolic subgroup (resp. opposite unipotent radical) corresponding to the non-negative (resp. negative) weights of $\mathrm{Ad}(\lambda)$.

The group scheme $\mathcal{H}_\lambda$ of Remark 2.2.10 can be explicitly described via its effect on the Lie algebra. The conjugation action of $\lambda(d)$ scales the weight i subspace ($i<0$) of $\mathrm{Lie}(\mathcal{U}_\lambda^-)$ by d^i. Consequently, $\mathcal{H}_\lambda$ is the unique flat group model over R that leaves $\mathcal{P}_\lambda^+$ unmodified while replacing each weight i root subgroup $U_i \subset \mathcal{U}_\lambda^-$ with its dilatation along the identity of depth $-i$.

Let $-N$ be the minimum weight of $\mathrm{Ad}(\lambda)$ on $\mathrm{Lie}(\mathcal{G})$. The depth-$N$ parabolic dilatation $\mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^N R}]}$ imposes d^N-divisibility on the entirety of $\mathcal{U}_\lambda^-$. Therefore, as subfunctors of $\mathcal{G}$, we have

$$\mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^N R}]} \subset \mathcal{H}_\lambda \subset \mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/dR}]},$$

with equality on the left if and only if all negative weights of λ are $-N$. In particular, if $\lambda = N\mu$ for a minuscule cocharacter μ, then $\mathcal{H}_\lambda = \mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^N R}]}$. Thus, when λ is miniscule, a modification bounded by λ is precisely the data of a reduction of structure of $\mathcal{T}_1|_{d=0}$ to $\mathcal{P}_\lambda^+$. In general, an $\mathcal{H}_\lambda$-torsor canonically induces a $\mathcal{P}_\lambda^+$-reduction only modulo d. Modulo d^N, the induced reduction is to a subgroup scheme intermediate between $(\mathcal{P}_\lambda^+)_{R/d^N R}$ and $\mathcal{G}_{R/d^N R}$.

Definition 2.2.12 ($(\mathcal{G},\mu)$-windows over frames)**.** Let $\underline{A} = (A, I)$ be a Breuil-Kisin frame for R equipped with a distinguished generator $E \in I$. An ***n*-truncated** $(\mathcal{G},\mu)$**-window** over $\underline{A}$ is a triple $(\mathcal{Q}, Q^-, \alpha)$ where $\mathcal{Q}$ is a $\mathcal{G}$-torsor over $A/^{\mathbf{L}}p^n A$, $Q^- \subset Q := \mathcal{Q}|_{R/^{\mathbf{L}}p^n R}$ is a reduction of structure group of Q to a $\mathcal{P}_\mu^-$-torsor, and $\alpha\colon \varphi^*\mathcal{Q}' \xrightarrow{\sim} \mathcal{Q}$ is an isomorphism of $\mathcal{G}$-torsors, where $\mathcal{Q}'$ is the modification of $\mathcal{Q}$ along Q^-. A(n ∞**-truncated)** $(\mathcal{G},\mu)$**-window** over $\underline{A}$ is the data of a compatible system of n-truncated $(\mathcal{G},\mu)$-windows over $\underline{A}$ for all $n \geqslant 1$.

Remark 2.2.13 (Description in terms of modifications)**.** In the notations of Definition 2.2.12, assume A and R are p-torsion free. Then $A/^{\mathbf{L}}p^n A$ and $R/^{\mathbf{L}}p^n R$ can be replaced with $A/p^n A$ and $R/p^n R$ in the above definition respectively, and an n-truncated $(\mathcal{G},\mu)$-window over $\underline{A}$ can be equivalently described as a pair $(\mathcal{E}, m)$ consisting of a $\mathcal{G}$-torsor $\mathcal{E}$ over $A/p^n A$ and a modification

$$m\colon \varphi^*\mathcal{E}[1/I] \xrightarrow{\sim} \mathcal{E}[1/I]$$

of type μ. A(n ∞-truncated) $(\mathcal{G},\mu)$-window over $\underline{A}$ can be described analogously. In practice, we prefer to use this description of windows.

Terminology 2.2.14. When R is a perfectoid ring, by a(n n-truncated) $(\mathcal{G},\mu)$-window over R, we always mean a(n n-truncated) $(\mathcal{G},\mu)$-window over the corresponding $\mathrm{A}_{\mathrm{inf}}$ frame.

Proposition 2.2.15 (Apertures over perfectoid rings are windows)**.** *Let R be a perfectoid ring and $n \in \mathbf{Z}_{\geqslant 1} \cup \{\infty\}$. There is canonical equivalence between the groupoid $\mathrm{BT}_n^{\mathcal{G},\mu}(R)$ and the groupoid of n-truncated $(\mathcal{G},\mu)$-windows over R.*

Proof. See [GM24, Lemma 9.2.3]. □

Remark 2.2.16 (Quotient description over perfectoid rings)**.** Let R be a p-torsion free perfectoid ring and ξ a generator of $\mathrm{Ker}(\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R)$. Denote by $\overline{\xi}$ the image of ξ in $W_n(R^\flat)$. By Proposition 2.2.15, the stack $\mathrm{BT}_n^{\mathcal{G},\mu}$

restricted to perfectoid R-algebras is the (p-completely étale) stackification of the prestack given by sending a perfectoid R-algebra S to the action groupoid of $\mathcal{G}(W_n(S^\flat))$ acting on the double coset $\mathcal{G}(W_n(S^\flat))\mu(\overline{\xi})\mathcal{G}(W_n(S^\flat))$ via φ-conjugation: $g \in \mathcal{G}(W_n(S^\flat))$ acts on an element C by $g \cdot C = g^{-1}C\varphi(g)$. One gets an analogous description for $\mathrm{BT}_\infty^{\mathcal{G},\mu}$ restricted to perfectoid R-algebras by taking the limit $n \to \infty$.

Example 2.2.17 (Breuil–Kisin–Fargues modules)**.** Let R be a p-torsion free perfectoid ring. When $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$, Proposition 2.2.15 combined with Example 2.2.4 and Remark 2.2.13 gives the familiar classification of p-divisible groups of height h and dimension d over R in terms of locally free modules M of rank h over $\mathrm{A}_{\mathrm{inf}}(R)$ equipped with an injective $\mathrm{A}_{\mathrm{inf}}(R)$-linear map $\Phi\colon \varphi^*M \to M$ whose cokernel is supported on R and locally free of rank d. See [Lau18] for details.

2.3. **$\mathcal{G}$-zips.** We review the theory of $\mathcal{G}$-zips as in [PWZ15] and their relationship with $(\mathcal{G}, \mu)$-apertures.

Setup 2.3.1. Let k be the residue field of $\mathcal{O}$ and φ the Frobenius lift on $\mathcal{O}/\mathbf{Z}_p$. Let $\mathcal{P}^+_{\varphi(\mu)} \subset \mathcal{G}$ be the smooth subgroup Because $\mathcal{G}$ is defined over $\mathbf{Z}_p$, the Frobenius φ acts naturally on the cocharacter lattice of $\mathcal{G}_{\mathcal{O}}$. We denote by $\varphi(\mu)$ the Galois twist of the cocharacter μ by φ. Let $\mathcal{P}^+_{\varphi(\mu)} \subset \mathcal{G}$ be the smooth subgroup scheme whose Lie algebra is identified with the sum of the weight-i spaces in $\mathrm{Lie}\,\mathcal{G}_{\mathcal{O}}$ for $\varphi(\mu)$ with $i \geqslant 0$. Let $\mathcal{U}^-_\mu$ (resp. $\mathcal{U}^+_{\varphi(\mu)}$) be the smooth subgroup scheme of $\mathcal{P}^-_\mu$ (resp. $\mathcal{P}^+_{\varphi(\mu)}$) whose Lie algebra is the sum of the weight-i spaces for μ (resp. for $\varphi(\mu)$) with $i < 0$ (resp. $i > 0$). Set $\mathcal{M}_\mu = \mathcal{P}^-_\mu/\mathcal{U}^-_\mu$ and $\mathcal{M}_{\varphi(\mu)} = \mathcal{P}^+_{\varphi(\mu)}/\mathcal{U}^+_{\varphi(\mu)}$. We have a canonical isomorphism of group schemes $\varphi^*\mathcal{M}_\mu \xrightarrow{\sim} \mathcal{M}_{\varphi(\mu)}$.

Definition 2.3.2 ($\mathcal{G}$-zips)**.** Let R be a k-algebra. An **F-zip with $\mathcal{G}$-structure** or **$\mathcal{G}$-zip** of type μ over R (see [PWZ15, Definition 1.4]) is a tuple

$$(\mathcal{Q}, \mathcal{Q}^-, \mathcal{Q}^+, \alpha)$$

where $\mathcal{Q}$ is a $\mathcal{G}$-torsor over R, $\mathcal{Q}^- \subset \mathcal{Q}$ is a reduction of structure to a $\mathcal{P}^-_\mu$-torsor, $\mathcal{Q}^+ \subset \mathcal{Q}$ is a reduction of structure to a $\mathcal{P}^+_{\varphi(\mu)}$-torsor and

$$\alpha\colon \varphi^*(\mathcal{Q}^-/\mathcal{U}^-_\mu) \xrightarrow{\sim} \mathcal{Q}^+/\mathcal{U}^+_{\varphi(\mu)}$$

is an isomorphism of $\mathcal{M}_{\varphi(\mu)}$-torsors. We denote the groupoid of such objects by $\mathcal{G}\text{-zip}_\mu(R)$.

Example 2.3.3. The ith de Rham cohomology group of any smooth projective scheme X over a k-algebra R with degenerating Hodge-to-de Rham spectral sequence, when equipped with its decreasing Hodge filtration, its increasing conjugate filtration, and Cartier isomorphism, yields an example of a GL_n-zip over R where $n = \mathrm{rank}_R\, \mathrm{H}^i_{\mathrm{dR}}(X/R)$.

Theorem 2.3.4 (Stack of $\mathcal{G}$-zips)**.** *The assignment $R \mapsto \mathcal{G}\text{-zip}_\mu(R)$ for k-algebras R is a smooth 0-dimensional Artin stack over k, which can be presented as $[E\backslash\mathcal{G}_k]$ where*

$$E := \{(a,b) \in \mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\varphi(\mu),k} \colon \varphi(a \bmod \mathcal{U}^-_{\mu,k}) = b \bmod \mathcal{U}^+_{\varphi(\mu),k}\}$$

and the action is given by $(a,b) \cdot g = agb^{-1}$.

Proof. See [PWZ15, Theorem 1.5]. □

The following theorem follows from the proof of [GM24, Theorem 9.3.2]:

Theorem 2.3.5 ($\mathcal{G}$-zip realization)**.** *There is a canonical smooth surjective map*

$$\mathrm{BT}_1^{\mathcal{G},\mu} \otimes_{\mathcal{O}} k \to \mathcal{G}\text{-zip}_\mu$$

that is a gerbe for a connected finite flat commutative group scheme killed by p.

Remark 2.3.6. Note that $\mathcal{G}\text{-zip}_\mu$ only depends on the geometric conjugacy class of μ_k.

Setup 2.3.7. In what follows, $\mathcal{G}$ is *reductive*. Choose a maximal torus $\mathcal{T}$ contained in a Borel subgroup $\mathcal{B} \subset \mathcal{G}$ such that μ factors through $\mathcal{T}_{\mathcal{O}}$ and is dominant. This is always possible by conjugating μ by an element of $\mathcal{G}(\mathcal{O})$.

Remark 2.3.8 (Ekedahl–Oort stratification)**.** The underlying topological space of the stack $\mathcal{G}\text{-zip}_\mu$, denoted $|\mathcal{G}\text{-zip}_\mu|$, is a finite poset consisting of the orbits of the E-action on $\mathcal{G}_k$. These orbits define the **Ekedahl–Oort stratification** on $\mathcal{G}\text{-zip}_\mu$ by smooth locally closed substacks. Let W be the Weyl group of $\mathcal{G}_k$ and let $W_{\mathcal{M}_\mu}$ be the Weyl group of $\mathcal{M}_\mu$. The strata are parameterized by the set ${}^{\mathcal{M}_\mu}W$ of minimal-length representatives for the quotient $W_{\mathcal{M}_\mu}\backslash W$. The closure relations in the topology of $|\mathcal{G}\text{-zip}_\mu|$ corresponds to a partial order on ${}^{\mathcal{M}_\mu}W$ which in general is strictly finer than the Bruhat order– see [PWZ11, §6.3]. See [PWZ15, Theorem 3.20, Remark 3.21] for more details.

Remark 2.3.9 (Explicit parametrization of strata)**.** Let us make the parametrization of Remark 2.3.8 more explicit following [PWZ11]. We identify the Weyl group W with $N_{\mathcal{G}_k}(\mathcal{T}_k)/\mathcal{T}_k$. Let $\mathcal{M}_{\mu,k} \subset \mathcal{P}^-_{\mu,k}$ and $\mathcal{M}_{\varphi(\mu),k} \subset \mathcal{P}^+_{\varphi(\mu),k}$ be the unique Levi subgroups containing $\mathcal{T}_k$ (which canonically identify with the Levi quotients defined previously). Let $I, J \subset \Delta$ denote their respective types with respect to $(\mathcal{B}_k, \mathcal{T}_k)$, so that $W_{\mathcal{M}_\mu}$ and $W_{\mathcal{M}_{\varphi(\mu)}}$ identify with the standard parabolic subgroups W_I and W_J of W.

Recall that a *frame* is a triple consisting of a Borel subgroup, a maximal torus, and a group element that ensures compatibility between the unipotent radicals and the Frobenius morphism [PWZ11, Definition 3.6]. For each $w \in W$, we choose a representative $\dot{w} \in N_{\mathcal{G}_k}(\mathcal{T}_k)$ compatible with the length function l, such that $(w_1w_2)^{\cdot} = \dot{w}_1\dot{w}_2$ whenever $l(w_1w_2) = l(w_1) + l(w_2)$. Defining $z := w_0w_{0,J}$, where w_0 and $w_{0,J}$ are the longest elements of W and W_J respectively, we obtain the relations $\mathcal{B}_k \subset \mathcal{P}^-_{\mu,k}$ and $\varphi(\mathcal{B}_k \cap \mathcal{M}_{\mu,k}) = \mathcal{B}_k \cap \mathcal{M}_{\varphi(\mu),k} = {}^z\mathcal{B}_k \cap \mathcal{M}_{\varphi(\mu),k}$. The triple $({}^z\mathcal{B}_k, \mathcal{T}_k, \dot{z}^{-1})$ then constitutes a frame.

This frame provides a canonical way to parametrize the E-orbits of $\mathcal{G}_k$. For any $w \in W$, the corresponding stratum X_w is defined as the E-orbit of the k-point $\dot{w}\dot{z}^{-1}$. The assignment $w \mapsto X_w$ restricts to a canonical bijection

$$ {}^{\mathcal{M}_\mu}W \xrightarrow{\sim} \{E\text{-orbits in } \mathcal{G}_k\}, $$

which aligns with the indexing in Remark 2.3.8. Similarly, using the cominimal length representatives $W^{\mathcal{M}_{\varphi(\mu)}} := W^J$, one obtains a bijection $W^{\mathcal{M}_{\varphi(\mu)}} \xrightarrow{\sim} \{E\text{-orbits in } \mathcal{G}_k\}$. For $w \in {}^{\mathcal{M}_\mu}W \cup W^{\mathcal{M}_{\varphi(\mu)}}$, the dimension of the stratum is given by $\dim(X_w) = l(w) + \dim(\mathcal{P}^-_{\mu,k})$. See [PWZ11, Theorem 7.5, Theorem 11.2] for proofs.

Definition 2.3.10 (The μ-ordinary stratum)**.** In the context of Remark 2.3.8, the unique open dense stratum of $\mathcal{G}\text{-zip}_\mu$ is called the **μ-ordinary stratum** or the **open zip stratum**, and we denote it by $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$.

Remark 2.3.11. The μ-ordinary stratum can be described as the zip stratum corresponding to the longest element of ${}^{\mathcal{M}_\mu}W$ or the image of E-orbit of the identity element. The latter statement is true because $\mathcal{B} \subseteq \mathcal{P}^+_{\varphi(\mu)}$ and $\mathcal{P}^-_\mu$ contains a Borel subgroup opposite to $\mathcal{B}$.

Remark 2.3.12 (p-rank stratification)**.** Assume μ is defined over $\mathbf{Z}_p$ so that $\mu = \varphi(\mu)$. By the presentation in Theorem 2.3.4, there is a natural map

$$ \pi\colon \mathcal{G}\text{-zip}_\mu \longrightarrow [\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k}\backslash\mathcal{G}_k], $$

where the stack quotient in the target is formed via the action $(a, b) \cdot g = agb^{-1}$. The open dense big cell

$$ \mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subseteq \mathcal{G}_k $$

defines an open dense substack of the target, and the induced stratification of $\mathcal{G}_k$ is much coarser than the Ekedahl–Oort stratification: its strata are indexed by the double cosets

$$ W_{\mathcal{M}_\mu}\backslash W/W_{\mathcal{M}_\mu}, $$

and, in the context of Siegel Shimura varieties, this recovers the classical p-rank stratification of the special fibre. The preimage of the open dense stratum under π is exactly the μ-ordinary stratum $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$. Indeed, by Lang's theorem, the E-orbit of $1 \in \mathcal{G}_k$ is exactly $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k}$. On the complement of the big cell, however, many distinct Ekedahl–Oort strata may lie in a single double-coset stratum, and π is far from being a topological bijection.

Remark 2.3.13 (Computing $\mathcal{G}$-zip realization over perfectoid rings)**.** Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G}, \mu)$-window over a p-torsion free perfectoid ring R. We explain how to access the $\mathcal{G}$-zip realization of the associated 1-truncated aperture obtained by base changing along $R \to R/pR$. Because m_1 is a modification of $\mathcal{E}_1$ of type μ, it induces a canonical reduction of structure $\mathcal{F}^- \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ to $\mathcal{P}^-_\mu$, called the **de Rham realization** and denoted $T_{\mathrm{dR}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}^-_\mu$. Similarly, the modification $(\sigma^{-1})^* m^{-1}$ gives a reduction of structure $\mathcal{F}^+ \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ to $\mathcal{P}^+_{\varphi(\mu)}$, called the **conjugate realization** and denoted $T_{\mathrm{conj}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}^+_{\varphi(\mu)}$. The isomorphism α of $\mathcal{M}_{\varphi(\mu)}$ is then induced by m_1. The tuple $(\mathcal{E}_1|_{R^\flat/tR^\flat}, \mathcal{F}^-, \mathcal{F}^+, \alpha)$ is a $\mathcal{G}$-zip of type μ over $R^\flat/tR^\flat \simeq R/pR$ and induces a morphism $\operatorname{Spec} R/pR \to \mathcal{G}\text{-zip}_\mu$.

Remark 2.3.14 (Computing $\mathcal{G}$-zip realization via quotient presentation)**.** Assume that $\mathcal{E}_1$ is trivialized, and the modification m_1 is represented by $\mu(t)M \in \mathcal{G}(R^\flat[1/t])$ for some matrix $M \in \mathcal{G}(R^\flat)$. Under the stack quotient presentation $\mathcal{G}\text{-zip}_\mu \simeq [\mathcal{G}_k/E]$ defined in Theorem 2.3.4, the $\mathcal{G}$-zip associated to $(\mathcal{E}_1, m_1)$ corresponds exactly to the reduction $\overline{M} \in \mathcal{G}(R^\flat/tR^\flat)$ of M modulo t, defined upto E-action. In particular, if $(\mathcal{E}_1, m_1)$ is μ-ordinary, $\overline{M}$ lands in the open dense big cell $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subset \mathcal{G}_k$ associated to the μ-ordinary stratum.

Remark 2.3.15 (Filtered de Rham realization)**.** Suppose $\mathcal{A} \in \mathrm{BT}^{\mathcal{G},\mu}_\infty(R)$. Then pulling back by the Hodge-filtered de Rham point $x^{\mathcal{N}}_{\mathrm{dR}}$ from Recollection 2.1.12, we get a $\mathcal{G}$-torsor over $\mathbf{A}^1/\mathbf{G}_m \times \operatorname{Spf} R$. The local triviality condition in the definition of aperture says that giving such a $\mathcal{G}$-torsor over $\mathbf{A}^1/\mathbf{G}_m \times \operatorname{Spf} R$ is equivalent to giving a $\mathcal{P}^-_\mu$-torsor over R. Thus, we get a canonical classifying map

$$\mathrm{BT}^{\mathcal{G},\mu}_\infty \to B\mathcal{P}^-_\mu$$

which we call the **filtered de Rham realization**.

2.4. **Hasse invariants.**

Group-theoretic Hasse invariants were constructed by Goldring and Koskivirta in [GK19]. These are sections of line bundles on $\mathcal{G}\text{-zip}_\mu$ whose nonvanishing loci cut out specific Ekedahl–Oort strata (Remark 2.3.8). Because this construction is intrinsic to $\mathcal{G}\text{-zip}_\mu$, it uniformly defines Hasse invariants for special fibers of arbitrary Shimura varieties via Theorems 2.3.5 and 3.1.5.

Setup 2.4.1. Maintain Setups 2.3.1 and 2.3.7 and the notations of Theorem 2.3.4.

Remark 2.4.2 (Line bundles)**.** Any character $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ inflates to a character of $\mathcal{P}^-_{\mu,k}$, and by restriction via the first projection, defines a character of the zip group E. This induces a line bundle on $\mathcal{G}\text{-zip}_\mu$, which we denote by $\mathcal{V}(\chi)$.

Definition 2.4.3 (Hasse invariants)**.** Let $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ be a character. A (μ-ordinary) **Hasse invariant** of weight χ is a global section

$$h \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$$

whose non-vanishing locus is exactly the open dense μ-ordinary stratum $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$.

Remark 2.4.4 (Hasse invariants are unique upto scaling)**.** Since the restriction to the open dense stratum

$$\mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi)) \to \mathrm{H}^0(\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu, \mathcal{V}(\chi))$$

is injective, it follows that $\dim_k \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi)) \leqslant 1$.

Theorem 2.4.5 (Existence of Hasse invariants)**.** *Let $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ be an $\mathcal{M}_{\mu,k}$-ample character (see [GK19, Definition N.5.1]). Then there exists an integer $N \geqslant 1$ such that the line bundle $\mathcal{V}(N\chi)$ admits a Hasse invariant*

$$h \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(N\chi))$$

cutting out the μ-ordinary locus.

Proof. See [GK19, §3.2]. □

Remark 2.4.6 (Hasse invariants via p-rank stratification)**.** Assume μ is defined over $\mathbf{Z}_p$ so that $\mu = \varphi(\mu)$. Following Remark 2.3.12, there is a natural projection

$$\pi \colon \mathcal{G}\text{-zip}_\mu \longrightarrow [\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k} \backslash \mathcal{G}_k]$$

where the target stack quotient is formed via the action $(a,b)\cdot g = agb^{-1}$. Any character $\eta \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ induces a character $(a,b) \mapsto \eta(a)^{-1}\eta(b)$ of $\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k}$, which defines a line bundle $\mathcal{L}(\eta)$ on the target stack. For an ample character η, a suitable representation of $\mathcal{G}_k$ provides a semi-invariant regular function $f \colon \mathcal{G}_k \to \mathbf{A}^1_k$ satisfying $f(agb^{-1}) = \eta(a)\eta(b)^{-1}f(g)$, whose non-vanishing locus is exactly the open dense big cell $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subseteq \mathcal{G}_k$. This function f defines a global section of $\mathcal{L}(\eta)$ cutting out the big cell. By definition of the zip group E (Theorem 2.3.4), the pullback $\pi^*\mathcal{L}(\eta)$ to $\mathcal{G}\text{-zip}_\mu$ is exactly the line bundle $\mathcal{V}(\chi)$, where $\chi = \eta \circ \varphi - \eta$. Since the preimage of the big cell under π is exactly the μ-ordinary stratum $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$ (Remark 2.3.12), the pullback $\pi^* f \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$ is exactly a Hasse invariant of weight χ.

Remark 2.4.7 (Hasse invariant of windows over frames)**.** Let $\underline{A} = (A, I)$ be a Breuil–Kisin frame for R such that A is p-torsion-free. Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G},\mu)$-window over $\underline{A}$. Analogously to Remark 2.3.13, the modification $m_1 \colon \varphi^*\mathcal{E}_1[1/I] \xrightarrow{\sim} \mathcal{E}_1[1/I]$ induces a canonical $\mathcal{G}$-zip of type μ over $A/(I,p) \simeq R/pR$, yielding a classifying morphism $z \colon \operatorname{Spec} R/pR \to \mathcal{G}\text{-zip}_\mu$. Given a Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$, we define the **Hasse invariant** $\mathrm{Ha}(\mathcal{E}_1, m_1)$ of the window as the pullback $z^*\mathrm{Ha} \in \mathrm{H}^0(\operatorname{Spec} R/pR, z^*\mathcal{V}(\chi))$. It therefore makes sense to speak of μ-ordinariness of $(\mathcal{G},\mu)$-windows over arbitrary frames.

Remark 2.4.8 (Well-definedness of p-adic valuation of Hasse invariant)**.** For a general Breuil–Kisin frame $\underline{A} = (A, I)$, identifying the Hasse invariant $\mathrm{Ha}(\mathcal{E}_1, m_1)$ with an element of $A/(I,p) \cong R/pR$ requires a choice of local trivialization for the line bundle $z^*\mathcal{V}(\chi)$. However, any two such choices differ by multiplication by a unit in $(R/pR)^\times$. Consequently, the principal ideal $(\mathrm{Ha}(\mathcal{E}_1, m_1))$ generated by the Hasse invariant in R/pR is canonically well-defined and independent of the chosen trivialization. Furthermore, if the base ring R is equipped with a rank-1 valuation ν_p extending the p-adic valuation on $\mathbf{Z}_p$, we define the truncated p-adic valuation of the Hasse invariant as $\min(1, \nu_p(\widetilde{h})) \in [0,1]$, where $\widetilde{h} \in R$ is any lift of $\mathrm{Ha}(\mathcal{E}_1, m_1)$. Because any two lifts differ by a multiple of p (which has valuation 1), and any two trivializations differ by a unit (which has valuation zero), this truncated valuation is canonically well-defined. In the special case of an $\mathrm{A}_{\mathrm{inf}}$ frame for a perfectoid ring R, $A/(I,p)$ identifies with $R^\flat/tR^\flat$ (where $t = p^\flat$), and this recovers the well-definedness of the similarly truncated t-adic valuation $\min(1, \nu_t(\widetilde{h})) \in [0,1]$.

Remark 2.4.9 (Hasse invariant under base change and Frobenius twist)**.** The formation of the Hasse invariant intrinsically commutes with base change. Specifically, if $R \to S$ is a ring map and $\mathcal{A}$ is a 1-truncated $(\mathcal{G},\mu)$-aperture over R with base change $\mathcal{A}_S$ over S, the classifying map to $\mathcal{G}\text{-zip}_\mu$ simply composes with $\operatorname{Spec} S/pS \to \operatorname{Spec} R/pR$. Consequently, $\mathrm{Ha}(\mathcal{A}_S)$ is precisely the image of $\mathrm{Ha}(\mathcal{A})$ under the map $R/pR \to S/pS$. As an obvious consequence, if R is a k-algebra and we base change along the absolute Frobenius $\varphi \colon R \to R$, the Frobenius pullback $\varphi^*\mathcal{A}$ is canonically a 1-truncated $(\mathcal{G}, \varphi(\mu))$-aperture and the Hasse invariant satisfies $\mathrm{Ha}(\varphi^*\mathcal{A}) = \mathrm{Ha}(\mathcal{A})^{\otimes p}$. In particular, the p-adic valuation gets multiplied by p.

2.5. **Cartier duality.** In order to establish a notion of Cartier duality for $(\mathcal{G},\mu)$-apertures, we require a group-theoretic operation on $\mathcal{G}$ that generalizes the map $g \mapsto (g^t)^{-1}$ on GL_n. This is provided by the Chevalley involution.

Assumption 2.5.1. $\mathcal{G}$ is *reductive.*

Definition 2.5.2 (Chevalley involution)**.** A **Chevalley involution** of $\mathcal{G}$ is an automorphism $C \colon \mathcal{G} \to \mathcal{G}$ of order 2 which restricts to the inversion map $t \mapsto t^{-1}$ on a maximal torus $\mathcal{T} \subset \mathcal{G}$, and maps a Borel subgroup $\mathcal{B}$ containing $\mathcal{T}$ to the opposite Borel subgroup $\mathcal{B}^-$ containing $\mathcal{T}$.

Theorem 2.5.3 (Existence of Chevalley involution)**.** *If $\mathcal{G}$ is a* split *reductive group, fixing a pinning (*épinglage*) $(\mathcal{T}, \mathcal{B}, \{X_\alpha\}_{\alpha\in\Delta})$ yields a unique automorphism C that satisfies:*

(1) $C(t) = t^{-1}$ for all $t \in \mathcal{T}$,
(2) $dC(X_\alpha) = -X_{-\alpha}$ for all $\alpha \in \Delta$, where $X_{-\alpha}$ is the unique basis of $\mathfrak{g}_{-\alpha}$ such that $[X_\alpha, X_{-\alpha}]$ is the corresponding simple coroot H_α.

This C is a Chevalley involution ($C^2 = \mathrm{id}_{\mathcal{G}}$). Because any two pinnings are conjugate by an element of the adjoint group $\mathcal{G}^{\mathrm{ad}}$, C is uniquely determined up to an inner automorphism.

Proof. See [DG70, Exp. XXIII, Théorème 4.1]. □

Remark 2.5.4 (Chevalley involution for non-split groups)**.** For a non-split reductive group $\mathcal{G}$ over $\mathbf{Z}_p$, its special fiber $\mathcal{G}_{\mathbf{F}_p}$ is quasi-split by Lang's theorem. By Hensel's lemma applied to the smooth and proper scheme of Borel subgroups, $\mathcal{G}$ itself is quasi-split over $\mathbf{Z}_p$. Thus, $\mathcal{G}$ is an outer form of a split group $\mathcal{G}_0$, obtained by Galois descent via pinned automorphisms. Because the canonical Chevalley involution on $\mathcal{G}_0$ commutes with all pinned automorphisms, it canonically descends to a Chevalley involution on $\mathcal{G}$ defined over $\mathbf{Z}_p$.

We thus obtain:

Theorem 2.5.5 (Cartier duality for apertures)**.** *Let $C \colon \mathcal{G} \to \mathcal{G}$ be a Chevalley involution. For any p-complete $\mathcal{O}$-algebra R and any $n \in \mathbf{Z}_{\geqslant 1} \cup \{\infty\}$, C induces an equivalence of groupoids*

$$\mathrm{BT}_n^{\mathcal{G},\mu}(R) \xrightarrow{\sim} \mathrm{BT}_n^{\mathcal{G},-w_0\mu}(R)$$

where w_0 is the longest element of the Weyl group of $\mathcal{G}$.

Example 2.5.6 (The case of GL_n)**.** When $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$, the standard pinning (diagonal torus, upper triangular Borel) yields exactly the inverse-transpose $C(g) = (g^\top)^{-1}$. This involution translates, on the level of representations, to taking the dual representation, which mirrors the classical Cartier duality of Barsotti-Tate groups via Dieudonné theory.

3. Shimura varieties and p-Hecke correspondences

3.1. **Shimura varieties.**

Setup 3.1.1. As in [MY26], an **unramified Shimura datum** is a triple $(G, \mathcal{G}, X)$ where (G, X) is a Shimura datum and $\mathcal{G}$ is a reductive model of G over $\mathbf{Z}_p$. We set $K_p = \mathcal{G}(\mathbf{Z}_p)$, and only consider level subgroups $K \subset G(\mathbf{A}_f)$ of the form K_pK^p for a neat compact open subgroup $K^p \subset G(\mathbf{A}_f^p)$. We call the tuple $(G, \mathcal{G}, X, K)$ an **unramified Shimura tuple**. We also fix a place $v|p$ of the reflex field E of (G, X). Over the associated Shimura variety Sh_K, we have a canonical étale $\mathcal{G}^c(\mathbf{Z}_p)$-local system $\mathbf{Et}_{K,p}$, where $\mathcal{G}^c$ is a reductive model over $\mathbf{Z}_p$ of the *cuspidal quotient* G^c– this is a technical construction for which we refer the reader to [MY26, §6.1] or [KSZ21, §1.5.6]. When G is adjoint, $G = G^c$. The Shimura datum gives an E_v-rational conjugacy class of cocharacters $\{\mu_v\}$. We in fact have a representative $\mu_v \colon \mathbf{G}_{\mathrm{m},\mathcal{O}_{E,v}} \to \mathcal{G}_{\mathcal{O}_{E,v}}$ for $\{\mu_v\}$. We may view each μ_v as a cocharacter of $\mathcal{G}^c_{\mathcal{O}_{E,v}}$. We can then form the formal stack $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$ over $\mathcal{O}_{E,v}$. For brevity, set $\mathcal{O} := \mathcal{O}_{E,v}$ and k be the residue field of $\mathcal{O}$.

Notation 3.1.2 (Shimura varieties)**.** For the remainder of this paper, we fix a neat tame level $K^p \subset \mathcal{G}(\mathbf{A}_f^p)$. Because this tame level remains unchanged throughout, we will suppress K^p from the notation of the Shimura varieties. Specifically, for any open compact subgroup $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, we denote the corresponding Shimura variety by Sh_{K_p} and its integral model(s) by $\mathscr{S}_{K_p}$. As usual, we denote formal completions by $\widehat{\mathscr{S}}_{K_p}$. Adic generic fibers are denoted by calligraphic letters $\mathcal{S}_{K_p}$. We use an asterisk in the superscript to denote the minimal compactification. For example, $\mathcal{S}^*_{K_p}$ is the analytification of the minimal compactification $\mathrm{Sh}^*_{K_p}$.

Notation 3.1.3 (Level structures at p)**.** Recall from Setup 3.1.1 that our Shimura datum provides a cocharacter μ_v defined over $\mathcal{O}_{E,v}$. Although $-\mu_v$ might not be defined over $\mathbf{Z}_p$, we can sum all the Galois conjugates as in Setup 3.3.1 to get a cocharacter $-\nu$ defined over $\mathbf{Z}_p$. We define the level subgroups at p involved in our tower:

- $\Gamma_0(p^n)$**-level**: This is the preimage of the parabolic subgroup $\mathcal{P}^+_{-\nu}$ under the reduction modulo p^n map $\mathcal{G}(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$. Using the element $\delta_n = (-\nu)(p^n) \in \mathcal{G}(\mathbf{Q}_p)$, this subgroup can be equivalently expressed as the intersection $\mathcal{G}(\mathbf{Z}_p) \cap \delta_n \mathcal{G}(\mathbf{Z}_p)\delta_n^{-1}$.
- $\Gamma_1(p^n)$**-level**: This is the preimage of the unipotent radical of $\mathcal{P}^+_{-\nu}$ under the reduction modulo p^n map.
- $\Gamma(p^n)$**-level**: This is the kernel of the full reduction map $\mathcal{G}(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$.

These are the parahoric, pro-p Iwahori, and principal congruence levels, respectively.

Remark 3.1.4 (p-Hecke correspondences)**.** Fix an element $h \in G(\mathbf{Q}_p)$. Associated with this element, we can set $\mathcal{G}(\mathbf{Z}_p)_h := \mathcal{G}(\mathbf{Z}_p) \cap h\mathcal{G}(\mathbf{Z}_p)h^{-1}$. This subgroup naturally gives rise to a Hecke correspondence:

$$\begin{tikzcd}
 & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)_h} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\
\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} & & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}
\end{tikzcd}$$

The two structure morphisms of this correspondence can be described as follows:

- The map t_h is the finite étale projection induced by the natural subgroup inclusion $\mathcal{G}(\mathbf{Z}_p)_h \subset \mathcal{G}(\mathbf{Z}_p)$.
- The map s_h is induced by the conjugation homomorphism $\mathcal{G}(\mathbf{Z}_p)_h \to \mathcal{G}(\mathbf{Z}_p)$ mapping $g \mapsto h^{-1}gh$, which geometrically corresponds to right multiplication by h^{-1}.

Let $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)} = [*/\underline{\mathcal{G}(\mathbf{Z}_p)}]$ be the classifying stack of étale $\mathcal{G}(\mathbf{Z}_p)$-local systems. The double coset $\mathcal{G}(\mathbf{Z}_p)h\mathcal{G}(\mathbf{Z}_p)$ carries a natural right action of $\mathcal{G}(\mathbf{Z}_p)$, allowing us to form the finite étale stack quotient over $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$:

$$\alpha_h \colon [\underline{\mathcal{G}(\mathbf{Z}_p)h\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}] \to [*/\underline{\mathcal{G}(\mathbf{Z}_p)}] = \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}\,.$$

By Setup 3.1.1, the Shimura variety $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$ carries a canonical étale local system that induces a *p-adic étale realization map* $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$. The pullback of the cover α_h over $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$ along this realization map is canonically isomorphic to the cover $t_h \colon \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)_h} \to \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$.

We recall the existence of universal $(\mathcal{G}^c, -\mu_v)$-apertures over the v-adic completion of the integral canonical models of Bakker, Shankar, and Tsimerman [BST25a].

Theorem 3.1.5 (Syntomic realization for integral canonical models)**.** *For an unramified Shimura tuple $(G, \mathcal{G}, X, K)$, and all p sufficiently large, an integral model $\mathscr{S}_K$ of Sh_K over $\mathcal{O}_{E,(v)}$ exists with v-adic completion $\widehat{\mathscr{S}_K}$, for any $v \mid p$, such that*

(1) (Serre–Tate property) There exists a formally étale map, called the ***syntomic realization****, of p-adic formal stacks over $\mathrm{Spf}\,\mathcal{O}_{E,v}$*

$$\varpi \colon \widehat{\mathscr{S}_K} \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v},$$

such that the induced $\mathcal{G}^c(\mathbf{Z}_p)$-local system on the generic fiber is isomorphic to $\mathbf{Et}_{K,p}$.

(2) (Pointwise criterion) For any mixed characteristic $(0,p)$ complete discrete valuation field F over $\mathcal{O}_{E,v}$ with perfect residue field and $x \in \mathrm{Sh}_K(F)$, the following are equivalent:
   (a) $x \in \mathscr{S}_K(\mathcal{O}_F)$;
   (b) $\mathbf{Et}_{K,p}|_x$ is a crystalline $\mathcal{G}^c(\mathbf{Z}_p)$-local system;
   (c) $\mathbf{Et}_{K,p}|_x$ is a potentially crystalline $\mathcal{G}^c(\mathbf{Z}_p)$-local system.

Proof. See [MY26, Theorem D]. ∎

Remark 3.1.6 (Exceptional Shimura varieties)**.** One of the key points of the above result is that it also applies to *exceptional* Shimura varieties, but only for an ineffectively large bound on p because the integral canonical models in [BST25a] are obtained by spreading out from the generic fiber. By an exceptional Shimura variety, we refer to a Shimura variety associated to a reductive group G over $\mathbf{Q}$ such that its adjoint group G^{ad} has $\mathbf{R}$-simple factors that are either compact or isomorphic to an exceptional Lie group admitting a Hermitian symmetric domain. By Cartan's classification, there are exactly two such non-compact real exceptional simple Lie groups:

- $E_{6(-14)}$: The real form of E_6 with maximal compact subgroup $\mathrm{Spin}(10) \times \mathrm{U}(1)$. It has $\mathbf{R}$-rank 2.
- $E_{7(-25)}$: The real form of E_7 with maximal compact subgroup $E_6 \times \mathrm{U}(1)$. It has $\mathbf{R}$-rank 3.

3.2. **Relative perfectness of mod-p syntomic realization.** It is well known that a relatively perfect homomorphism of $\mathbf{F}_p$-algebras is formally étale but the converse is not necessarily true. It is shown in [DO25] that relative perfectness of a map of $\mathbf{F}_p$-algebras is equivalent to the map being what's called b-nil formally étale. The following is a straightforward generalization of this to stacks:

Definition 3.2.1 (B-nil formal smoothness, unramifiedness, and étaleness)**.** A morphism $f: \mathcal{X} \to \mathcal{Y}$ of (possibly nonalgebraic) stacks over $\mathbf{F}_p$ is called **b-nil formally smooth** (resp. **b-nil formally unramified**, **b-nil formally étale**) if for every affine scheme S over $\mathbf{F}_p$ and every closed subscheme $S_0 \subset S$ defined by an ideal sheaf $\mathcal{I} \subset \mathcal{O}_S$ such that $\mathcal{I}^{[p]} = 0$ ($\mathcal{I}^{[p]}$ is the ideal sheaf generated by pth powers of local sections of $\mathcal{I}$), the canonical restriction functor

$$\Phi: \mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$$

is essentially surjective (resp. fully faithful, an equivalence of groupoids).

Remark 3.2.2. Smooth (resp. unramified, étale) morphisms are b-nil formally smooth (resp. unramified, étale). Futhermore, these notions are stable under base change.

Proposition 3.2.3 (B-nil formally étale $\Leftrightarrow$ relatively perfect)**.** *Let $f: \mathcal{X} \to \mathcal{Y}$ be a morphism of stacks over $\mathbf{F}_p$ that is representable by Deligne–Mumford stacks. Then f is b-nil formally étale if and only if f is relatively perfect (i.e., the relative Frobenius $F_{\mathcal{X}/\mathcal{Y}}: \mathcal{X} \to \mathcal{X}^{(p)} := \mathcal{X} \times_{\mathcal{Y}, F_{\mathcal{Y}}} \mathcal{Y}$ is an isomorphism).*

Proof. First, assume f is relatively perfect. Let $S_0 \hookrightarrow S$ be a closed embedding of affine schemes defined by an ideal $\mathcal{I}$ with $\mathcal{I}^{[p]} = 0$. The absolute Frobenius $F_S: S \to S$ factors canonically as $S \to S_0 \hookrightarrow S$. Applying the purely categorical argument of [DO25, Lemma 3.17] to the groupoids of points $\mathcal{X}(S)$ and $\mathcal{Y}(S)$ directly establishes that the canonical functor $\mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$ is an equivalence.

Conversely, assume f is b-nil formally étale. Let $V \to \mathcal{Y}$ be a map from an affine scheme. The projection $W := \mathcal{X} \times_{\mathcal{Y}} V \to V$ is b-nil formally étale. The assumption implies that W is a Deligne–Mumford stack. Choose an étale atlas $U \to W$ where U is an affine scheme. The composition $U \to W \to V$ is a composition of an étale map and a b-nil formally étale map, so it is a b-nil formally étale morphism of affine schemes. By [DO25, Theorem 3.21], its relative Frobenius $F_{U/V}: U \to U^{(p)}$ is an isomorphism. Because the map $U \to W$ is étale, the relative Frobenius commutes with base change along it, meaning the commutative square formed by $U \to W$, $U^{(p)} \to W^{(p)}$, $F_{U/V}$, and $F_{W/V}$ is Cartesian. Since $F_{U/V}$ is an isomorphism and $U^{(p)} \to W^{(p)}$ is an étale cover, descent implies $F_{W/V}$ is an isomorphism. As $F_{W/V}$ is the base change of $F_{\mathcal{X}/\mathcal{Y}}$ to V, it follows that $F_{\mathcal{X}/\mathcal{Y}}$ is an isomorphism, making f relatively perfect. $\square$

Proposition 3.2.4. *Let $f: \mathcal{X} \to \mathcal{Y}$ be a formally étale morphism of stacks such that $\mathcal{Y} \simeq \lim \mathcal{Y}_n$ is the inverse limit of Artin stacks $\{\mathcal{Y}_n\}$. Assume further that $\mathcal{X}$ and $\mathcal{Y}_n$ are Noetherian Artin stacks with quasi-affine diagonal or they are quasicompact, separated, and tame Deligne–Mumford stacks. Then f satisfies the extension property with respect to arbitrary locally nilpotent thickenings[1]. That is, for every affine scheme S and every closed*

[1] Following [Sta26, Definition 00IL], an ideal I is called **locally nilpotent** if every element $x \in I$ is nilpotent. Equivalently, every finitely generated subideal of I is nilpotent, making I the filtered colimit of its nilpotent subideals.

subscheme $S_0 \subset S$ defined by a locally nilpotent ideal, the canonical functor

$$\mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$$

is an equivalence of categories.

Proof. Let R be a ring and $I \subset R$ a locally nilpotent ideal. We set $R_0 := R/I$. By [BH17, Corollary 1.5], the canonical map

$$\mathcal{X}(R) \xrightarrow{\sim} \lim \mathcal{X}(R/I^m)$$

is an equivalence. Similarly, for each n, $\mathcal{Y}_n(R) \xrightarrow{\sim} \lim_m \mathcal{Y}_n(R/I^m)$. Taking the inverse limit over n, we deduce that $\mathcal{Y}$ also satisfies this property:

$$\mathcal{Y}(R) \simeq \lim_n \mathcal{Y}_n(R) \simeq \lim_n \lim_m \mathcal{Y}_n(R/I^m) \simeq \lim_m \lim_n \mathcal{Y}_n(R/I^m) \simeq \lim_m \mathcal{Y}(R/I^m).$$

For each $m \geqslant 1$, the ideal $I/I^m \subset R/I^m$ is nilpotent. By formal étaleness, the functor

$$\mathcal{X}(R/I^m) \to \mathcal{X}(R_0) \times_{\mathcal{Y}(R_0)} \mathcal{Y}(R/I^m)$$

is an equivalence of categories. Taking the inverse limit over all $m \geqslant 1$ yields the desired equivalence

$$\mathcal{X}(R) \xrightarrow{\sim} \mathcal{X}(R_0) \times_{\mathcal{Y}(R_0)} \mathcal{Y}(R). \qquad \square$$

Corollary 3.2.5 (Serre–Tate for locally nilpotent thickenings)**.** *The syntomic realization $\varpi \colon \widehat{\mathscr{S}_K} \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v}$ satisfies the Serre–Tate property with respect to arbitrary locally nilpotent thickenings.*

Corollary 3.2.6 (Mod-p syntomic realization is relatively perfect)**.** *The map*

$$\varpi_k \colon \widehat{\mathscr{S}_K} \otimes k \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$$

of stacks over k is relatively perfect.

Proof. Let $\operatorname{Spec} A \hookrightarrow \widehat{\mathscr{S}_K} \otimes k$ be an affine open. Then the composite map of stacks $\operatorname{Spec} A \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$ is representable by affine schemes. Indeed, $\operatorname{Spec} A \to \mathrm{BT}_n^{\mathcal{G}^c, -\mu_v} \otimes k$ is an affine morphism since $\mathrm{BT}_n^{\mathcal{G}^c, -\mu_v}$ has affine diagonal, and the inverse limit of affine morphisms is representable by affine morphisms. Now the desired result is immediate by applying Proposition 3.2.3 to $\operatorname{Spec} A \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$. $\square$

3.3. **The μ-ordinary locus.** We briefly recall the μ-ordinary locus, following [MY26, §3.6, §7.4].

Setup 3.3.1. Maintain Setups 2.2.1 and 2.3.1. Suppose μ is chosen within its conjugacy class to factor through a maximal torus $T \subset \mathcal{G}$. Let $d = [\mathcal{O}[1/p] : \mathbf{Q}_p]$, and let $\nu = \sum_{i=0}^{d-1} \varphi^i(\mu)$, which defines a cocharacter of $\mathcal{G}$ defined over $\mathbf{Z}_p$. We let $\mathcal{P}_\nu^+ \subset \mathcal{G}$ denote the parabolic subgroup whose Lie algebra consists of the non-negative weight spaces for the adjoint action of ν, and let $\mathcal{M}_\nu$ be its Levi quotient (the centralizer of ν). By construction, μ is central in $\mathcal{M}_\nu$.

Definition 3.3.2 (The μ-ordinary locus)**.** Recall from Remark 2.3.8 that the stack $\mathcal{G}\text{-zip}_\mu$ admits a unique open dense stratum $\mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}$. For any $n \in \mathbf{N} \cup \{\infty\}$, the **$\mu$-ordinary locus** of the stack of apertures is the unique open formal substack $\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}} \subset \mathrm{BT}_n^{\mathcal{G},\mu}$ characterized by the property:

$$\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}} \otimes k = (\mathrm{BT}_n^{\mathcal{G},\mu} \otimes k) \times_{\mathcal{G}\text{-zip}_\mu} \mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}.$$

Proposition 3.3.3. *The natural map $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{G},\mu}$ is an open embedding onto $\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}}$.*

Proof. See [MY26, Proposition 3.6.12]. $\square$

There is a canonical projection $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ given by quotienting by the unipotent radical of $\mathcal{P}_\nu^+$ at the level of syntomic torsors.

Theorem 3.3.4. *The formal stack $\mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ is étale over $\mathcal{O}$ and is non-canonically isomorphic to the formal classifying stack of the finite group $\mathcal{M}_\nu(\mathbf{Z}/p^n\mathbf{Z})$ and $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ presents the source as the relative classifying stack of a certain (noncommutative) finite flat group scheme over $\mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$.*

Proof. See [MY26, Proposition 3.6.8]. □

Remark 3.3.5 (The case when μ is defined over $\mathbf{Z}_p$)**.** For our applications, we are primarily concerned with the case where μ is defined over $\mathbf{Z}_p$, which implies $\mu = \varphi(\mu)$. Let us explicate Theorem 3.3.4 in this setting by detailing the structure of the fiber of the map $\mathrm{BT}_n^{\mathcal{P}_\mu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$. Recall that $\mathcal{U}_\mu^+$ denotes the unipotent radical of $\mathcal{P}_\mu^+$. The Levi quotient $\mathcal{M}_\mu$ acts on $\mathcal{U}_\mu^+$ by conjugation. A point in $\mathrm{BT}_n^{\mathcal{M}_\mu,\mu}(R)$ corresponds to an $\mathcal{M}_\mu$-torsor $\mathcal{Q}$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$. Twisting the unipotent radical by this torsor yields a vector group $\mathcal{U} := \mathcal{Q} \times^{\mathcal{M}_\mu} \mathcal{U}_\mu^+$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$. Moreover, since μ is minuscule, the Hodge–Tate weights of $\mathcal{U}$ are concentrated in $\{0,1\}$. The fiber of $\mathrm{BT}_n^{\mathcal{P}_\mu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$ over $\mathcal{Q}$ is canonically identified with the assignment $C \mapsto \mathrm{Map}_{R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}}(C^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}, B\mathcal{U})$ which is precisely the classifying stack of the n-truncated Barsotti–Tate group over $\mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$ corresponding to the vector bundle F-gauge $\mathcal{U}$ via [GM24, Theorem A].

Definition 3.3.6 (The global μ-ordinary locus)**.** The **μ-ordinary locus** of the integral canonical model is the unique open formal subscheme $\widehat{\mathscr{S}}_K^{\mathrm{ord}} \subset \widehat{\mathscr{S}}_K$ whose special fiber is the preimage of the μ-ordinary stratum under the $\mathcal{G}^c$-zip realization map ζ (cf. Proposition 3.6.5):

$$\widehat{\mathscr{S}}_K^{\mathrm{ord}} \otimes k := \zeta^{-1}(\mathcal{G}^c\text{-zip}^{\mathrm{ord}}_{-\mu_v}).$$

By Corollary 3.6.9, this locus canonically extends to a formal affine open subscheme $\widehat{\mathscr{S}}_K^{*\mathrm{ord}}$ of the (formal) minimal compactification $\widehat{\mathscr{S}}_K^{*}$ as it coincides with the non-vanishing locus of the extended Hasse invariant.

Remark 3.3.7. To summarize, we have a diagram with Cartesian squares

$$\begin{tikzcd}
\widehat{\mathscr{S}}_K^{*\mathrm{ord}} \arrow[d, hook] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[l, hook'] \arrow[r] \arrow[d, hook] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \arrow[d, hook] \\
\widehat{\mathscr{S}}_K^{*} & \widehat{\mathscr{S}}_K \arrow[l, hook'] \arrow[r, "\varpi"'] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}
\end{tikzcd}$$

where the horizontal arrows on the right are the syntomic realization maps, the horizontal arrows on the left are open embeddings into the minimal compactifications, and the vertical maps are open embeddings.

Proposition 3.3.8 (Canonical Frobenius lift on $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}$)**.** *Let $q = |k|$ be the cardinality of the residue field of $\mathcal{O}$. There is a canonical endomorphism $\widetilde{\Phi}_{\mathrm{ap}} \colon \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}$ of p-adic formal stacks over* Spf $\mathcal{O}$ *lifting the relative q-Frobenius endomorphism.*

Proof. See [MY26, Proposition 3.6.11]. □

We use the relative perfectness result from the previous section to give a formal explanation of existence of canonical Frobenius lifts on the μ-ordinary loci of Shimura varieties (cf. [MY26, Theorem 7.4.21] and [BST25b, Proposition 6.6]):

Theorem 3.3.9 (Canonical Frobenius lift on the μ-ordinary locus)**.** *Let $q = |k|$ be the cardinality of the residue field k of $\mathcal{O}$. There exists a unique Cartesian square of p-adic formal stacks over* Spf $\mathcal{O}$*:*

$$\begin{tikzcd}
\widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[r, "\widetilde{\Phi}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[d, "\varpi"] \\
\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \arrow[r, "\widetilde{\Phi}_{\mathrm{ap}}"] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}
\end{tikzcd}$$

such that the vertical arrows are induced by syntomic realizations and the bottom horizontal arrow $\widetilde{\Phi}_{\mathrm{ap}}$ is the canonical Frobenius lifting of Proposition 3.3.8. Consequently, $\widetilde{\Phi}$ is a lift of the relative q-Frobenius endomorphism.

Proof. Following the argument of Theorem 5.1.2 (and [MY26, Lemma 7.4.10]), set $h = \nu(p)$. In what follows, $(-)^{\mathrm{an}}$ denotes the analytification functor over E_v. Recall the Hecke correspondence diagram associated to $K_h := K \cap hKh^{-1} = K \cap \nu(p)K\nu(p)^{-1}$:

$$\begin{tikzcd} & \mathrm{Sh}_{K_h} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\ \mathrm{Sh}_K & & \mathrm{Sh}_K . \end{tikzcd}$$

Over $\mathcal{S}_K^{\mathrm{ord}}$, the p-adic étale realization map factors through $\mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)}$. The description in Remark 3.1.4 therefore shows that $t_h^{\mathrm{an}}|_{\mathcal{S}_K^{\mathrm{ord}}}$ is canonically isomorphic to the base-change of

$$[\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}} \times_{\mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)} \simeq [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}}$$

along $\mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)}$. Since $\mathcal{P}_\nu^+(\mathbf{Z}_p)\nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p) = \nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p)$, we have a canonical section:

$$\mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)} \xrightarrow{\sim} [\underline{\nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}} \to [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}}.$$

This implies that we have a section

$$\mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Sh}^{\mathrm{an}}_{K_h} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}_K^{\mathrm{an}}} \mathcal{S}_K^{\mathrm{ord}}.$$

Composing this section with the map $s_h^{\mathrm{an}} \colon \mathrm{Sh}^{\mathrm{an}}_{K_h} \to \mathrm{Sh}^{\mathrm{an}}_K$ gives a map $\Phi' \colon \mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Sh}_K^{\mathrm{an}}$ such that the diagram

$$\begin{tikzcd} \mathcal{S}_K^{\mathrm{ord}} \arrow[r, "\Phi'"] \arrow[d, "\varpi^{\mathrm{ad}}"'] & \mathrm{Sh}_K^{\mathrm{an}} \arrow[d] \\ \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}, \mathrm{ad}} \arrow[r, "\widetilde{\Phi}^{\mathrm{ad}}_{\mathrm{ap}}"] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \end{tikzcd}$$

commutes. By [MY26, Proposition 4.3.1], this identifies Φ' as the adic generic fiber of a morphism $\widetilde{\Phi} \colon \widehat{\mathscr{S}}_K^{\mathrm{ord}} \to \widehat{\mathscr{S}}_K$ which factors through $\widehat{\mathscr{S}}_K^{\mathrm{ord}}$.

It remains to check that the square

$$\begin{tikzcd} \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[r, "\widetilde{\Phi}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[d, "\varpi"] \\ \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}} \arrow[r, "\widetilde{\Phi}_{\mathrm{ap}}"] & \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}} \end{tikzcd}$$

is Cartesian. Since all the members of the square are smooth over $\mathrm{Spf}\,\mathcal{O}$, by [MY26, Proposition 4.3.1], it suffices to check it at adic generic fibers, which is definitional. Because this square is Cartesian and the mod-p syntomic realization ϖ_k is relatively perfect (Corollary 3.2.6), $\widetilde{\Phi}$ modulo p must be the q-Frobenius, as $\widetilde{\Phi}_{\mathrm{ap}}$ modulo p is the q-Frobenius. □

3.4. **The Vinberg monoid.**

Setup 3.4.1. Let R be a discrete valuation ring of mixed characteristic $(0, p)$ with uniformiser π and fraction field K. Let $\mathcal{G}$ be a split reductive group scheme over R. Let $\mathcal{T}$ be the abstract Cartan of $\mathcal{G}$—this may be defined as $\mathcal{B}/\mathrm{R_u}(\mathcal{B})$ for any R-Borel $\mathcal{B} \subset \mathcal{G}$. The key point is that this does not depend on the choice of $\mathcal{B}$. Define $\mathcal{G}_+ := \mathcal{T} \times_R^{Z_\mathcal{G}} \mathcal{G}$ and $\mathcal{T}_{\mathrm{ad}} := \mathcal{T}/Z_\mathcal{G}$, where $Z_\mathcal{G}$ denotes the center of $\mathcal{G}$. Let $\Delta \subset \mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})$ be the set of simple roots. Note that Δ is canonically defined as a *subset* of $\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})$. Let $\mathcal{T}_{\mathrm{ad}} \hookrightarrow \mathcal{T}_{\mathrm{ad}}^+ := \mathrm{Spec}\, R[\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}]$ be the obvious toric embedding where $\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}$ is the submonoid generated by Δ. Here, $\mathcal{T}_{\mathrm{ad}}^+$ is viewed as an R-monoid with unit group $\mathcal{T}_{\mathrm{ad}}$.

Let $\mathbf{X}^\bullet(\mathcal{T})^+$ be the submonoid of dominant characters, and let $\mathbf{X}^\bullet(\mathcal{T})^+_{\mathrm{pos}}$ be the submonoid generated by $\mathbf{X}^\bullet(\mathcal{T})^+$ and $\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}$. We equip $\mathbf{X}^\bullet(\mathcal{T})$ with the partial order $\preceq$ defined by $\lambda \preceq \mu$ if $\mu - \lambda \in \mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}$. We write $\lambda \prec \mu$ if $\lambda \preceq \mu$ and $\lambda \neq \mu$.

Construction 3.4.2 (Multi-filtration and construction of the Vinberg monoid)**.** The *Vinberg monoid* is a certain reductive monoid scheme $V_{\mathcal{G}}$ over R equipped with an R-monoid homomorphism $\mathfrak{a}\colon V_{\mathcal{G}} \to \mathcal{T}^+_{\mathrm{ad}}$ called the *abelianization*. It is true that $\mathcal{T}^+_{\mathrm{ad}}$ is the largest commutative monoid quotient of $V_{\mathcal{G}}$ in a precise sense.

The properties relevant to us are recorded in Theorem 3.4.4. Let us briefly recall the construction of the Vinberg monoid given in [XZ19, §3.2] over an algebraically closed field. The same works over $\mathbf{Z}$ and hence for any split reductive group over an arbitrary base. This is also recorded in [Nat26, §2].

The coordinate ring $\mathcal{O}(\mathcal{G})$ admits a canonical multi-filtration indexed by $\mathbf{X}^\bullet(\mathcal{T})^+_{\mathrm{pos}}$, induced by the $\mathcal{G} \times \mathcal{G}$ action on $\mathcal{O}(\mathcal{G})$ via left and right translation, given by setting $\mathrm{fil}^{\mathcal{G}}_\nu \mathcal{O}(\mathcal{G})$, for $\nu \in \mathbf{X}^\bullet(\mathcal{T})^+_{\mathrm{pos}}$, as the maximal $\mathcal{G} \times \mathcal{G}$-submodule of $\mathcal{O}(\mathcal{G})$ such that all its weights $(\lambda, \lambda') \in \mathbf{X}^\bullet(\mathcal{T}) \times \mathbf{X}^\bullet(\mathcal{T})$ satisfy $\lambda \preceq -w_0(\nu)$ and $\lambda' \preceq \nu$. Here, w_0 is the longest element of the Weyl group of $\mathcal{G}$. Each piece $\mathrm{fil}^{\mathcal{G}}_\nu \mathcal{O}(\mathcal{G})$ is a finite free R-module and the associated graded algebra is given by

$$\mathrm{gr}\,\mathcal{O}(\mathcal{G}) := \bigoplus_{\nu \in \mathbf{X}^\bullet(\mathcal{T})^+_{\mathrm{pos}}} \frac{\mathrm{fil}^{\mathcal{G}}_\nu \mathcal{O}(\mathcal{G})}{\sum_{\lambda \prec \nu} \mathrm{fil}^{\mathcal{G}}_\lambda \mathcal{O}(\mathcal{G})} = \bigoplus_{\nu \in \mathbf{X}^\bullet(\mathcal{T})^+} \mathrm{S}_{-w_0(\nu)}(\mathcal{G}) \otimes_R \mathrm{S}_\nu(\mathcal{G}),$$

where $\mathrm{S}_\nu(\mathcal{G})$ denotes the Schur module of highest weight ν, i.e., the induced $\mathcal{G}$-module from the character $-\nu$ of $\mathcal{B}$ [XZ19, Lemma 3.2.1 (4)].

The Vinberg monoid $V_{\mathcal{G}}$ is defined as the spectrum of the multi-Rees algebra associated to this filtration:

$$V_{\mathcal{G}} := \mathrm{Spec}\left(\bigoplus_{\nu \in \mathbf{X}^\bullet(\mathcal{T})^+_{\mathrm{pos}}} \mathrm{fil}^{\mathcal{G}}_\nu \mathcal{O}(\mathcal{G}) \right),$$

endowed with the natural (co)multiplication map. It is an affine monoid scheme over R of finite type which admits a monoid homomorphism $\mathfrak{a}\colon V_{\mathbf{G}} \to \mathcal{T}^+_{\mathrm{ad}} = \mathrm{Spec}\, R[\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}]$ induced by the inclusion of the ring $R[\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}]$ into the Rees algebra.

Remark 3.4.3 (Base change compatibility of Vinberg monoid)**.** The formation of $(V_{\mathcal{G}}, \mathfrak{a})$ commutes with base change on R. Indeed, for any ring R', the injection of filtered R'-algebras $R' \otimes_R \sum_\nu \mathrm{fil}^{\mathcal{G}}_\nu \mathcal{O}(\mathcal{G}) \hookrightarrow \sum_\nu \mathrm{fil}^{\mathcal{G}_{R'}}_\nu \mathcal{O}(\mathcal{G}_{R'})$ induces an isomorphism on associated graded algebras since the formation of Schur modules commutes with base change. Thus, it is actually an isomorphism of filtered rings.

Theorem 3.4.4. *There is a reductive monoid scheme $V_{\mathcal{G}}$ over R which fits into a Cartesian diagram*

$$\begin{tikzcd} \mathcal{G}_+ \arrow[r, hook] \arrow[d] & V_{\mathcal{G}} \arrow[d, "\mathfrak{a}"] \\ \mathcal{T}_{\mathrm{ad}} \arrow[r, hook] & \mathcal{T}^+_{\mathrm{ad}} \end{tikzcd}$$

such that

- *$\mathcal{G}_+$ is the unit group of $V_{\mathcal{G}}$,*
- *the $\mathcal{T} \times_R \mathcal{G} \times_R \mathcal{G}$ action on $\mathcal{G}_+$ given by $\mathcal{T}$ acting on the first component and $\mathcal{G}$ acting on itself by left and right multiplication extends to an action on $V_{\mathcal{G}}$,*
- *$\mathfrak{a}$ is faithfully flat and equivariant.*

Construction 3.4.5. Let λ be a dominant cocharacter of $\mathcal{G}$ defined over R. This defines an R-point $\lambda(\pi)\colon \mathrm{Spec}\, R \to \mathcal{T}^+_{\mathrm{ad}}$. We can base change the Vinberg monoid along this point to obtain an affine R-scheme:

$$V_{\mathcal{G},\lambda(\pi)} := V_{\mathcal{G}} \times_{\mathcal{T}^+_{\mathrm{ad}},\lambda(\pi)} \mathrm{Spec}\, R.$$

This has a natural two-sided action of $\mathcal{G}$ such that its generic fiber is biequivariantly isomorphic to $\mathcal{G}_K$. As such, the R-points $V_{\mathcal{G},\lambda(\pi)}(R)$ for any π-torsion free ring R naturally sits inside $\mathcal{G}(K)$ and is stable under the two-sided action of $\mathcal{G}(R)$.

Proposition 3.4.6. *We have*

$$V_{\mathcal{G},\lambda(\pi)}(R) = \bigsqcup_{\gamma \preceq \lambda} \mathcal{G}(R)\gamma(\pi)\mathcal{G}(R)$$

as subsets of $\mathcal{G}(K)$.

Proof. [LM] **[TODO: such things will probably be proved in Lee–Madapusi]** □

Example 3.4.7. Let $\mathcal{G} = \mathrm{SL}_2$ and $\lambda(z) = \mathrm{diag}(z^n, z^{-n})$ for $n \geqslant 1$. The Vinberg monoid V_{SL_2} identifies with $\mathrm{Mat}_{2\times 2}$ with abelianization $\mathfrak{a} = \det$. Thus $V_{\mathrm{SL}_2,\lambda(\pi)}(R) = \{M \in \mathrm{Mat}_{2\times 2}(R) \mid \det(M) = \pi^{2n}\}$, with embedding into the generic fiber $\mathrm{SL}_2(K)$ given by $M \mapsto \pi^{-n}M$. For an unramified DVR R, the Smith normal form classification of matrices with determinant π^{2n} yields elementary divisors (π^{n-c}, π^{n+c}) for $0 \leqslant c \leqslant n$. In $\mathrm{SL}_2(K)$, these give exactly the $\mathrm{SL}_2(R)$ double cosets of $\gamma_c(\pi) = \mathrm{diag}(\pi^c, \pi^{-c})$. This recovers Proposition 3.4.6, as the dominant cocharacters $\gamma \preceq \lambda$ are precisely γ_c for $0 \leqslant c \leqslant n$.

Construction 3.4.8 (A parabolic monoid)**.** Let $\mathcal{P} \supset \mathcal{B}$ be a standard parabolic subgroup of $\mathcal{G}$, with Levi decomposition $\mathcal{P} = \mathcal{M} \ltimes \mathcal{U}$, where $\mathcal{M}$ is the standard Levi factor containing $\mathcal{T}$ and $\mathcal{U}$ is the unipotent radical. Denote by $V_{\mathcal{P}}$ the Zariski closure of $\mathcal{T} \times^{Z_{\mathcal{G}}} \mathcal{P} \subset \mathcal{G}^+$ inside $V_{\mathcal{G}}$. For a dominant cocharacter λ defined over R, define $V_{\mathcal{P},\lambda(\pi)}$ as the fiber over the R-point $\lambda(\pi)$ along the abelianization map. This is exactly the schematic closure of $\mathcal{P}_K$ sitting inside the generic fiber $\mathcal{G}_K$ of $V_{\mathcal{G},\lambda(\pi)}$.

Construction 3.4.9 (Categorical quotient by the unipotent radical)**.** We define the affine categorical quotient by the unipotent radical $\mathcal{U}$ of $\mathcal{P}$:

$$W := V_{\mathcal{P}} /\!\!/ (\mathcal{U} \times \mathcal{U}).$$

Explicitly, the group $\mathcal{G} \times \mathcal{G}$ naturally acts on $V_{\mathcal{G}}$ by left and right multiplication: $(g_1, g_2) \cdot v := g_1 v g_2^{-1}$. Since $V_{\mathcal{P}}$ is the closure of $\mathcal{P}_K$, it is stable under the restricted action of $\mathcal{P} \times \mathcal{P}$. In particular, the unipotent subgroup $\mathcal{U} \times \mathcal{U}$ acts morphically on the affine scheme $\mathcal{X} = V_{\mathcal{P}}$. The categorical quotient $\mathcal{X} /\!\!/ \mathcal{H}$ by the group scheme $\mathcal{H} = \mathcal{U} \times \mathcal{U}$ is defined as the spectrum of the subring of elements $x \in \mathcal{O}(\mathcal{X})$ which map to $x \otimes 1$ under the coaction map $\mathcal{O}(\mathcal{X}) \to \mathcal{O}(\mathcal{X}) \otimes \mathcal{O}(\mathcal{H})$. Similarly, we set

$$W_{\lambda(\pi)} := V_{\mathcal{P},\lambda(\pi)} /\!\!/ (\mathcal{U} \times \mathcal{U})$$

for a dominant cocharacter λ. Note that $W_{\lambda(\pi)}$ is a flat affine R-scheme with a two-sided action of $\mathcal{M}$ such that its generic fiber is biequivariantly isomorphic to $\mathcal{M}_K$.

Remark 3.4.10. It may be tempting to expect $V_{\mathcal{M}}$ to be a quotient of W by the central torus $Z_{\mathcal{M}}/Z_{\mathcal{G}}$. However, this is false.

To see this, let us consider the example of $\mathcal{G} = \mathrm{SL}_2$ with $\mathcal{P}$ the upper triangular Borel subgroup and $\mathcal{M} = \mathcal{T} \cong \mathbf{G}_m$ the diagonal torus. The Vinberg monoid of $\mathcal{G}$ is simply the monoid of all 2×2 matrices, $V_{\mathcal{G}} = \mathrm{Mat}_{2\times 2} \cong \mathbf{A}^4_k$. The parabolic monoid $V_{\mathcal{P}}$ is the schematic closure of $\mathcal{P}$ in $V_{\mathcal{G}}$:

$$V_{\mathcal{P}} = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{Mat}_{2\times 2} \right\}.$$

We then compute the relative monoid by taking the affine categorical quotient by the left and right actions of $\mathcal{U} \times \mathcal{U}$. For $u_1 = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ and $u_2 = \left(\begin{smallmatrix} 1 & y \\ 0 & 1 \end{smallmatrix}\right)$, the action $u_1 v u_2^{-1}$ maps the entry b to $b + xd - ya$. The invariant ring eliminates b, leaving exactly $k[a, d]$. Thus,

$$W = V_{\mathcal{P}} /\!\!/ (\mathcal{U} \times \mathcal{U}) = \operatorname{Spec} k[a, d] \cong \mathbf{A}^2_k.$$

On the other hand, the Vinberg monoid $V_{\mathcal{M}}$ of the torus $\mathcal{M} \cong \mathbf{G}_m$ is just the torus itself, $V_{\mathcal{M}} \cong \mathbf{G}_m$.

Definition 3.4.11 (The set $S_{\mathcal{M}}(\lambda)$)**.** Let λ be a $\mathcal{G}$-dominant cocharacter. A cocharacter λ' of $\mathcal{T}$ belongs to the set $S_{\mathcal{M}}(\lambda)$ if and only if it satisfies the following three conditions:

(1) **$\mathcal{M}$-dominance:** λ' lies in the $\mathcal{M}$-dominant chamber, meaning $\langle \alpha, \lambda' \rangle \geqslant 0$ for all simple roots α of $\mathcal{M}$.
(2) **$\mathcal{G}$-boundedness:** Let $(\lambda')^{\mathcal{G}}_{\mathrm{dom}}$ denote the unique $\mathcal{G}$-dominant cocharacter in the Weyl orbit of λ'. We require that $(\lambda')^{\mathcal{G}}_{\mathrm{dom}} \preceq \lambda$.
(3) **$\mathcal{M}$-maximality:** For every strictly positive $\mathcal{M}$-coroot combination β, the shifted cocharacter $\lambda' + \beta$ violates at least one of the first two conditions.

Proposition 3.4.12. *For each $\lambda' \in S_{\mathcal{M}}(\lambda)$, let $V_{\mathcal{M},\lambda'(\pi)}$ be the base-changed Vinberg monoid of the Levi subgroup $\mathcal{M}$. Both $W_{\lambda(\pi)}$ and $V_{\mathcal{M},\lambda'(\pi)}$ are flat affine R-schemes whose generic fibers are biequivariantly isomorphic to $\mathcal{M}_K$. Inside the K-algebra $\mathcal{O}(\mathcal{M}_K)$, we have an equality of sub-R-algebras:*

$$\mathcal{O}(W_{\lambda(\pi)}) = \bigcap_{\lambda' \in S_{\mathcal{M}}(\lambda)} \mathcal{O}(V_{\mathcal{M},\lambda'(\pi)}).$$

Proof. The coordinate ring $\mathcal{O}(\mathcal{M})$ is a flat R-algebra equipped with the intrinsic $\mathcal{M}$-filtration $\mathrm{fil}^{\mathcal{M}}_{\eta}\mathcal{O}(\mathcal{M})$ indexed by $\mathcal{M}$-dominant weights $\eta \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{M}}$. The associated graded pieces $\mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M})$ are finite free R-modules (given by tensor products of Schur modules). Because the graded pieces are projective R-modules, the filtration admits a (non-canonical) R-module splitting:

$$\mathcal{O}(\mathcal{M}) \cong \bigoplus_{\eta \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{M}}} \mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M}),$$

under which $\mathrm{fil}^{\mathcal{M}}_{\eta}\mathcal{O}(\mathcal{M}) \cong \bigoplus_{\eta' \preceq_{\mathcal{M}} \eta} \mathrm{gr}_{\eta'}\,\mathcal{O}(\mathcal{M})$. We can analyze the various R-algebras by tracking the π-adic lattices they define within this split direct sum over $\mathcal{O}(\mathcal{M})[1/\pi]$.

For each $\lambda' \in S_{\mathcal{M}}(\lambda)$, the absolute Levi monoid $V_{\mathcal{M},\lambda'(\pi)}$ is controlled by the intrinsic $\mathcal{M}$-filtration base-changed by $\lambda'(\pi)$. Its coordinate ring is the fractional R-subalgebra:

$$\mathcal{O}(V_{\mathcal{M},\lambda'(\pi)}) = \sum_{\eta \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{M}}} \mathrm{fil}^{\mathcal{M}}_{\eta}\mathcal{O}(\mathcal{M}) \cdot \pi^{\langle \eta, \lambda' \rangle}.$$

Under the splitting, the component $\mathrm{gr}_{\eta'}\,\mathcal{O}(\mathcal{M})$ is scaled by the R-ideal generated by $\pi^{\langle \eta, \lambda' \rangle}$ for all $\eta \succeq_{\mathcal{M}} \eta'$. Since λ' is an $\mathcal{M}$-dominant cocharacter, $\eta \succeq_{\mathcal{M}} \eta'$ implies $\langle \eta, \lambda' \rangle \geqslant \langle \eta', \lambda' \rangle$. Thus, the generating minimum is precisely $\pi^{\langle \eta', \lambda' \rangle}$, simplifying the algebra to:

$$\mathcal{O}(V_{\mathcal{M},\lambda'(\pi)}) \cong \bigoplus_{\eta \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{M}}} \mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M}) \cdot \pi^{\langle \eta, \lambda' \rangle}.$$

By computing the intersection of these fractional R-algebras over the finite set $S_{\mathcal{M}}(\lambda)$, we can evaluate the π-adic lattice component-wise by simply taking the maximum valuation:

$$\bigcap_{\lambda' \in S_{\mathcal{M}}(\lambda)} \mathcal{O}(V_{\mathcal{M},\lambda'(\pi)}) \cong \bigoplus_{\eta \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{M}}} \mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M}) \cdot \pi^{\max_{\lambda' \in S_{\mathcal{M}}(\lambda)} \langle \eta, \lambda' \rangle}.$$

It remains to identify this intersection with $\mathcal{O}(W_{\lambda(\pi)})$. The coordinate ring of the Vinberg monoid $V_{\mathcal{G},\lambda(\pi)}$ is the fractional R-subalgebra of $\mathcal{O}(\mathcal{G}_K)$ given by $\sum_{\mu \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{G}}} \mathrm{fil}^{\mathcal{G}}_{\mu}\mathcal{O}(\mathcal{G}) \cdot \pi^{\langle \mu, \lambda \rangle}$. The parabolic monoid $V_{\mathcal{P},\lambda(\pi)}$ is defined as the schematic closure of $\mathcal{P}_K$ inside $V_{\mathcal{G},\lambda(\pi)}$, so its coordinate ring is the image under the restriction map $\mathcal{O}(\mathcal{G}) \twoheadrightarrow \mathcal{O}(\mathcal{P})$. The categorical quotient $W_{\lambda(\pi)} := V_{\mathcal{P},\lambda(\pi)} /\!\!/ (\mathcal{U} \times \mathcal{U})$ is formed by taking the unipotent invariants. Since $\mathcal{P} \cong \mathcal{M} \ltimes \mathcal{U}$, the composition $\mathcal{O}(\mathcal{G}) \to \mathcal{O}(\mathcal{P}) \to \mathcal{O}(\mathcal{M})$ precisely isolates the $\mathcal{U} \times \mathcal{U}$-invariant functions. Thus, we define the relative filtration $\mathrm{fil}^{\mathcal{G} \to \mathcal{M}}_{\mu}\mathcal{O}(\mathcal{M})$ as the image of $\mathrm{fil}^{\mathcal{G}}_{\mu}\mathcal{O}(\mathcal{G})$ under this canonical restriction map. The coordinate ring of $W_{\lambda(\pi)}$ is exactly governed by this relative filtration:

$$\mathcal{O}(W_{\lambda(\pi)}) = \sum_{\mu \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{G}}} \mathrm{fil}^{\mathcal{G} \to \mathcal{M}}_{\mu}\mathcal{O}(\mathcal{M}) \cdot \pi^{\langle \mu, \lambda \rangle}.$$

The associated graded pieces of $\mathrm{fil}^{\mathcal{G}}_{\mu}\mathcal{O}(\mathcal{G})$ are canonically isomorphic to $\mathrm{S}_{-w_0^{\mathcal{G}}(\mu)}(\mathcal{G}) \otimes_R \mathrm{S}_{\mu}(\mathcal{G})$, where $\mathrm{S}_{\mu}(\mathcal{G}) = H^0(\mathcal{G}/\mathcal{B}, \mathcal{L}_{\mu})$ is the Schur module of highest weight μ. The restriction $\mathcal{O}(\mathcal{G}) \to \mathcal{O}(\mathcal{M})$ corresponds to the pullback of global sections along the closed immersion of flag varieties $\mathcal{M}/(\mathcal{B} \cap \mathcal{M}) \hookrightarrow \mathcal{G}/\mathcal{B}$. Because this pullback is surjective, we obtain canonical surjections $\mathrm{S}_{\mu}(\mathcal{G}) \twoheadrightarrow \mathrm{S}_{\mu}(\mathcal{M})$ of finite free R-modules. Thus, the graded piece $\mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M})$ appears in the relative filtration $\mathrm{fil}^{\mathcal{G}\to\mathcal{M}}_{\mu}\mathcal{O}(\mathcal{M})$ exactly when η appears as an $\mathcal{M}$-highest weight in the restriction of $\mathrm{S}_{\mu}(\mathcal{G})$, which occurs if and only if $\mu \succeq_{\mathcal{G}} \eta$. Under the splitting, the component for gr_{η} in $\mathcal{O}(W_{\lambda(\pi)})$ is generated by $\pi^{\langle \mu, \lambda \rangle}$ across all such $\mathcal{G}$-dominant weights μ. The minimal valuation is therefore $\min_{\mu \succeq_{\mathcal{G}} \eta} \langle \mu, \lambda \rangle$.

This minimum valuation exactly coincides with the supremum over the maximal bounded $\mathcal{M}$-dominant cocharacters:

$$\min_{\substack{\mu \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathcal{G}} \\ \mu \succeq_{\mathcal{G}} \eta}} \langle \mu, \lambda \rangle = \max_{\lambda' \in S_{\mathcal{M}}(\lambda)} \langle \eta, \lambda' \rangle.$$

This confirms that the π-adic lattice on each graded piece $\mathrm{gr}_{\eta}\,\mathcal{O}(\mathcal{M})$ of $\mathcal{O}(W_{\lambda(\pi)})$ matches the intersection bound exactly. Because both fractional R-subalgebras of $\mathcal{O}(\mathcal{M})[1/\pi]$ decompose identically under the splitting, they must coincide, yielding the desired equality of R-algebras. □

Corollary 3.4.13. *W is finitely presented over R.*

Remark 3.4.14 (Points of the categorical quotient)**.** The above result implies that there is a natural map

$$\bigsqcup_{\lambda' \in S_{\mathcal{M}}(\lambda)} V_{\mathcal{M},\lambda(\pi)} \to W_{\lambda(\pi)}$$

which identifies all the components of the generic fiber of the source with the generic fiber of the target. In particular, taking R-points gives exactly the disjoint union:

$$W_{\lambda(\pi)}(R) = \bigsqcup_{\lambda' \in S_{\mathcal{M}}(\lambda)} V_{\mathcal{M},\lambda'(\pi)}(R)$$

as subsets of $\mathcal{M}(K)$. By Proposition 3.4.6 applied to $\mathcal{M}$, the points $V_{\mathcal{M},\lambda'(\pi)}(R)$ decompose into $\mathcal{M}(R)$ double cosets bounded by λ', yielding a Cartan decomposition for $W_{\lambda(\pi)}(R)$.

Remark 3.4.15 (Descent to non-split groups)**.** Suppose $\mathcal{G}$ is a non-split reductive group scheme over $\mathbf{Z}_p$ and λ is a cocharacter defined over $\mathbf{Z}_p$. By the structure theory of reductive groups over local rings, $\mathcal{G}$ splits over a finite étale extension R of $\mathbf{Z}_p$. Let $\Gamma = \mathrm{Gal}(R/\mathbf{Z}_p)$ be the Galois group.

The Vinberg monoid $V_{\mathcal{G}}$ and the related constructions can be defined over $\mathbf{Z}_p$ by Galois descent. Over R, the Vinberg monoid $V_{\mathcal{G}_R}$ is constructed as above. The canonical action of Γ on $\mathcal{G}_R$ via the descent datum induces an action on its coordinate ring $\mathcal{O}(\mathcal{G}_R)$. This action preserves the abstract Cartan $\mathcal{T}_R$ and its set of simple roots Δ. Consequently, it stabilizes the dominant chamber $\mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathrm{pos}}$ and the partial order $\preceq$. Because the multi-filtration $\mathrm{fil}^{\mathcal{G}_R}_{\nu}\mathcal{O}(\mathcal{G}_R)$ is defined intrinsically via the $\mathcal{G}_R \times \mathcal{G}_R$-action and this partial order, an element $\sigma \in \Gamma$ maps $\mathrm{fil}^{\mathcal{G}_R}_{\nu}\mathcal{O}(\mathcal{G}_R)$ isomorphically onto $\mathrm{fil}^{\mathcal{G}_R}_{\sigma(\nu)}\mathcal{O}(\mathcal{G}_R)$. Thus, the multi-Rees algebra

$$\mathcal{O}(V_{\mathcal{G}_R}) = \bigoplus_{\nu \in \mathbf{X}^{\bullet}(\mathcal{T})^{+}_{\mathrm{pos}}} \mathrm{fil}^{\mathcal{G}_R}_{\nu}\mathcal{O}(\mathcal{G}_R)$$

inherits a Γ-action compatible with its algebra structure. By the effectiveness of descent for affine schemes, taking Γ-invariants yields an affine monoid scheme $V_{\mathcal{G}}$ over $\mathbf{Z}_p$ such that $V_{\mathcal{G}} \times_{\mathbf{Z}_p} R \cong V_{\mathcal{G}_R}$.

The abelianization map $\mathfrak{a}_R\colon V_{\mathcal{G}_R} \to \mathcal{T}^{+}_{\mathrm{ad},R}$ is Γ-equivariant, where Γ acts on $\mathcal{T}^{+}_{\mathrm{ad},R}$ via its canonical action on the based root datum. This allows $\mathfrak{a}_R$ to descend to a morphism $\mathfrak{a}\colon V_{\mathcal{G}} \to \mathcal{T}^{+}_{\mathrm{ad}}$ over $\mathbf{Z}_p$. Furthermore, because the cocharacter λ is defined over $\mathbf{Z}_p$, the element $\lambda(p)$ defines a $\mathbf{Z}_p$-point of $\mathcal{T}^{+}_{\mathrm{ad}}$. Base-changing $V_{\mathcal{G}}$ along this point yields an affine $\mathbf{Z}_p$-scheme $V_{\mathcal{G},\lambda(p)}$ that recovers $V_{\mathcal{G}_R,\lambda(p)}$ after base change to R.

Similarly, if $\mathcal{P} \subset \mathcal{G}$ is a parabolic subgroup defined over $\mathbf{Z}_p$ with Levi quotient $\mathcal{M}$, the Γ-action preserves $\mathcal{P}_R$ and $\mathcal{M}_R$. The categorical quotient $W_{\lambda(p),R} := V_{\mathcal{P}_R,\lambda(p)} /\!\!/ (\mathcal{U}_R \times \mathcal{U}_R)$ inherits a Γ-action and descends to a flat affine $\mathbf{Z}_p$-scheme $W_{\lambda(p)}$. Finally, the subring equality of Proposition 3.4.12 descends to $\mathbf{Z}_p$. While individual cocharacters $\lambda' \in S_{\mathcal{M}}(\lambda)$ might not be defined over $\mathbf{Z}_p$, the set $S_{\mathcal{M}}(\lambda)$ is stable under the action of Γ. Consequently, the intersection $\bigcap_{\lambda' \in S_{\mathcal{M}}(\lambda)} \mathcal{O}(V_{\mathcal{M}_R, \lambda'(p)})$ is a Γ-stable R-subalgebra of $\mathcal{O}(\mathcal{M}_K)$, where $K := R[1/p]$. Because taking invariants under the finite group Γ commutes with intersections of submodules inside the common fraction field, evaluating the Γ-invariants on both sides of the equality over R yields the desired equality for $\mathcal{O}(W_{\lambda(p)})$ as a $\mathbf{Z}_p$-subalgebra of $\mathcal{O}(\mathcal{M}_{\mathbf{Q}_p})$.

3.5. **Stacks of p-isogenies.** Following [LM], we explain the notion of quasi-isogenies using the Vinberg monoid, which serves as a local model for the space of modifications (cf. Remark 2.1.21).

Construction 3.5.1 (A $V_{\mathcal{G},\lambda(p)}$-bundle)**.** Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be two $\mathcal{G}$-torsors over $\mathfrak{X}^{\mathrm{syn}}$. The product $\mathcal{P}_1 \times_{\mathfrak{X}^{\mathrm{syn}}} \mathcal{P}_2$ naturally forms a $(\mathcal{G} \times \mathcal{G})$-torsor. We form the associated $V_{\mathcal{G},\lambda(p)}$-fiber bundle over $\mathfrak{X}^{\mathrm{syn}}$:

$$V_{\lambda(p)}(\mathcal{P}_1, \mathcal{P}_2) := (\mathcal{P}_1 \times_{\mathfrak{X}^{\mathrm{syn}}} \mathcal{P}_2) \times^{\mathcal{G} \times \mathcal{G}} V_{\mathcal{G},\lambda(p)}.$$

Definition 3.5.2 (Quasi-isogeny bounded by λ)**.** A **quasi-isogeny bounded by** λ (or **λ-modification**) $\mathcal{P}_1 \dashrightarrow \mathcal{P}_2$ over $\mathfrak{X}^{\mathrm{syn}}$ is a global section $s\colon \mathfrak{X}^{\mathrm{syn}} \to V_{\lambda(p)}(\mathcal{P}_1, \mathcal{P}_2)$.

We can define the moduli stack of such bounded quasi-isogenies between apertures.

Definition 3.5.3 (Stack of p-isogenies)**.** Let λ be a dominant cocharacter of $\mathcal{G}$ defined over $\mathbf{Z}_p$. We define the p-adic formal prestack $\mathrm{QIsog}^{\mathcal{G},\mu}_{\lambda(p)}$ on p-complete rings R as the category whose objects over R are triples $(\mathcal{A}_1, \mathcal{A}_2, s)$, where $\mathcal{A}_1, \mathcal{A}_2 \in \mathrm{BT}^{\mathcal{G},\mu}_\infty(R)$ are $(\mathcal{G},\mu)$-apertures over R, and $s\colon \mathcal{A}_1 \dashrightarrow \mathcal{A}_2$ is a quasi-isogeny bounded by λ.

Remark 3.5.4 (As a stack of syntomic sections)**.** The definition of $\mathrm{QIsog}^{\mathcal{G},\mu}_{\lambda(p)}$ can be naturally reformulated using syntomic sections. For an arbitrary formal prestack $\mathcal{Y}$ over $\mathfrak{X}^{\mathrm{syn}}$ of a p-adic formal Artin stack $\mathfrak{X}$, we define the formal prestack of **syntomic sections** $\Gamma^{\mathrm{syn}}(\mathcal{Y})$ over $\mathfrak{X}$ by assigning to any test ring R mapping to $\mathfrak{X}$ the groupoid of sections of the pullback $\mathcal{Y} \times_{\mathfrak{X}^{\mathrm{syn}}} R^{\mathrm{syn}} \to R^{\mathrm{syn}}$. It is shown in [GM24, Theorem 8.8.4] that $\Gamma^{\mathrm{syn}}(Y)$ is represented by a p-adic formal Artin stack if $\mathcal{Y}$ is what is called *1-bounded*, together with few other technical assumptions. In these terms, the stack of p-isogenies $\mathrm{QIsog}^{\mathcal{G},\mu}_{\lambda(p)}$ is precisely the stack of syntomic sections $\Gamma^{\mathrm{syn}}(V_{\lambda(p)}(\mathcal{P}_1, \mathcal{P}_2))$ relative to the base $\mathfrak{X} = \mathrm{BT}^{\mathcal{G},\mu}_\infty \times \mathrm{BT}^{\mathcal{G},\mu}_\infty$, where $\mathcal{P}_1$ and $\mathcal{P}_2$ are the universal $\mathcal{G}$-torsors representing the universal $(\mathcal{G},\mu)$-apertures.

Remark 3.5.5 (Mixed cocharacters)**.** Definition 3.5.3 naturally extends to the case where $\mathcal{A}_1$ and $\mathcal{A}_2$ are $(\mathcal{G},\mu_1)$ and $(\mathcal{G},\mu_2)$-apertures, respectively, for potentially distinct cocharacters μ_1 and μ_2. In this case, it still makes sense to talk about quasi-isogenies bounded by λ between them. We warn the reader that this gives a nontrivial notion only in characteristic p.

Theorem 3.5.6 (Representability of stack of quasi-isogenies)**.** *The p-adic formal prestack $\mathrm{QIsog}^{\mathcal{G},\mu}_{\lambda(p)}$ is represented by a 0-dimensional finitely presented p-adic formal Artin stack over $\mathrm{Spf}\,\mathcal{O}$. In particular, there is a p-Hecke correspondence diagram of p-adic formal Artin stacks:*

$$\begin{tikzcd} & \mathrm{QIsog}^{\mathcal{G},\mu}_{\lambda(p)} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ \mathrm{BT}^{\mathcal{G},\mu}_\infty & & \mathrm{BT}^{\mathcal{G},\mu}_\infty \end{tikzcd}$$

Proof. See [LM]. □

Theorem 3.5.7 (Relative Frobenius quasi-isogeny)**.** *Let R be an $\mathbf{F}_p$-algebra. For any $(\mathcal{G},\mu)$-aperture $\mathcal{A} \in \mathrm{BT}^{\mathcal{G},\mu}_\infty(R)$, its pullback $\mathcal{A}^{(p)}$ along the absolute Frobenius on R is naturally a $(\mathcal{G},\varphi(\mu))$-aperture. There is a* *canonical quasi-isogeny bounded by μ, called the* ***relative Frobenius quasi-isogeny:***

$$\mathrm{Frob}_{\mathcal{A}} \colon \mathcal{A} \dashrightarrow \mathcal{A}^{(p)}.$$

Furthermore, if $\mathcal{A}$ is represented by a $\mathcal{G}$-torsor $\mathcal{E}$ over R^{syn} with its 1-truncation $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ equipped with its conjugate realization $T_{\mathrm{conj}}(\mathcal{E}_1)$, which is a reduction of structure to $\mathcal{P}^{+}_{\varphi(\mu)}$, then the quasi-isogeny $\mathrm{Frob}_{\mathcal{A}}$ is precisely the parahoric pushforward of this reduction $T_{\mathrm{conj}}(\mathcal{E}_1)$.

Proof. We can always perform the parahoric pushforward routine using the conjugate realization to produce a functorial quasi-isogeny $\mathcal{A} \dashrightarrow \mathcal{A}'$ bounded by μ. What needs to be shown is that there is a functorial isomorphism $\mathcal{A}' \simeq \mathcal{A}^{(p)}$. Since $\mathrm{BT}^{\mathcal{G},\mu}_{\infty} \otimes \mathbf{F}_p$ is smooth, we may assume R is a semiperfect ring by quasisyntomic descent. There is a universal map $R^{\flat} \to R$ from a perfect $\mathbf{F}_p$-algebra to R. For $R^{\flat}$, everything can be checked by hand using the the Witt frame $(W(R^{\flat}), (p))$. By base changing, we get the result over R too. □

Theorem 3.5.8 ($\Gamma_0(p^n)$-level as a space of p-isogenies)**.** *Recall from Notation 3.1.3 that ν is the sum of all Galois conjugates of μ_v. For any $n \geqslant 1$, there is a Cartesian diagram of formal stacks:*

$$\begin{tikzcd}
\widehat{\mathscr{S}}_{\Gamma_0(p^n)} \arrow[r] \arrow[d, "s_n"'] & \mathrm{QIsog}^{\mathcal{G},-\mu_v}_{-n\nu(p)} \arrow[d, "p_1"] \\
\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \arrow[r, "\varpi"'] & \mathrm{BT}^{\mathcal{G},-\mu_v}_{\infty}
\end{tikzcd}$$

where ϖ is the syntomic realization and **[TODO: compatibility after étale realization]** *Furthermore, $p_2 \circ \widetilde{\pi}$ classifies the universal aperture over the Hecke-translated Shimura variety.*

Proof. [LM] **[TODO: this is probably not hard to prove regardless.]** □

Remark 3.5.9 (Cartier duality and quasi-isogenies)**.** Let $C \colon \mathcal{G} \to \mathcal{G}$ be a Chevalley involution. Recall from Theorem 2.5.5 that C induces an equivalence between the stacks of apertures $\mathrm{BT}^{\mathcal{G},\mu}_{\infty} \xrightarrow{\sim} \mathrm{BT}^{\mathcal{G},-w_0\mu}_{\infty}$. This duality extends naturally to quasi-isogenies: it transforms a quasi-isogeny $\mathcal{A}_1 \dashrightarrow \mathcal{A}_2$ of $(\mathcal{G},\mu)$-apertures bounded by λ into a quasi-isogeny $\mathcal{A}_2^C \dashrightarrow \mathcal{A}_1^C$ of $(\mathcal{G},-w_0\mu)$-apertures bounded by λ in the opposite direction.

3.6. **Minimal compactifications.** To extend our perfectoidness results from the open Shimura variety to its minimal compactification, we will use Hartog's extension principle to uniquely extend functions and morphisms across the boundary.

Notation 3.6.1. For any open compact subgroup $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, we denote the minimal (Baily–Borel–Satake) compactification of the generic fiber by $\mathrm{Sh}^{*}_{K_p}$. This is a normal projective variety over the reflex field.

Remark 3.6.2 (Automorphic vector bundles)**.** By composing the syntomic realization $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{\infty}$ with the filtered de Rham realization $\mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{\infty} \to B\mathcal{P}^{-}_{-\mu_v}$ (Remark 2.3.15), we obtain a canonical classifying map $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to B\mathcal{P}^{-}_{-\mu_v}$. Any representation of $\mathcal{P}^{-}_{-\mu_v}$ thus induces an automorphic vector bundle over $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ (which algebraizes to $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$). In particular, an ample character $\chi \in \mathbf{X}^{\bullet}(\mathcal{M}_{\mu_v})$, which inflates to a character of $\mathcal{P}^{-}_{-\mu_v}$, induces an automorphic line bundle, denoted ω^{χ}. These automorphic bundles are canonically defined over $\overline{\mathbf{Q}}$ and by the classical Baily-Borel theorem, because the character χ is ample, the graded algebra of automorphic forms on the generic fiber is finitely generated, and the sections of $(\omega^{\chi})^{\otimes n}$ for $n \gg 0$ define an open immersion of the Shimura variety into the resulting projective variety. Thus, for all sufficiently large p, this projective embedding actually extends integrally and in particular ω^{χ} is ample.

Remark 3.6.3 (Automorphic vector bundles coming from the Levi)**.** In the context of Remark 3.6.2, the cocharacter μ_v associated to the Shimura datum is only well-defined up to conjugacy. As a result, the precise parabolic subgroup $\mathcal{P}^{-}_{-\mu_v}$ depends on the specific choice of the representative μ_v. If we were to construct an automorphic vector bundle from an arbitrary representation of $\mathcal{P}^{-}_{-\mu_v}$, the resulting bundle would depend on this non-canonical choice. However, the Levi quotient $\mathcal{M}_{\mu_v}$ of $\mathcal{P}^-_{-\mu_v}$ is canonically independent of the choice of representative (up to canonical isomorphism). Therefore, by restricting our attention to representations of $\mathcal{P}^-_{-\mu_v}$ that arise as inflations of representations of the Levi quotient $\mathcal{M}_{\mu_v}$ (such as the character χ), the induced automorphic vector bundles are canonical and genuinely independent of the choice of μ_v.

Construction 3.6.4 (Minimal compactification)**.** As discussed in Remark 3.6.2, for sufficiently large p, we can construct the integral model of the minimal compactification as

$$\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} := \operatorname{Proj}\left(\bigoplus_{n\geqslant 0} \mathrm{H}^0(\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}, (\omega^\chi)^{\otimes n})\right).$$

This construction yields a normal projective scheme over $\mathcal{O}_{E,(v)}$ into which $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$ naturally embeds as an open dense subscheme. By standard properties of the minimal compactification, this resulting scheme $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ is canonically independent of the choice of ample character χ. By construction, ω^χ extends to $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ and we denote it by the same notation. For any deeper level structure $K_p \subset \mathcal{G}(\mathbf{Z}_p)$ at p—such as the parahoric level $\Gamma_0(p^n)$, the pro-p Iwahori level $\Gamma_1(p^n)$, or the principal congruence subgroup $\Gamma(p^n)$—we define the integral model $\mathscr{S}^*_{K_p}$ of the minimal compactification simply as the normalization of $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ in the function field of $\mathrm{Sh}^*_{K_p}$. This normalization procedure yields a tower of normal projective integral models extending the tower of (open) integral canonical models $\mathscr{S}_{K_p}$.

The following is true essentially by construction:

Proposition 3.6.5 (Comparison of automorphic line bundles)**.** *Let $\zeta\colon \mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k \to \mathcal{G}^c\text{-zip}_{-\mu_v}$ be the map classifying the universal $\mathcal{G}^c$-zip. Then the restriction of the automorphic line bundle ω^χ to the special fiber $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ is canonically isomorphic to $\zeta^*\mathcal{V}(\chi)$ (cf. Remark 2.4.2).*

Proof. Indeed, $(\mathcal{Q}, \mathcal{Q}^-, \mathcal{Q}^+, \alpha) \mapsto \mathcal{Q}^-$ gives a classifying map $\mathcal{G}^c\text{-zip}_{-\mu_v} \to B\mathcal{P}^-_{-\mu_v}$ whose composition with ζ is the mod p reduction of the filtered de Rham realization. □

Let $Z_{K_p} := \mathscr{S}^*_{K_p} \setminus \mathscr{S}_{K_p}$ denote the boundary of the compactification. Since the transition morphisms in the tower $\mathscr{S}^*_{K_p} \to \mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ are finite, the codimension of the boundary is independent of the choice of K_p.

Assumption 3.6.6. We assume that the boundary $Z := \mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \setminus \mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$ has codimension $\geqslant 2$ in $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$. This is satisfied precisely when no $\mathbf{Q}$-simple factor of G^{ad} giving rise to a non-compact factor of $G(\mathbf{R})$ has $\mathbf{Q}$-rank 1. For instance, in the Siegel case, this is equivalent to excluding the modular curves. For exceptional Shimura varieties, the $\mathbf{Q}$-rank of any non-compact $\mathbf{Q}$-simple factor is bounded above by its $\mathbf{R}$-rank, which is at most 2 (for type E_6) or 3 (for type E_7). Our assumption therefore safely includes exceptional Shimura varieties, provided that their $\mathbf{Q}$-rank is not 1.

Because $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ is a normal scheme, the codimension $\geqslant 2$ assumption allows us to use classical extension principles, such as the algebraic Hartog's lemma:

Proposition 3.6.7 (Algebraic Hartog's lemma)**.** *Let R be a normal Noetherian ring, and let $Z \subset \operatorname{Spec} R$ be a closed subset everywhere of codimension $\geqslant 2$. Then the restriction map*

$$R \to \mathrm{H}^0(\operatorname{Spec} R \setminus Z, \mathcal{O}_{\operatorname{Spec} R})$$

is an isomorphism.

Proposition 3.6.8 (Extension of Hasse invariant)**.** *For ample characters χ, the Hasse invariant* Ha *uniquely extends to a global section of the automorphic line bundle ω^χ over the special fiber of the minimal compactification $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$.*

Proof. By Proposition 3.6.5, the restriction of ω^χ to the open Shimura variety $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ canonically identifies with $\zeta^*\mathcal{V}(\chi)$, meaning Ha defines a regular section of ω^χ over this open locus. Since $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ is normal and the boundary has codimension $\geqslant 2$ (Assumption 3.6.6), Proposition 3.6.7 implies that Ha extends uniquely to a global section over $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$. □

The following is a generalization of [GK19, Corollary I.2.6] for the μ-ordinary stratum and in particular also recovers [BST25b, Lemma 6.4]:

Corollary 3.6.9 (Affineness of extended μ-ordinary locus)**.** *The μ-ordinary locus of $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$, defined as the non-vanishing locus of the extended Hasse invariant* Ha*, is an affine open subscheme.*

3.7. **p-Hecke correspondences over the μ-ordinary locus.** We study spaces of isogenies between μ-ordinary apertures. This will be used to define the anticanonical tower. In the Hodge type case, a version of this is worked out in [Mad25, §2] in the language of abelian schemes.

Setup 3.7.1. Maintain Setup 3.3.1. Let λ be a dominant cocharacter of $\mathcal{G}$ defined over $\mathbf{Z}_p$. We carry over notation from Section 3.4. Also, we write $S_\nu(\lambda)$ instead of $S_{\mathcal{M}_\nu}(\lambda)$.

Remark 3.7.2 (Quasi-isogenies preserve μ-ordinariness)**.** Since μ-ordinariness is an open condition, it can be checked on geometric points. Let κ be an algebraically closed field of characteristic p. Using the Witt frame, a $(\mathcal{G},\mu)$-apertures over k is just the data of a $\mathcal{G}$-torsor over $W(\kappa)$ equipped with a semilinear modification over $W(\kappa)[1/p]$ bounded by μ. This data gives an isocrystal with $\mathcal{G}$-structure over κ in the sense of Kottwitz [Kot85] (see also [MY26, §3.3]). A quasi-isogeny of $(\mathcal{G},\mu)$-apertures over κ then induces an isomorphism of respective isocrystals. Therefore, it suffices to observe that μ-ordinariness of a $(\mathcal{G},\mu)$-aperture over κ is witnessed by its isocrystal. This is implied by [Wor13, Proposition 7.2] (see also [MY26, Lemma 3.6.14]).

Lemma 3.7.3. *Let $\mathcal{Q}_1, \mathcal{Q}_2$ be $(\mathcal{P}_\nu^+,\mu)$-apertures over a p-complete ring R. Let $\mathcal{Q}_1^+, \mathcal{Q}_2^+$ be $\mathcal{Q}_1, \mathcal{Q}_2$ viewed as $(\mathcal{G},\mu)$-apertures. The natural map of stacks over R^{syn}*

$$(\mathcal{Q}_1 \times \mathcal{Q}_2) \times^{\mathcal{P}_\nu^+ \times \mathcal{P}_\nu^+} V_{\mathcal{P}_\nu^+,\lambda(p)} \to (\mathcal{Q}_1^+ \times \mathcal{Q}_2^+) \times^{\mathcal{G}\times\mathcal{G}} V_{\mathcal{G},\lambda(p)}$$

induces an isomorphism of stacks of syntomic sections.

Proof. Since $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu}$ is smooth, we may assume R is p-completely smooth. The desired isomorphism can be checked quasisyntomic locally. Therefore, we may assume R is perfectoid (see Example 2.1.8). Using the $\mathrm{A}_{\mathrm{inf}}$ frame, the desired result comes down to showing that if $m_1, m_2 \in \mathcal{P}_\nu^+(\mathrm{A}_{\mathrm{inf}}(R)[1/\xi])$ and $v \in \mathcal{G}(\mathrm{A}_{\mathrm{inf}}(R)[1/p])$ are such that $vm_1 = m_2\varphi(v)$ in $\mathcal{G}(\mathrm{A}_{\mathrm{inf}}(R)[1/p, 1/\xi])$ then $v \in \mathcal{P}_\nu^+(\mathrm{A}_{\mathrm{inf}}(R)[1/p])$.

Choose a faithful representation $\mathcal{G} \hookrightarrow \mathrm{GL}_n$ over $\mathbf{Z}_p$ and a basis adapted to the weight spaces of ν, so that $\mathcal{P}_\nu^+$ acts via block upper-triangular matrices. The diagonal blocks correspond to the strictly decreasing weights $w_1 > w_2 > \cdots > w_k$ of ν. Because m_1 and m_2 are μ-ordinary modifications, their l-th diagonal blocks are of the form $\xi^{w_l}U_{1,l}$ and $\xi^{w_l}U_{2,l}$ respectively, where $U_{1,l}, U_{2,l}$ are invertible matrices over $\mathrm{A}_{\mathrm{inf}}(R)[1/p]$. Assume for the sake of contradiction that $v \notin \mathcal{P}_\nu^+(\mathrm{A}_{\mathrm{inf}}(R)[1/p])$, meaning v has some non-zero block strictly below the diagonal. Choose a block index pair (i,j) with $i > j$ such that $v_{i,j} \neq 0$ and $i - j$ is maximal. By this maximality, $v_{i,k} = 0$ for all $k < j$, and $v_{k,j} = 0$ for all $k > i$. Examining the (i,j) block of the equation $vm_1 = m_2\varphi(v)$, the block upper-triangular structure of m_1 and m_2 along with our maximality assumption reduces the matrix multiplication sums to a single term on each side:

$$v_{i,j}(m_1)_{j,j} = (m_2)_{i,i}\varphi(v_{i,j}).$$

Substituting the explicit forms of the diagonal blocks, we obtain:

$$v_{i,j}\xi^{w_j}U_{1,j} = \xi^{w_i}U_{2,i}\varphi(v_{i,j}).$$

Let $d = w_j - w_i$. Because $i > j$, the weights strictly decrease down the diagonal, so $d > 0$. We can rewrite the equation as:

$$v_{i,j} = \xi^{-d} U_{2,i} \varphi(v_{i,j}) U_{1,j}^{-1}.$$

Since $v \in \mathrm{GL}_n(\mathrm{A}_{\mathrm{inf}}(R)[1/p])$, the entry $v_{i,j}$ has no ξ-adic poles. The equation then implies that $\varphi(v_{i,j})$ must be divisible by ξ^d in $\mathrm{A}_{\mathrm{inf}}(R)[1/p]$. Consequently, $v_{i,j}$ itself must be divisible by $\varphi^{-1}(\xi)^d$. By pulling back the equation under φ^{-1}, we see that $v_{i,j}$ being divisible by $\varphi^{-1}(\xi)^d$ implies $\varphi(v_{i,j})$ is further divisible by $\varphi(\varphi^{-1}(\xi))^d = \xi^d$ (which we already knew) but iterating this logic means $v_{i,j}$ is divisible by $\varphi^{-2}(\xi)^d$, and so forth. Inductively, $v_{i,j}$ must be divisible by $\prod_{k=1}^{N} \varphi^{-k}(\xi)^d$ for all $N \geqslant 1$. However, the elements $\varphi^{-k}(\xi) = p - [\widetilde{p}^{1/p^k}]$ generate pairwise coprime ideals in $\mathrm{A}_{\mathrm{inf}}(R)[1/p]$ corresponding to distinct untilts. Since $v_{i,j}$ is bounded p-adically, it cannot be infinitely divisible by such a sequence of non-units unless $v_{i,j} = 0$, which contradicts our assumption. Thus, v must be block upper-triangular, concluding that $v \in \mathcal{P}_\nu^+(\mathrm{A}_{\mathrm{inf}}(R)[1/p])$. □

Lemma 3.7.4. *Let $\mathcal{Q}_1, \mathcal{Q}_2$ be $(\mathcal{M}_\nu, \mu)$-apertures over a p-complete ring R. The natural map of stacks over R^{syn}*

$$(\mathcal{Q}_1 \times \mathcal{Q}_2) \times^{\mathcal{M}_\nu \times \mathcal{M}_\nu} \left(\bigsqcup_{\lambda' \in S_\nu(\lambda)} V_{\mathcal{M}_\nu, \lambda'(p)} \right) \to (\mathcal{Q}_1 \times \mathcal{Q}_2) \times^{\mathcal{M}_\nu \times \mathcal{M}_\nu} W_{\lambda(p)}$$

induced by Remark 3.4.14 yields an isomorphism of stacks of syntomic sections.

Proof. Since μ is central in $\mathcal{M}_\nu$, we may choose a (non-canonical) identification of $\mathrm{BT}_\infty^{\mathcal{M}_\nu, \mu}$ with the formal classifying stack of the profinite group $\mathcal{M}_\nu(\mathbf{Z}_p)$ by [GM24, Proposition 10.3.4]. Then the desired result is clear by putting $R = \mathbf{Z}_p$ in the point description in Remark 3.4.14. □

Definition 3.7.5 (μ-ordinary quasi-isogeny components)**.** Recall that we have the stack of quasi-isogenies:

$$\begin{tikzcd} & \mathrm{QIsog}_{\lambda(p)}^{\mathcal{G},\mu} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ \mathrm{BT}_\infty^{\mathcal{G},\mu} & & \mathrm{BT}_\infty^{\mathcal{G},\mu} \end{tikzcd}.$$

By the previous two lemmas, there is a canonical diagram of stacks of syntomic sections:

$$\begin{tikzcd} \mathrm{QIsog}_{\lambda(p)}^{\mathcal{G},\mu}|_{\mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}} \times \mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}}} \arrow[r, "\simeq", "\text{Lemma 3.7.3}"'] & \Gamma^{\mathrm{syn}}\left((\mathcal{Q}_1 \times \mathcal{Q}_2) \times^{\mathcal{P}_\nu^+ \times \mathcal{P}_\nu^+} V_{\mathcal{P}_\nu^+, \lambda(p)}\right) \arrow[d] \\ & \Gamma^{\mathrm{syn}}\left((\mathcal{Q}_1/\mathcal{U}_\nu^+ \times \mathcal{Q}_2/\mathcal{U}_\nu^+) \times^{\mathcal{M}_\nu \times \mathcal{M}_\nu} W_{\lambda(p)}\right) \arrow[d, "\text{Lemma 3.7.4}"', "\simeq"] \\ \bigsqcup_{\lambda' \in S_\nu(\lambda)} \mathrm{QIsog}_{\lambda'(p)}^{\mathcal{M}_\nu,\mu} & \Gamma^{\mathrm{syn}}\left((\mathcal{Q}_1/\mathcal{U}_\nu^+ \times \mathcal{Q}_2/\mathcal{U}_\nu^+) \times^{\mathcal{M}_\nu \times \mathcal{M}_\nu} \bigsqcup_{\lambda' \in S_\nu(\lambda)} V_{\mathcal{M}_\nu, \lambda'(p)}\right) \arrow[l, "\simeq"'] \end{tikzcd}$$

For each $\lambda' \in S_\nu(\lambda)$, we define the **$\mu$-ordinary quasi-isogeny component** $\mathrm{QIsog}_{\lambda'(p)}^{\mathcal{G},\mu,\mathrm{ord}}$ as the open and closed substack of $\mathrm{QIsog}_{\lambda(p)}^{\mathcal{G},\mu}|_{\mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}} \times \mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}}}$ given by the preimage of $\mathrm{QIsog}_{\lambda'(p)}^{\mathcal{M}_\nu,\mu}$ under this map.

Theorem 3.7.6 (Decomposition of μ-ordinary quasi-isogenies)**.** *The space of quasi-isogenies bounded by $\lambda(p)$ over the μ-ordinary locus decomposes into disjoint components:*

$$\mathrm{QIsog}_{\lambda(p)}^{\mathcal{G},\mu}|_{\mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}} \times \mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}}} \simeq \bigsqcup_{\lambda' \in S_\nu(\lambda)} \mathrm{QIsog}_{\lambda'(p)}^{\mathcal{G},\mu,\mathrm{ord}}.$$

Definition 3.7.7 (The canonical locus)**.** For any $n \geqslant 1$, the open and closed substack $\mathrm{QIsog}_{n\nu(p)}^{\mathcal{G},\mu,\mathrm{ord}}$ of $\mathrm{QIsog}_{n\nu(p)}^{\mathcal{G},\mu}|_{\mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}} \times \mathrm{BT}_\infty^{\mathcal{G},\mu,\mathrm{ord}}}$ corresponding to $n\nu \in S_\nu(n\nu)$ is called the **canonical locus of level** n.

Notation 3.7.8. For brevity, when we write bounds or cocharacters such as $-\nu$, $-\mu$, or $-\lambda$, it is implicitly understood that we take their dominant representatives, i.e., $-w_0\nu$, $-w_0\mu$, and $-w_0\lambda$.

Definition 3.7.9 (The anticanonical locus)**.** For any $n \geqslant 1$, the open and closed substack $\mathrm{QIsog}^{\mathcal{G},\mu,\mathrm{ord}}_{n(-\nu)(p)}$ of $\mathrm{QIsog}^{\mathcal{G},\mu}_{n(-\nu)(p)}\big|_{\mathrm{BT}^{\mathcal{G},\mu,\mathrm{ord}}_\infty \times \mathrm{BT}^{\mathcal{G},\mu,\mathrm{ord}}_\infty}$ corresponding to $-n\nu \in S_\nu(-n\nu)$ is called the **anticanonical locus of level** n.

Remark 3.7.10 (Cartier duality swaps canonical and anticanonical loci)**.** The Chevalley involution induces a contravariant Cartier duality between $\mathrm{BT}^{\mathcal{G},\mu}_\infty$ and $\mathrm{BT}^{\mathcal{G},-\mu}_\infty$, which takes quasi-isogenies bounded by $\lambda(p)$ to quasi-isogenies bounded by $(-\lambda)(p)$ in the opposite direction. Under this duality, an anticanonical quasi-isogeny $x \dashrightarrow y$ of $(\mathcal{G},\mu)$-apertures Cartier dualizes to a canonical quasi-isogeny $y^\vee \dashrightarrow x^\vee$ of $(\mathcal{G},-\mu)$-apertures. This establishes a canonical isomorphism swapping the anticanonical locus for μ with the canonical locus for $-\mu$.

Remark 3.7.11. Note that there is no anticanonical locus in the space of quasi-isogenies bounded by $n\nu(p)$, but that is to be expected. In the special case when $-w_0\nu = \nu$ (such as the Siegel case), the anticanonical locus is found within the same stack of quasi-isogenies as the canonical locus.

3.8. **Embeddings of exceptional Shimura varieties.** We will use the following result to reduce the question of perfectoidness of infinite level exceptional Shimura varieties to only having to consider Shimura varieties with a totally real reflex field. Shimura varieties with totally real reflex fields can then be embedded into Shimura varieties with reflex field $\mathbf{Q}$ by a Weil restriction argument.

Theorem 3.8.1 (Exceptional embedding)**.** *Let (G,X) be a $\mathbf{Q}$-simple adjoint Shimura datum where $G \simeq \mathrm{Res}_{F_0/\mathbf{Q}} H_0$, with F_0 a totally real number field and H_0 an absolutely simple adjoint algebraic group over F_0 of outer type E_6. Let K_0/F_0 denote its diagram field, which is an imaginary quadratic extension of F_0. Let $\pi\colon \widetilde{H}_0 \to H_0$ be the simply connected cover of H_0 over F_0, and define the norm-one torus $T := \mathrm{Res}^1_{K_0/F_0}\mathbf{G}_m$. Fixing an identification $Z(\widetilde{H}_0) \simeq T[3]$, we define the reductive F_0-group*

$$L_0 := (\widetilde{H}_0 \times T)/Z(\widetilde{H}_0),$$

where $Z(\widetilde{H}_0)$ is embedded anti-diagonally. Let $\mathbf{L} := \mathrm{Res}_{F_0/\mathbf{Q}} L_0$. Then the following hold:

(1) We have $\mathbf{L}^{\mathrm{ad}} \simeq G$, and the Shimura datum (G,X) lifts to a Shimura datum $(\mathbf{L}, X_L)$.
(2) There exists a finite totally real extension F/F_0, an adjoint algebraic group G_F over F of type E_7, and a closed embedding of F-groups $(L_0)_F \hookrightarrow G_F$, whose geometric fiber recovers the standard inclusion $(E_6^{\mathrm{sc}} \times \mathbf{G}_m)/\mu_3 \subset E_7^{\mathrm{ad}}$ (see Remark 3.8.3).
(3) Setting $\mathbf{G} := \mathrm{Res}_{F/\mathbf{Q}} G_F$, there exists a Shimura datum $(\mathbf{G}, Y)$ with a totally real reflex field, admitting a closed embedding of Shimura data:

$$(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y).$$

Remark 3.8.2 (Failure of naive Weil restriction trick)**.** The reader may be wondering why we are not employing a routine Weil restriction argument. Specifically, suppose we have a Shimura variety with local reflex field E_v, and we wish to embed it into a Shimura variety with local reflex field $\mathbf{Q}_p$. A standard approach would be to choose a totally real number field F admitting E_v as a p-adic completion (so $F_u \simeq E_v$ for some place $u \mid p$ of F). Then, by Weil restriction along $F/\mathbf{Q}$ of the Shimura datum, one obtains an embedding of Shimura varieties where the target has local reflex field $\mathbf{Q}_p$. Such an embedding is useful for proving perfectoidness because a closed adic subspace of a perfectoid space is automatically perfectoid. This strategy works for Shimura varieties of pre-abelian type but fails for exceptional Shimura varieties because we do not know if the target Shimura variety admits an integral canonical model at p even if the source Shimura variety admits an integral canonical model. The key point is that the embedding in Theorem 3.8.1 is global and hence spreads out to a closed embedding of integral canonical models for all sufficiently large primes p.

Remark 3.8.3 (The standard inclusion of E_6 in E_7)**.** Let G_{ad} be the split adjoint algebraic group of type E_7 over a field k of characteristic zero. The Dynkin diagram of type E_7 contains a unique node (node 7 under the Bourbaki numbering [Bou68, Plate VI]) whose removal yields the Dynkin diagram of type E_6. The choice of this node defines a maximal parabolic subgroup $P \subset G_{\mathrm{ad}}$ whose Levi factor M is isomorphic to the quotient

$$M \simeq (E_6^{\mathrm{sc}} \times \mathbf{G}_m)/\mu_3,$$

where E_6^{sc} is the simply connected group of type E_6, and the finite center $Z(E_6^{\mathrm{sc}}) \simeq \mu_3$ is embedded diagonally (a central element is paired with the inverse scalar by which it acts on the 27-dimensional minuscule module). This gives the canonical closed immersion $(E_6^{\mathrm{sc}} \times \mathbf{G}_m)/\mu_3 \hookrightarrow E_7^{\mathrm{ad}}$.

Under this immersion, it is the 133-dimensional adjoint representation $\mathfrak{e}_7 = \mathrm{Lie}(G_{\mathrm{ad}})$ that decomposes under the action of M as

$$\mathfrak{e}_7 \simeq \mathfrak{e}_6 \oplus \mathfrak{gl}_1 \oplus V_{27} \oplus V_{27}^{\vee},$$

where V_{27} is the 27-dimensional minuscule representation of E_6^{sc}, on which the $\mathbf{G}_m$-factor of M acts with weight 1, and $V_{27}^{\vee}$ is its dual representation, on which the $\mathbf{G}_m$-factor acts with weight -1.

The rest of the section is dedicated to proving Theorem 3.8.1. We first construct the lifted Shimura datum on $\mathbf{L}$.

Construction 3.8.4 (The lifted Shimura datum $(\mathbf{L}, X_L)$)**.** Let H_0 be an absolutely simple adjoint algebraic group over F_0 of outer type E_6, and let $\widetilde{H}_0$ be its simply connected cover. Recall that the quotient of the weight lattice $P(E_6)$ by the root lattice $Q(E_6)$ (which corresponds by duality to the quotient of the coweight lattice $P^{\vee}(E_6)$ by the coroot lattice $Q^{\vee}(E_6)$) is isomorphic to $\mathbf{Z}/3\mathbf{Z}$; see [Bou68, Plate V]. Consequently, the center $Z(\widetilde{H}_0)$ is a group scheme of multiplicative type of order 3, whose characters are permuted by the diagram involution of E_6 over the CM field K_0. The minuscule coweight $\omega_1^{\vee}$ of E_6 (corresponding to the special node 1 under the Bourbaki numbering [Bou68, Plate V]) does not lie in the coroot lattice $Q^{\vee}(E_6)$. Therefore, the associated Hodge cocharacter $\mu_h \colon \mathbf{G}_m \to H_{0,\mathbf{C}}$ does not lift to $\widetilde{H}_{0,\mathbf{C}}$. To resolve this obstruction, we define the norm-one torus $T = \mathrm{Res}^1_{K_0/F_0} \mathbf{G}_m$, which at each real place v of F_0 satisfies $T_v(\mathbf{R}) \simeq U(1)$ as K_0/F_0 is an imaginary quadratic extension of a totally real field. The center of $\widetilde{H}_0$ is canonically isomorphic to the group scheme of 3rd roots of unity $T[3]$ inside T. The cocharacter lattice of $L_{0,\mathbf{C}}$ is the fiber product

$$X_*(L_{0,\mathbf{C}}) = \{(\lambda^{\vee}, \frac{1}{3}\chi^{\vee}) \in P^{\vee}(E_6) \oplus \frac{1}{3}X_*(T_{\mathbf{C}}) \mid \lambda^{\vee} \pmod{Q^{\vee}(E_6)} \equiv \chi^{\vee} \pmod{3 \cdot X_*(T_{\mathbf{C}})}\}$$

under the canonical identification of $P^{\vee}(E_6)/Q^{\vee}(E_6) \simeq \mathbf{Z}/3\mathbf{Z} \simeq X_*(T_{\mathbf{C}})/3X_*(T_{\mathbf{C}})$. Since $X_*(T_{v,\mathbf{C}}) \simeq \mathbf{Z}$ for each archimedean place v, the projection $\mathbf{Z} \to \mathbf{Z}/3\mathbf{Z}$ is surjective, allowing us to choose a cocharacter $\chi^{\vee} \in X_*(T_{\mathbf{C}})$ whose class modulo 3 matches the image of $\omega_1^{\vee}$ in $\mathbf{Z}/3\mathbf{Z}$. The pair $\mu_L := (\omega_1^{\vee}, \frac{1}{3}\chi^{\vee})$ is then a well-defined cocharacter in $X_*(L_{0,\mathbf{C}})$, which represents the lifted Hodge cocharacter.

Lemma 3.8.5. *The conjugacy class X_L generated by the cocharacter μ_L from Construction 3.8.4 defines a valid Shimura datum $(\mathbf{L}, X_L)$ over $\mathbf{Q}$ that lifts the adjoint Shimura datum (G, X).*

Proof. Let $h_L = \prod_v h_{L,v} \colon \mathbf{S} \to L_{0,\mathbf{R}} \simeq \prod_v L_{0,v,\mathbf{R}}$ be the product homomorphism associated with the cocharacters $\mu_{L,v} = (\omega_1^{\vee}, \chi^{\vee})_v$. Under the natural projection map $\mathrm{pr} \colon L_0 \to H_0$, the composition $\mathrm{pr}_{\mathbf{R}} \circ h_L$ coincides with the Deligne homomorphism $h_H \colon \mathbf{S} \to H_{0,\mathbf{R}}$ defining the adjoint Shimura datum (G, X). Since pr induces an isomorphism of adjoint groups, the Hodge types on $\mathrm{Lie}(L_{0,\mathbf{R}})$ and the positivity of the Cartan involution on $L_{0,\mathbf{R}}^{\mathrm{der}}$ are automatically inherited from (G, X). The only axiom remaining to verify is the compactness of the connected center at infinity: as K_0/F_0 is a totally complex quadratic extension of a totally real field, the center $Z(L_0)^0 \simeq T = \mathrm{Res}^1_{K_0/F_0} \mathbf{G}_m$ is anisotropic over $\mathbf{R}$, so its real points $T(\mathbf{R}) \simeq U(1)^{[F_0:\mathbf{Q}]}$ are compact. The conjugacy class X_L thus defines a valid Shimura datum $(\mathbf{L}, X_L)$, proving part (1) of Theorem 3.8.1. □

Remark 3.8.6 (The Tits algebra obstruction)**.** Let k be a field of characteristic zero and let H be a connected semisimple group over k. Over the algebraic closure $\overline{k}$, the irreducible representations of $H_{\overline{k}}$ are classified by their dominant weights. However, over k itself, two obstructions prevent a dominant weight λ of $H_{\overline{k}}$ from arising as the highest weight of a representation defined over k. First, the weight λ must be stable under the action of the absolute Galois group $\mathrm{Gal}(\overline{k}/k)$ on the Dynkin diagram of H. Second, even if λ is Galois-stable, the representation may not be realizable on a vector space over k, but rather on a module over a central simple algebra A_λ over k. This algebra A_λ, or more precisely its class in the Brauer group $\mathrm{Br}(k)$, is called the **Tits algebra** associated to λ and H; see Tits' foundational work [Tit71] or the treatment in [Knu+98, Chapter VI, Section 27]. The exponent of $[A_\lambda]$ in $\mathrm{Br}(k)$ divides the index of connection of the root system, which is the order of the quotient of the weight lattice by the root lattice. In the case of an absolutely simple group H_0 of type E_6 over a CM field K_0, the center of the simply connected cover $\widetilde{H}_0$ is isomorphic to μ_3. For the minuscule weight ω_1, the associated minuscule representation has dimension 27, and its Tits algebra class $[A_{\omega_1}]$ lies in $\mathrm{Br}(K_0)[3]$. The representation can be defined as an actual 27-dimensional vector space over K_0 if and only if the Brauer class $[A_{\omega_1}]$ is trivial.

To proceed with an embedding into E_7, the 27-dimensional minuscule representation V_{ω_1} of the simply connected group $\widetilde{H}_{0,K_0}$ must be defined over K_0. This is obstructed precisely by the Tits algebra $[A_{\omega_1}] \in \mathrm{Br}(K_0)[3]$.

Lemma 3.8.7 (Annihilation of the Tits algebra obstruction)**.** *There exists a finite, totally real extension F/F_0 such that the base-changed minuscule Tits algebra splits completely: $[A_{\omega_1} \otimes_{K_0} K_0F] = 0$ in the Brauer group $\mathrm{Br}(K_0F)$.*

Proof. For the absolutely simple algebraic group $\widetilde{H}_{0,K_0}$, the Galois-stable minuscule weight ω_1 (representing the 27-dimensional minuscule representation) defines a Brauer class $[A_{\omega_1}] \in \mathrm{Br}(K_0)[3]$ (see Remark 3.8.6). Let Σ be the finite set of finite places of K_0 at which the local invariant $\mathrm{inv}_w(A_{\omega_1}) \in \mathbf{Q}/\mathbf{Z}$ is non-zero. Since K_0 is a CM field, it has no real completions, and all archimedean completions are complex, where the local Brauer group is trivial. For each place v of F_0 lying beneath a place $w \in \Sigma$, we construct a local cubic extension of the local field $F_{0,v}$ by choosing an irreducible cubic polynomial over $F_{0,v}$. Using weak approximation, we find a global monic polynomial $g(x) \in F_0[x]$ of degree 3 that is sufficiently close to this local polynomial at each v, and close to a polynomial with three distinct real roots at all infinite places of F_0. The global extension $F^{(v)} = F_0[x]/(g(x))$ is thus a totally real cubic extension of F_0 in which each such place v is inert, i.e., the local degree $[F^{(v)}_v : F_{0,v}] = 3$. Let F be the compositum of the $F^{(v)}$ for all v below Σ. Then F/F_0 is a totally real extension. For any place u of the CM compositum K_0F lying over a place $w \in \Sigma$, the local extension degree satisfies

$$[(K_0F)_u : (K_0)_w] = [K_0F_v : K_0] \equiv 0 \pmod 3$$

since the local degree of the cubic field $F^{(v)}$ is 3, which is prime to the degree $[(K_0)_w : F_{0,v}] \leqslant 2$. By the local compatibility of the invariant map, the local invariant is given by

$$\mathrm{inv}_u(A_{\omega_1} \otimes_{K_0} K_0F) = [(K_0F)_u : (K_0)_w] \cdot \mathrm{inv}_w(A_{\omega_1}) \equiv 0 \pmod 1$$

since the order of $[A_{\omega_1}]$ divides 3. By the Albert-Brauer-Hasse-Noether theorem, a global Brauer class is zero if and only if all of its local invariants vanish. Thus, $[A_{\omega_1} \otimes_{K_0} K_0F] = 0$ in $\mathrm{Br}(K_0F)$, establishing the existence of the totally real extension F/F_0 as required for Theorem 3.8.1(2). $\square$

Remark 3.8.8 (Algebraic models for exceptional groups and Galois descent)**.** While root data classify reductive groups over algebraically closed fields, constructing explicit homomorphisms between forms of E_6 and E_7 over a general field F relies on the Galois descent of non-associative algebras.

The classical starting point is an **Albert algebra** J over a field E, namely a 27-dimensional exceptional simple Jordan algebra (modelled as the space of 3×3 Hermitian matrices over the octonions). By [Gar01b, Section 1 and Theorem 1.4], the reduced structure group of J is a simply connected algebraic group of inner type ${}^1E_6$, meaning that $\mathrm{Gal}(E_{\mathrm{sep}}/E)$ acts trivially on its Dynkin diagram. In contrast, our group $\widetilde{H}_{0,F}$ is of outer type ${}^2E_6$, with diagram involution given by the quadratic extension K_0F/F.

To realize outer forms of E_6 over F, one employs 56-dimensional structurable algebras known as **Brown algebras** [Gar01b, Section 2]. Given an Albert algebra defined over K_0F, Galois descent along the non-trivial involution of K_0F/F produces a Brown algebra over F. The automorphism group of this descended algebra is a simply connected group of outer type ${}^2E_6$.

Any 56-dimensional Brown algebra induces a 56-dimensional **Freudenthal triple system** over F on the same underlying vector space, defined by a skew-symmetric bilinear form and a trilinear product derived from the algebra multiplication. The automorphism group of a non-degenerate Freudenthal triple system is a simply connected group of type E_7. Because the triple system structure is defined using the Brown algebra operations, the automorphism group of the Brown algebra canonically embeds into the automorphism group of the Freudenthal triple system.

Over $\mathbf{R}$, this un-twisted Freudenthal triple system construction produces real forms of E_7 with trivial Tits algebra. In particular, it yields the real form $E_{7(-25)}$ (associated with the Hermitian symmetric domain E VII), without requiring the non-split quaternion algebra twists that arise in alternative constructions.

Construction 3.8.9 (The E_7 embedding over F)**.** By the choice of F in Lemma 3.8.7, the minuscule Tits algebra of H_0 splits over the compositum K_0F. Consequently, the 27-dimensional minuscule representation V_{ω_1} of the simply connected cover $\widetilde{H}_{0,K_0F}$ is defined over K_0F, where it carries the structure of a central simple Albert algebra J [Knu+98, Chapter IX].

Following [Gar01b, Example 2.4], we use the non-trivial Galois involution of K_0F/F to descend the structurable algebra $\mathcal{B}(J, K_0F \times K_0F)$ to a 56-dimensional Brown algebra $\mathcal{B} := \mathcal{B}(J, K_0F)$ over F; after base change to K_0F, this algebra decomposes as $K_0F \oplus J \oplus J \oplus K_0F$. By [Gar01b, Theorem 2.9], the connected automorphism group $\mathrm{Aut}^+(\mathcal{B})$ is a simply connected algebraic group of outer type E_6 associated with the extension K_0F/F. Because we descended the algebra via the exact Galois cocycle defining $H_{0,F}$, we obtain a canonical identification $\mathrm{Aut}^+(\mathcal{B}) \simeq \widetilde{H}_{0,F}$.

Let $\mathfrak{M}(\mathcal{B})$ be the 56-dimensional Freudenthal triple system over F associated with the Brown algebra $\mathcal{B}$ [Gar01b, Section 4]. By [Gar01b, Theorem 4.15], the automorphism group $\mathrm{Aut}(\mathfrak{M}(\mathcal{B}))$ is a simply connected group of type E_7; let G_F denote its adjoint quotient over F. Because G_F acts faithfully on the 56-dimensional vector space $\mathfrak{M}(\mathcal{B})$, the 2-torsion Brauer class associated with the minuscule representation is trivial in $\mathrm{Br}(F)$. In other words, the Tits algebra of G_F is trivial, distinguishing $\mathfrak{M}(\mathcal{B})$ from a generalized Freudenthal triple system defined over a non-split quaternion algebra [Gar01a, Section 3]. At any real place $\tau\colon F \hookrightarrow \mathbf{R}$, an F-form of E_7 with trivial Tits algebra is isomorphic to either the split form $E_{7(7)}$ or the form $E_{7(-25)}$. Since the subgroup $\mathrm{Aut}^+(\mathcal{B}_\tau) \simeq E_{6(-14)}$ is non-split over $\mathbf{R}$, the group $(G_F)_\tau$ cannot be split, and hence $(G_F)_\tau \simeq E_{7(-25)}$ for every τ.

Within G_F, the natural embedding of $\mathrm{Aut}^+(\mathcal{B}) \simeq \widetilde{H}_{0,F}$ extends to a canonical closed immersion of the reductive F-group

$$(L_0)_F = (\widetilde{H}_{0,F} \times \mathrm{Res}^1_{K_0F/F}\,\mathbf{G}_m)/Z(\widetilde{H}_{0,F}) \hookrightarrow G_F$$

as the centralizer of a 1-dimensional anisotropic torus (which over the real places τ yields the maximal pseudo-compact symmetric subgroup $E_{6(-14)} \cdot U(1)$ inside $E_{7(-25)}$). This completes the construction for part (2) of Theorem 3.8.1.

Lemma 3.8.10. *Applying Weil restriction along the totally real extension $F/\mathbf{Q}$, the group $\mathbf{G} := \mathrm{Res}_{F/\mathbf{Q}}\,G_F$ admits a Shimura datum $(\mathbf{G}, Y)$ with a totally real reflex field, and there is a closed embedding of Shimura data $(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y)$.*

Proof. The closed embedding of F-groups $(L_0)_F \hookrightarrow G_F$ induces a closed embedding of $\mathbf{Q}$-groups:

$$j\colon \mathbf{L} = \mathrm{Res}_{F_0/\mathbf{Q}}\,L_0 \to \mathrm{Res}_{F/\mathbf{Q}}(L_0)_F \hookrightarrow \mathrm{Res}_{F/\mathbf{Q}}\,G_F = \mathbf{G}.$$

Since $F/\mathbf{Q}$ is a totally real extension, we have an isomorphism of real Lie groups

$$\mathbf{G}_{\mathbf{R}} \simeq \prod_{\tau\colon F\hookrightarrow \mathbf{R}} (G_F)_\tau(\mathbf{R}).$$

By our construction, the real form of G_F at each embedding τ is precisely the non-compact exceptional form of type $E_{7(-25)}$, which is the isometry group of the exceptional Hermitian symmetric domain E VII $\simeq E_{7(-25)}/(E_6 \cdot U(1))$ of dimension 27 (see, for instance, [Hel01, Chapter X, §6]). The product of these domains defines a Hermitian symmetric space Y for $\mathbf{G}(\mathbf{R})$. At each real place, the image of $L_{0,v}(\mathbf{R})$ under the embedding corresponds to the subgroup of type $E_{6(-14)} \cdot U(1)$, which stabilizes the holomorphic embedding of the domain E III $\hookrightarrow$ E VII. Thus, j induces a closed embedding of Shimura data $(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y)$. Finally, by Deligne's description of reflex fields of adjoint Shimura data [Del71, Proposition 3.8], we conclude that $E(\mathbf{G}, Y)$ is totally real. This establishes part (3) of Theorem 3.8.1. □

4. Canonical subgroups

In this section, we prove the overconvergence of canonical subgroups integrally and produce lifts of Frobenii. A standing assumption is that μ is defined over $\mathbf{Z}_p$.

Setup 4.1. Maintain Setup 2.3.1. In what follows, $\mathcal{G}$ is *reductive*, μ is defined over $\mathbf{Z}_p$ so that $\varphi(\mu) = \mu$, and R is always a p-torsion free perfectoid ring unless otherwise stated. In particular, we say 'ordinary' instead of 'μ-ordinary'. We denote the tilt of p as $t \in R^\flat$, and the Frobenius on $R^\flat$ by σ. Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G}, \mu)$-window over R. Because μ is minuscule, the maximal parabolic $\mathcal{P}_\mu^+$ corresponds to a fundamental weight χ_{fun} of $\mathcal{G}$. We fix a Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$ of fundamental weight $\chi = \chi_{\mathrm{fun}}$, supplied by Remark 2.4.6. This defines a Hasse invariant on $\mathrm{BT}_1^{\mathcal{G},\mu} \otimes k$ by Theorem 2.3.5 which we also denote by Ha. For instance, $\nu_t(\mathrm{Ha}(\mathcal{E}_1))$ denotes the t-adic valuation of the Hasse invariant of the $\mathcal{G}$-zip corresponding to $(\mathcal{E}_1, m_1)$ as in Remark 2.3.13 (we set $\nu_t(0) = \infty$).

Remark 4.2 (Hasse invariants in the GL_n case)**.** Let us make this explicit in the case of $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$, where $\mu_d(x) = \mathrm{diag}(x, \dots, x, 1, \dots, 1)$ with d copies of x. The associated parabolic subgroup $\mathcal{P}_{\mu_d}^+$ consists of block upper-triangular matrices with block sizes d and $h-d$, and its Levi quotient is $\mathcal{M}_{\mu_d} \cong \mathrm{GL}_d \times \mathrm{GL}_{h-d}$. The fundamental weight χ_{fun} corresponding to the unique simple root of GL_h not in $\mathcal{M}_{\mu_d}$ is the character $\mathcal{M}_{\mu_d} \to \mathbf{G}_{\mathrm{m}}$ given by the determinant of the top-left GL_d-block. In the context of a p-divisible group of height h and dimension d over a characteristic p ring, the reduction of structure of its de Rham cohomology to $\mathcal{P}_{\mu_d}^+$ corresponds to the Hodge filtration, and the line bundle $\mathcal{V}(\chi_{\mathrm{fun}})$ evaluates to the top exterior power of the Lie algebra, $\det(\mathrm{Lie}\, G) = \bigwedge^d \mathrm{Lie}\, G$. Under this identification, the Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi_{\mathrm{fun}}))$ corresponds exactly to the map $\det(\mathrm{Lie}\, G^{(p)}) \to \det(\mathrm{Lie}\, X)$ induced by the Verschiebung homomorphism $V\colon G^{(p)} \to G$. Thus, the chosen Hasse invariant is precisely the classical Hasse invariant, and $\nu_p(\mathrm{Ha}(G))$ recovers what is often called the Hodge height.

4.1. Canonical subgroups for BT_1.

Recall that we have canonical reductions $T_{\mathrm{dR}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}_\mu^-$ and $T_{\mathrm{conj}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}_\mu^+$ as in Remark 2.3.13. Denote by $\mathcal{E}_1^- \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ the $\mathcal{P}_\mu^-$-torsor induced by $T_{\mathrm{dR}}(\mathcal{E}_1)$.

Notation 4.1.1. Let $\mathbf{Z}_p^{\mathrm{cycl}}$ be the ring of integers of the (p-adic completion of) maximal cyclotomic extension of $\mathbf{Q}_p$. Note that $\mathbf{Z}_p^{\mathrm{cycl}}$ contains elements of p-adic valuation $\frac{a}{(p-1)p^n}$ for any integers $a, n \geqslant 0$.

Remark 4.1.2. In what follows, $\mathbf{Z}_p^{\mathrm{cycl}}$ can be replaced by any (possibly finite) totally ramified extension of an unramified ring of integers as long as all the p-adic valuations make sense.

The goal of this section is to prove

Theorem 4.1.3 (Canonical subgroup for 1-truncated apertures). *Let $\mathcal{E}_1 \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{F}_p$ be a $\mathcal{G}$-torsor representing a* 1*-truncated* $(\mathcal{G}, \mu)$*-aperture* $\mathcal{A}\colon \mathfrak{X} \to \mathrm{BT}_1^{\mathcal{G},\mu}$ *for a flat* p*-adic formal scheme* $\mathfrak{X}$ *over* $\mathrm{Spf}\, \mathbf{Z}_p^{\mathrm{cycl}}$*. Let* $x_{\mathrm{dR}}\colon \mathfrak{X}_{\mathbf{F}_p} \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{F}_p$ *be the mod* p *version of the de Rham section from Recollection 2.1.12. Let* $(\mathcal{Q} := (x_{\mathrm{dR}})^*\mathcal{E}_1, \mathcal{Q}^+, \mathcal{Q}^-, \alpha) \in \mathcal{G}\text{-zip}_\mu(\mathfrak{X}_{\mathbf{F}_p})$ *be the* $\mathcal{G}$*-zip corresponding to* $\mathcal{A}|_{\mathfrak{X}_{\mathbf{F}_p}}$*. If* $\frac{p}{p+1} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{\mathfrak{X}_{\mathbf{F}_p}}))$*, then there exists a unique reduction* $\mathcal{E}_1^- \subset \mathcal{E}_1$ *to* $\mathcal{P}_\mu^+$ *such that* $(x_{\mathrm{dR}})^*\mathcal{E}_1^-$ *agrees with* $\mathcal{Q}^+$ *modulo* $p^{1-\varepsilon}$*.*

See Section 4.3 to see how the above relates to classical results on canonical subgroups of Barsotti–Tate groups. We will first prove it for perfectoid rings and then use a moduli argument combined with quasisyntomic descent to deduce the general case.

Remark 4.1.4 (Big cell in relative Grassmannian). Set $\mathrm{Gr}^+_{\mathcal{E}_1,\mu} := \mathcal{E}_1/\mathcal{P}_\mu^+$. Then there is a canonical open dense 'big cell'

$$\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat} \subset \mathrm{Gr}^+_{\mathcal{E}_1,\mu}|_{R^\flat/tR^\flat}$$

given by the image of the open embedding

$$\mathcal{E}_1^- \times^{\mathcal{P}_\mu^-} (\mathcal{P}_\mu^-\mathcal{P}_\mu^+/\mathcal{P}_\mu^+) \hookrightarrow \mathcal{E}_1^- \times^{\mathcal{P}_\mu^-} (\mathcal{G}/\mathcal{P}_\mu^+) \xrightarrow{\sim} \mathrm{Gr}^+_{\mathcal{E}_1,\mu}|_{R^\flat/tR^\flat}.$$

This open subscheme $\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat}$ is canonically a torsor for the unipotent group scheme $\mathcal{E}_1^- \times^{\mathcal{P}_\mu^-} \mathcal{U}_\mu^-$. Observe that $(\mathcal{E}_1, m_1)$ is ordinary if and only if $T_{\mathrm{conj}}(\mathcal{E}_1)$ factors through $\mathcal{C}_{\mathrm{big}}$.

Definition 4.1.5 (m_1-stable reduction of structure). A reduction of structure $\mathcal{E}_1^- \subset \mathcal{E}_1$ to a smooth subgroup scheme $\mathcal{H} \subset \mathcal{G}$ is said to be m_1**-stable** if the restriction of the modification m_1 to $\varphi^*\mathcal{E}_1^-[1/t]$ factors through $\mathcal{E}_1^-[1/t]$, thereby inducing an isomorphism $m_1^-\colon \varphi^*\mathcal{E}_1^-[1/t] \xrightarrow{\sim} \mathcal{E}_1^-[1/t]$ of $\mathcal{H}$-torsors over $R^\flat[1/t]$.

Proposition 4.1.6. $(\mathcal{E}_1, m_1)$ *is ordinary if and only if there exists a* m_1*-stable reduction* $\mathcal{E}_1^- \subset \mathcal{E}_1$ *to* $\mathcal{P}_\mu^+$ *lifting* $T_{\mathrm{conj}}(\mathcal{E}_1)$*.*

Proof. If $(\mathcal{E}_1, m_1)$ is ordinary then the desired result is clear because such a window reduces structure to a 1-truncated $(\mathcal{P}_\mu^+, \mu)$-window. Conversely, suppose $\mathcal{E}_1^- \subset \mathcal{E}_1$ is a m_1-stable reduction of structure as in the proposition inducing a modification $m_1^-\colon \varphi^*\mathcal{E}_1^-[1/t] \xrightarrow{\sim} \mathcal{E}_1^-[1/t]$. By assumption, the reduction of structure as in Remark 2.2.10 of $\mathcal{E}_1^-|_{R^\flat/tR^\flat}$ induced by m_1^- agrees with $T_{\mathrm{conj}}(\mathcal{E}_1)$. This implies that m_1^- has type μ as a modification of $\mathcal{P}_\mu^+$-torsors which is to say that $(\mathcal{E}_1^-, m_1^-)$ is a 1-truncated $(\mathcal{P}_\mu^+, \mu)$-window. □

We now prove the key overconvergence result:

Proposition 4.1.7 (Overconvergence for 1-truncated windows). *If* $\frac{p}{p+1} > \varepsilon \geqslant \nu_t(\mathrm{Ha}\,\mathcal{E}_1)$*, then there exists a unique* m_1*-stable reduction* $\mathcal{E}_1^+ \subset \mathcal{E}_1$ *to* $\mathcal{P}_\mu^+$ *such that* $\mathcal{E}_1^+$ *agrees with* $T_{\mathrm{conj}}(\mathcal{E}_1)$ *modulo* $t^{1-\varepsilon}$*.*

Proof. The modification m_1 induces an isomorphism

$$\mathrm{Gr}^+_{\varphi^*\mathcal{E}_1,\mu}[1/t] \to \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t]$$

which when composed with the relative Frobenius yields an endomorphism

$$\Psi\colon \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t] \to \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t].$$

Choose an $R^\flat$-point of $\mathcal{G}\text{-zip}_\mu$ lifting the canonical map $\mathrm{Spec}\, R^\flat/tR^\flat \to \mathcal{G}\text{-zip}_\mu$. This is possible, for e.g., by smoothness of $\mathcal{G}\text{-zip}_\mu$. This gives an $R^\flat$-point y_0 of $\mathrm{Gr}^+_{\mathcal{E}_1,\mu}$ lifting $T_{\mathrm{conj}}(\mathcal{E}_1)$ and also allows us to define an open dense 'big cell' $\mathcal{C}_{\mathrm{big}} \subset \mathrm{Gr}^+_{\mathcal{E}_1,\mu}$ as in Remark 4.1.4 lifting $\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat}$. Since $\mathrm{Ha}(\mathcal{E}_1)$ has positive t-adic valuation it becomes invertible when t is inverted which implies that $y_0[1/t] \in \mathcal{C}_{\mathrm{big}}[1/t]$. We wish to show that there exists $x \in \mathrm{Gr}^+_{\mathcal{E}_1,\mu}(R^\flat[1/t])$ such that $\Psi(x) = x$ and $x \equiv y_0 \pmod{t^{1-\varepsilon}}$.

After p-completely étale localization on R, we may assume that $\mathcal{E}_1$ is a trivial torsor. Let the modification m_1 be represented by $\mu(t)M \in \mathcal{G}(R^\flat[1/t])$ where $M \in \mathcal{G}(R^\flat)$. By Remarks 2.3.14 and 2.4.6, $M \in \mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$ since the Hasse invariant has positive t-adic valuation. Denote by $\mathfrak{u}_\mu^-$ the Lie algebra of the unipotent group $\mathcal{U}_\mu^-$.

By centering coordinates of the big cell $\mathcal{C}_{\mathrm{big}}$ around y_0, we can identify $\mathcal{C}_{\mathrm{big}}(R^\flat[1/t])$ with $\mathfrak{u}_\mu^-(R^\flat[1/t])$ so that $y_0[1/t]$ corresponds to 0. The fixed point condition is equivalent to the containment

$$\exp(Z)^{-1}\mu(t)M\exp(\sigma(Z)) \in \mathcal{P}_\mu^+$$

where $\exp\colon \mathfrak{u}_\mu^- \xrightarrow{\sim} \mathcal{U}_\mu^-$ is the exponential map isomorphism; we denote the inverse of exp by log. Since $\mu(t)\exp(Z)\mu(t)^{-1} = \exp(t^{-1}Z)$ and $\mu(t) \in \mathcal{P}_\mu^+(R^\flat[1/t])$, the above is equivalent to verifying

$$M\exp(\sigma(Z)) \in \exp(t^{-1}Z)\mathcal{P}_\mu^+$$

Using the projection $\pi_-\colon \mathcal{P}_\mu^-\mathcal{P}_\mu^+ \twoheadrightarrow \mathcal{P}_\mu^-\mathcal{P}_\mu^+/\mathcal{P}_\mu^+ \xrightarrow{\sim} \mathcal{U}_\mu^-$, we get the operator

$$\mathbf{F}(Z) = t\log\pi_-(M\exp(\sigma(Z)))$$

which gives the fixed point equation $Z = \mathbf{F}(Z)$. A priori, it is not clear that this operator is well-defined since we don't know if $M\exp(\sigma(Z)) \in \mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$, but we will show this is the case in a t-adic ball of radius $1-\varepsilon$.

Denote by $\Delta \in \mathcal{O}(\mathcal{G}_{R^\flat})$ the regular function descending to the Hasse invariant Ha that cuts out the big cell $\mathcal{P}_\mu^-\mathcal{P}_\mu^+$. The assumption is $\nu_t(\Delta(M)) \leqslant \varepsilon$. We solve the fixed point equation iteratively in the t-adic ball $\mathcal{B}_{1-\varepsilon} := \{Z \in \mathfrak{u}_\mu^-(R^\flat[1/t]) \mid \nu_t(Z) \geqslant 1-\varepsilon\}$. The map $X \mapsto \Delta(M\exp(X))$ is a polynomial function on the Lie algebra $\mathfrak{u}_\mu^-(R^\flat)$. For $Z \in \mathcal{B}_{1-\varepsilon}$, the Frobenius multiplies the valuation by p, so $\nu_t(\sigma(Z)) \geqslant p(1-\varepsilon)$. Evaluating this polynomial yields:

$$\Delta(M\exp(\sigma(Z))) = \Delta(M) + P(\sigma(Z))$$

where P is a polynomial without a constant term, whose coefficients lie in $R^\flat$ because $M \in \mathcal{G}(R^\flat)$. We obtain the valuation bound:

$$\nu_t(\Delta(M\exp(\sigma(Z))) - \Delta(M)) \geqslant \nu_t(\sigma(Z)) \geqslant p(1-\varepsilon).$$

Since $\varepsilon < \frac{p}{p+1} < 1$, we have $p(1-\varepsilon) > \varepsilon \geqslant \nu_t(\Delta(M))$, and thus:

$$\nu_t(\Delta(M\exp(\sigma(Z)))) = \min(\nu_t(\Delta(M)), \nu_t(\Delta(M\exp(\sigma(Z))) - \Delta(M))) = \nu_t(\Delta(M))$$

and hence log is well defined and $\mathbf{F}$ is a well defined operator on $\mathcal{B}_{1-\varepsilon}$.

To find an explicit formula for $\mathbf{F}(Z) = t\log\pi_-(M\exp(\sigma Z))$, consider the fundamental representation V of $\mathcal{G}$ corresponding to the maximal parabolic $\mathcal{P}_\mu^+$. Let $v_0 \in V$ be its highest weight vector, so $\mathcal{P}_\mu^+$ is exactly the stabilizer of the line spanned by v_0. Because our chosen Hasse invariant has minimal weight $\chi = \chi_{\mathrm{fun}}$, the regular function Δ is exactly the principal minor Δ_{fun} up to a unit. The representation V is naturally graded by weight spaces relative to the Levi: $V = V_0 \oplus V_1 \oplus \cdots$, where V_0 is the line spanned by v_0, $V_1 = \mathfrak{u}_\mu^- v_0$, and elements of $\mathfrak{u}_\mu^-$ map $V_i \to V_{i+1}$. Let $\Pi\colon V \to V_0 \oplus V_1$ be the canonical projection. Since $\log U_y \in \mathfrak{u}_\mu^-$ shifts degrees by $+1$, projecting the exponential truncates higher-order terms, so $\Pi(U_y v_0) = v_0 + \log U_y \cdot v_0$. For any matrix $y \in \mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$ with Birkhoff factorization $y = U_y P_y$, the action on v_0 yields $yv_0 = \Delta(y)U_y v_0$. Projecting this gives $\Pi(yv_0) = \Delta(y)(v_0 + \log U_y \cdot v_0)$. Applying this to M, we have $\Pi(Mv_0) = \Delta(M)(v_0 + \log U \cdot v_0)$, where $\log U = \log\pi_-(M)$. Because M is integral, Mv_0 is an integral vector, meaning $\Delta(M)\log U$ is integral. Thus, the assumption $\nu_t(\Delta(M)) \leqslant \varepsilon$ guarantees the t-adic poles of $\log U$ are bounded by precisely $\nu_t(\Delta(M)) \leqslant \varepsilon$.

Now apply this to $y := M\exp(\sigma Z)$. We have $yv_0 = M(v_0 + \sigma Z v_0) = Mv_0 + M\sigma Z v_0$. Applying Π, we obtain $\Pi(yv_0) = \Pi(Mv_0) + \Pi(M\sigma Z v_0)$. Since $\sigma Z \in \mathfrak{u}_\mu^-$, the vector $\sigma Z v_0$ lies in V_1. Because M is an integral matrix, $M\sigma Z v_0$ is an integral vector in V, and its projection $\Pi(M\sigma Z v_0) \in V_0 \oplus V_1$ can be uniquely written as $a(\sigma Z)v_0 + X(\sigma Z)v_0$ for some scalar $a(\sigma Z)$ and some Lie algebra element $X(\sigma Z) \in \mathfrak{u}_\mu^-$. Because M is integral, a and X are integral linear maps. Equating the expressions for $\Pi(yv_0)$, we have:

$$\Delta(y)(v_0 + \log U_y \cdot v_0) = \Delta(M)(v_0 + \log U \cdot v_0) + a(\sigma Z)v_0 + X(\sigma Z)v_0$$

Matching the V_0 components (the coefficients of v_0) yields $\Delta(y) = \Delta(M) + a(\sigma Z)$. This confirms the polynomial expansion from earlier. Matching the V_1 components (identifying V_1 with $\mathfrak{u}_\mu^-$), we get $\Delta(y)\log U_y = \Delta(M)\log U + X(\sigma Z)$. Dividing by $\Delta(y)$ gives:

$$\log U_y = \frac{\Delta(M)\log U + X(\sigma Z)}{\Delta(M) + a(\sigma Z)} = \log U + \frac{X(\sigma Z) - a(\sigma Z)\log U}{\Delta(M) + a(\sigma Z)}$$

Define $B(\sigma Z) := X(\sigma Z) - a(\sigma Z)\log U$. Since $\nu_t(\log U) \geqslant -\nu_t(\Delta(M))$, the linear map $B\colon \mathfrak{u}_\mu^- \to \mathfrak{u}_\mu^-$ introduces t-adic poles of order at most $\nu_t(\Delta(M)) \leqslant \varepsilon$. We obtain the exact formula:

$$\mathbf{F}(Z) = t\log U_y = t\log U + t\frac{B(\sigma Z)}{\Delta(M) + a(\sigma Z)}$$

The initial point of the iteration is $Z_1 = \mathbf{F}(0) = t\log U$. As shown earlier, its poles are bounded by $\nu_t(\Delta(M)) \leqslant \varepsilon$, so $\nu_t(Z_1) \geqslant 1 - \nu_t(\Delta(M)) \geqslant 1 - \varepsilon$, proving $\mathbf{F}(0) \in \mathcal{B}_{1-\varepsilon}$.

For $Z_1, Z_2 \in \mathcal{B}_{1-\varepsilon}$, let $\Delta_i := \Delta(M) + a(\sigma Z_i)$ for $i = 1, 2$. The difference is:

$$\mathbf{F}(Z_1) - \mathbf{F}(Z_2) = t\left(\frac{B(\sigma Z_1)}{\Delta_1} - \frac{B(\sigma Z_2)}{\Delta_2}\right) = t\frac{\Delta(M)B(\sigma(Z_1 - Z_2)) + B(\sigma Z_1)a(\sigma Z_2) - B(\sigma Z_2)a(\sigma Z_1)}{\Delta_1\Delta_2}$$

Recall that $\nu_t(\Delta_i) = \nu_t(\Delta(M))$, so the denominator has valuation $2\nu_t(\Delta(M)) \leqslant 2\varepsilon$. In the numerator, B has poles bounded by $\nu_t(\Delta(M))$, meaning $\nu_t(B(W)) \geqslant \nu_t(W) - \nu_t(\Delta(M))$. Thus $\nu_t(\Delta(M)B(\sigma(Z_1 - Z_2))) \geqslant \nu_t(\Delta(M)) + p\nu_t(Z_1 - Z_2) - \nu_t(\Delta(M)) = p\nu_t(Z_1 - Z_2)$. The cross terms can be rewritten as $B(\sigma Z_1)a(\sigma(Z_2 - Z_1)) + a(\sigma Z_1)B(\sigma(Z_1 - Z_2))$. Since a is integral, $\nu_t(a(W)) \geqslant \nu_t(W)$. Because $Z_i \in \mathcal{B}_{1-\varepsilon}$, we have $\nu_t(\sigma Z_i) \geqslant p(1-\varepsilon)$. This implies $\nu_t(B(\sigma Z_1)) \geqslant p(1-\varepsilon) - \nu_t(\Delta(M))$, and $\nu_t(a(\sigma(Z_2 - Z_1))) \geqslant p\nu_t(Z_1 - Z_2)$, making their product's valuation at least $p(1-\varepsilon) - \nu_t(\Delta(M)) + p\nu_t(Z_1 - Z_2)$. Similarly, the second term has valuation $\geqslant p(1-\varepsilon) + p\nu_t(Z_1 - Z_2) - \nu_t(\Delta(M))$. Because $\varepsilon < \frac{p}{p+1}$, we have $p(1-\varepsilon) - \nu_t(\Delta(M)) \geqslant p(1-\varepsilon) - \varepsilon > 0$. Thus, all terms in the numerator have valuation at least $p\nu_t(Z_1 - Z_2)$, meaning the fraction introduces poles of order at most $2\nu_t(\Delta(M)) \leqslant 2\varepsilon$. We obtain

$$\nu_t(\mathbf{F}(Z_1) - \mathbf{F}(Z_2)) \geqslant 1 - 2\nu_t(\Delta(M)) + p\nu_t(Z_1 - Z_2) \geqslant 1 - 2\varepsilon + p\nu_t(Z_1 - Z_2).$$

For $\mathbf{F}$ to be a strict contraction on $\mathcal{B}_{1-\varepsilon}$, we require this distance to strictly exceed $\nu_t(Z_1 - Z_2)$, which implies $(p-1)\nu_t(Z_1 - Z_2) > 2\varepsilon - 1$. Since $\nu_t(Z_1 - Z_2) \geqslant 1 - \varepsilon$, it suffices to verify $(p-1)(1-\varepsilon) > 2\varepsilon - 1$, meaning $\varepsilon < \frac{p}{p+1}$. Furthermore, to map the ball into itself, the initial point must lie within it. Since $\mathbf{F}(0) \in \mathcal{B}_{1-\varepsilon}$ and the distance to $\mathbf{F}(0)$ is strictly less than the radius, $\mathbf{F}$ maps the ball $\mathcal{B}_{1-\varepsilon}$ to itself and is a strict contraction on it. To conclude, $\mathbf{F}$ has a *unique* fixed point in the t-adic ball $\mathcal{B}_{1-\varepsilon}$, which implies that the reduction agrees with $T_{\mathrm{conj}}(\mathcal{E}_1)$ modulo $t^{1-\varepsilon}$. □

Remark 4.1.8 (Sharpness of the bound)**.** The bound $t^{1-\varepsilon}$ is sharp in the generic case where $\nu_t(\Delta(M)) = \varepsilon$. In the proof, the canonical subgroup corresponds to the limit Z of the sequence Z_n obtained by iterating the operator $\mathbf{F}$, starting at $Z_0 = 0$. The first step of the iteration is $Z_1 = \mathbf{F}(0) = t\log U$. As shown, $\Delta(M)\log U$ is an integral matrix, and $\nu_t(\Delta(M)) = \varepsilon$, so the t-adic poles of $\log U$ are bounded by ε. However, for a generic matrix M on the locus where $\nu_t(\Delta(M)) = \varepsilon$, there is no reason for $\Delta(M)\log U$ to be divisible by any positive powers of t. Thus, generically, the exact valuation is $\nu_t(Z_1) = 1 - \varepsilon$. Because $\mathbf{F}$ is a strict contraction, the distance between subsequent iterates is strictly greater than the initial step size. Indeed, using $\nu_t(Z_2 - Z_1) \geqslant 1 - 2\varepsilon + p(1-\varepsilon)$ and the assumption $\varepsilon < \frac{p}{p+1}$, a quick check shows $\nu_t(Z_2 - Z_1) > 1 - \varepsilon$. By the non-Archimedean triangle inequality, terms of strictly higher valuation do not alter the exact valuation of the sum $Z = Z_1 + (Z_2 - Z_1) + \cdots$. Therefore, generically, $\nu_t(Z) = \nu_t(Z_1) = 1 - \varepsilon$. This implies that the fixed point lies at a distance of exactly $1 - \varepsilon$ from 0 (which corresponds to $T_{\mathrm{conj}}(\mathcal{E}_1)$).

Remark 4.1.9 (Dependence on the choice of Hasse invariant)**.** If one instead works with a Hasse invariant of a different weight χ, the overconvergence bound becomes strictly worse. Because μ is minuscule, the associated parabolic $\mathcal{P}_\mu^+$ is maximal, meaning the Picard group of the flag variety $\mathcal{G}/\mathcal{P}_\mu^+$ has rank 1 and its ample cone is generated by the line bundle associated to χ_{fun}. Consequently, any ample character χ on the Levi subgroup $\mathcal{M}_\mu$ must restrict to a strictly positive integer multiple of χ_{fun} on the derived group. Thus, for χ to be ample, it must be of the form $k\chi_{\mathrm{fun}} + \eta$ for some integer $k \geqslant 1$ and a character η of the whole group $\mathcal{G}$. The generalized minor Δ cutting out the big cell would then be exactly Δ_{fun}^k (up to a unit), so its valuation would be $\varepsilon' := k\nu_t(\Delta_{\mathrm{fun}}(M))$. In this case, the bound on the poles of the iteration operator is still determined by $\Delta_{\mathrm{fun}}(M)^2$, which translates to $\frac{2}{k}\varepsilon'$. The strict contraction inequality would then require $(p-1)(1-\varepsilon') > \frac{2}{k}\varepsilon' - 1$, leading to the condition $\varepsilon' < \frac{kp}{kp+2-k}$. However, translating this bound back to the valuation of the fundamental Hasse invariant $\varepsilon = \varepsilon'/k$, the condition restricts the locus to $\varepsilon < \frac{p}{kp+2-k}$. Since the function $k \mapsto \frac{p}{kp+2-k}$ is strictly decreasing for $p \geqslant 2$, choosing $k = 1$ (the fundamental weight) yields $\frac{p}{p+1}$, which provides the largest and therefore optimal radius of overconvergence.

Construction 4.1.10 (ε-neighbourhood of ordinary locus)**.** Fix an element $p^\varepsilon \in \mathbf{Z}_p^{\mathrm{cycl}}$ with p-adic valuation $0 \leqslant \varepsilon < 1$. We define the p-adic formal stack $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ over $\operatorname{Spf}\mathbf{Z}_p^{\mathrm{cycl}}$. For any p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra S, the groupoid $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}(S)$ consists of tuples $(f, g, \alpha, \widetilde{\mathrm{Ha}}, s)$ where:

- $f \in \mathrm{BT}_1^{\mathcal{G},\mu}(S)$ is a 1-truncated $(\mathcal{G},\mu)$-aperture over S, yielding a map $f_{\mathrm{zip}} \colon \operatorname{Spec} S/pS \to \mathcal{G}\text{-zip}_\mu$,
- $g \colon \operatorname{Spf} S' \to \operatorname{Spf} S$ is a p-completely étale cover,
- $\alpha \colon \mathcal{O}_{S'/pS'} \xrightarrow{\sim} (g \otimes \mathbf{F}_p)^*(f_{\mathrm{zip}})^*\mathcal{V}(\chi)$ is a line bundle trivialization over $\operatorname{Spec} S'/pS'$,
- $\widetilde{\mathrm{Ha}} \in S'$ is a lift of $\alpha^{-1}(g \otimes \mathbf{F}_p)^* \mathrm{Ha} \in S'/pS'$ to characteristic zero,
- $s \in S'$ is an element witnessing $s\widetilde{\mathrm{Ha}} = p^\varepsilon$.

A morphism $(f_1, g_1, \alpha_1, \widetilde{\mathrm{Ha}}_1, s_1) \to (f_2, g_2, \alpha_2, \widetilde{\mathrm{Ha}}_2, s_2)$ is the data of an equivalence class of tuples $(\gamma, g_{12}, h_1, h_2)$ where:

- $\gamma \colon f_1 \xrightarrow{\sim} f_2$ is an isomorphism in $\mathrm{BT}_1^{\mathcal{G},\mu}(S)$,
- $g_{12} \colon \operatorname{Spf} S'' \to \operatorname{Spf} S$ is a p-completely étale cover,
- $h_i \colon \operatorname{Spf} S'' \to \operatorname{Spf} S'_i$ are p-completely étale morphisms factoring the covers (so $g_1 \circ h_1 = g_{12} = g_2 \circ h_2$),

such that the local witnesses are strictly compatible under pullback. Explicitly, γ induces an isomorphism of line bundles $\gamma^*_{\mathrm{zip}} \colon (f_{2,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{\sim} (f_{1,\mathrm{zip}})^*\mathcal{V}(\chi)$ over $\operatorname{Spec} S/pS$. Pulling back to $\operatorname{Spec} S''/pS''$, the trivializations are related by a unit $u \in (S''/pS'')^\times$ uniquely determined by the composition

$$u \colon \mathcal{O}_{S''/pS''} \xrightarrow{(h_2\otimes\mathbf{F}_p)^*\alpha_2} (g_{12}\otimes\mathbf{F}_p)^*(f_{2,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{(g_{12}\otimes\mathbf{F}_p)^*\gamma^*_{\mathrm{zip}}} (g_{12}\otimes\mathbf{F}_p)^*(f_{1,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{((h_1\otimes\mathbf{F}_p)^*\alpha_1)^{-1}} \mathcal{O}_{S''/pS''}.$$

If $\widetilde{u} \in (S'')^\times$ is any lift of u to characteristic zero, the pulled-back local lifts of the Hasse invariant satisfy $h_2^*\widetilde{\mathrm{Ha}}_2 \equiv \widetilde{u}h_1^*\widetilde{\mathrm{Ha}}_1 \pmod p$, meaning there exists some $\delta \in S''$ such that $h_2^*\widetilde{\mathrm{Ha}}_2 = \widetilde{u}h_1^*\widetilde{\mathrm{Ha}}_1 + p\delta$. The compatibility of witnesses requires the strict equality

$$h_2^*s_2 = h_1^*s_1\widetilde{u}^{-1}(1 + h_1^*s_1\widetilde{u}^{-1}p^{1-\varepsilon}\delta)^{-1}$$

in S''. Because $1 - \varepsilon > 0$ and S'' is p-complete, this geometric series converges perfectly. Two such tuples $(\gamma, g_{12}, h_1, h_2)$ and $(\gamma', g'_{12}, h'_1, h'_2)$ are equivalent if $\gamma = \gamma'$ and there exists a further common p-completely étale refinement dominating both g_{12} and g'_{12} where the connecting maps commute.

Proposition 4.1.11. *There exists a smooth atlas* $\operatorname{Spf} A_0 \to \mathrm{BT}_1^{\mathcal{G},\mu}$ *and a nonzerodivisor* $f \in A_0$ *such that for* $A := A_0 \otimes_{\mathbf{Z}_p} \mathbf{Z}_p^{\mathrm{cycl}}$*, there is a Cartesian diagram*

$$\begin{tikzcd}
\operatorname{Spf} A\langle s\rangle/(sf - p^\varepsilon) \arrow[r] \arrow[d] & \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon} \arrow[d] \\
\operatorname{Spf} A \arrow[r] & \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}
\end{tikzcd}$$

In particular, $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ *is a regular p-adic formal Artin stack over* $\operatorname{Spf}\mathbf{Z}_p^{\mathrm{cycl}}$ *with affine diagonal.*

Proof. We show that the natural forgetful morphism $\pi \colon \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon} \to \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ is representable by affine formal schemes. Given a morphism from a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R to $\mathrm{BT}_1^{\mathcal{G},\mu}$, we must show that the fiber product $\mathcal{X} := \operatorname{Spf} R \times_{\mathrm{BT}_1^{\mathcal{G},\mu}} \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ is an affine formal scheme. Because relative affine formal schemes satisfy effective descent in the p-completely étale topology, we may pass to a p-completely étale cover of $\operatorname{Spf} R$ to assume the pullback of $\mathcal{V}(\chi)$ is trivialized, identifying the Hasse invariant with an element $\mathrm{Ha}_R \in R/pR$. Then $\mathcal{X}$ is represented by the p-adic completion of $\operatorname{Spec} R[s]/(s\widetilde{\mathrm{Ha}}_R - p^\varepsilon)$, where $\widetilde{\mathrm{Ha}}_R \in R$ is any lift of Ha_R.

By Theorem 2.2.3, $\mathrm{BT}_1^{\mathcal{G},\mu}$ is smooth over $\mathbf{Z}_p$, so we can choose a $\mathbf{Z}_p$-smooth affine atlas $\operatorname{Spf} A_0 \to \mathrm{BT}_1^{\mathcal{G},\mu}$ over which the pullback of $\mathcal{V}(\chi)$ is trivialized. We can choose a lift $f \in A_0$ of the Hasse invariant Ha_{A_0}. Because the ordinary locus is dense, Ha_{A_0} is a nonzerodivisor in A_0/pA_0. This implies that f is a nonzerodivisor in A_0. Let $A := A_0 \otimes_{\mathbf{Z}_p} \mathbf{Z}_p^{\mathrm{cycl}}$. By the representability shown above, the fiber product of $\operatorname{Spf} A \to \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ along π is exactly $\operatorname{Spf} A\langle s\rangle/(sf - p^{\varepsilon})$, which gives the desired Cartesian diagram. Finally, Ha_{A_0} being a nonzerodivisor in A_0/pA_0 implies that $sf - p^{\varepsilon}$ and p form a regular sequence in $A_0[s]$. Since A_0 is a regular ring (being smooth over $\mathbf{Z}_p$), this regular sequence ensures that the quotient $A_0\langle s\rangle/(sf - p^{\varepsilon})$ is regular, which in turn implies the asserted regularity of the stack. □

Remark 4.1.12 (Relation to admissible formal blowups)**.** The construction of $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ can be viewed as taking an admissible formal blowup in the sense of Raynaud (specifically blowing up the ideal generated by (f, p^{ε})), but extended from formal schemes to formal stacks. The condition $sf - p^{\varepsilon} = 0$ in the local charts imposes the locus where the valuation of the Hasse invariant is bounded by ε.

Remark 4.1.13 (Descent to smaller base rings)**.** Although we defined the formal stack $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ over $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$, it naturally descends to a formal stack over any p-adic ring of integers where the valuation ε makes sense (i.e., any p-adic ring of integers containing an element of p-adic valuation exactly ε).

The following result can be thought of as an analogue of the classical (and easy) fact that if G is an n-truncated Barsotti-Tate group over a p-torsion free $\mathbf{Z}_p$-algebra R then a finite flat subgroup scheme $H \subset G$ is determined by $H[1/p]$ as a subgroup of $G[1/p]$.

Lemma 4.1.14. *Let $\mathfrak{X}$ be a p-quasisyntomic flat p-adic formal scheme and $\mathcal{P} \subset \mathcal{G}$ a parabolic subgroup scheme. Let $\mathcal{E} \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ be a $\mathcal{G}$-torsor and $T_{\text{ét}}(\mathcal{E})$ its étale realization, a pro-étale $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$-local system on $\mathfrak{X}^{\mathrm{ad}}$. The canonical functor from the groupoid of all reduction of structures of $\mathcal{E}$ to $\mathcal{P}$ to the groupoid of all reduction of structures of $T_{\text{ét}}(\mathcal{E})$ to $\mathcal{P}(\mathbf{Z}/p^n\mathbf{Z})$ is fully faithful.*

Proof. By quasisyntomic descent, we may assume $\mathfrak{X} = \operatorname{Spf} R$ for a p-torsion free qrsp ring R. Its absolute prismatic cohomology is a p-torsion free δ-ring Δ_R with a generalized Cartier divisor (d) such that (p, d) is a regular sequence. By Recollection 2.1.18, evaluating mod-p^n F-gauges with $\mathcal{G}$-structure on prisms is a fully faithful functor and gives a $\mathcal{G}$-bundle in mod-p^n prismatic F-crystals over $\mathfrak{X}$ so that $\mathcal{E}$ corresponds to a $\mathcal{G}$-torsor $\mathcal{E}_{\Delta}$ over Δ_R/p^n equipped a semilinear modification along $d \neq 0$. By definition, the étale realization functor factors through $\mathcal{E}_{\Delta}$. A reduction of structure to $\mathcal{P}$ is a section of the relative Grassmannian $\mathcal{E}_{\Delta}/\mathcal{P} \to \operatorname{Spec} \Delta_R/p^n$. The locus where any two sections agree is closed, defined by an ideal $J \subset \Delta_R/p^n$. Since the étale realization $T_{\text{ét}}(\mathcal{E})$ corresponds to the φ-invariants over $\Delta_R[1/d]/p^n$, sections agreeing on $T_{\text{ét}}(\mathcal{E})$ must agree after inverting d. Thus, a power of d annihilates J. As (p, d) is a regular sequence, d is a non-zero-divisor in Δ_R/p^n, forcing $J = 0$. This proves the desired result. □

Proof of Theorem 4.1.3. If the base ring is perfectoid, existence and uniqueness follow directly from Propositions 2.2.15 and 4.1.7. For a general p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, the condition on the Hasse invariant defines a map to the stack $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$. Because reduction of structure satisfies effective descent in the quasisyntomic topology, it suffices to construct a unique such reduction in the universal case.

By the density of the value group of $\mathbf{Z}_p^{\mathrm{cycl}}$, we may approximate ε from above and assume without loss of generality that there is an element $\pi \in \mathbf{Z}_p^{\mathrm{cycl}}$ with p-adic valuation exactly ε. This element π generates a finite, totally ramified extension $K = \mathbf{Q}_p(\pi)$.

By Proposition 4.1.11, we can choose an affine atlas $\operatorname{Spf} A$ for $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ given by

$$A = A_0\langle s\rangle/(s\widetilde{\mathrm{Ha}}_A - \pi),$$

where A_0 is a p-completely smooth $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, and $\widetilde{\mathrm{Ha}}_A \in A_0$ is a characteristic zero lift of the Hasse invariant. In particular, $(p, \widetilde{\mathrm{Ha}}_A)$ is a regular sequence. The Čech nerve of $\operatorname{Spf} A \to \mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ is the base change of that of $\operatorname{Spf} A_0 \to \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ along $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu} \to \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. This implies that the Čech nerve of $\operatorname{Spf} A \to \mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ is consists of terms of the form $A_0^{(n)}\langle s\rangle/(s\,\mathrm{pr}_1^*\,\widetilde{\mathrm{Ha}}_A - \pi)$ where $A_0^{(n)}$ are Čech nerve terms of the latter, and in particular p-completely smooth. Since smooth morphisms are in particular quasisyntomic, it therefore suffices to solve the problem for rings of the form $B := B'\langle t\rangle/(tf - \pi)$ where $B' = B_0 \otimes_{\mathbf{Z}_p} \mathcal{O}_K$ with B_0 a p-completely smooth ring and $f \in B_0$ maps to a nonzerodivisor in B_0/pB_0. Note that even though B is not a $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, the desired statement still makes sense.

By choosing a monic Eisenstein polynomial $E(u) \in \mathbf{Z}_p[u]$ such that $E(\pi) = 0$, we can write $B \simeq B_0\langle t\rangle/(E(tf))$. By p-complete étale localization, we may choose a smooth framing $B_0 = \mathbf{Z}_p\langle T_1^{\pm1}, \dots, T_n^{\pm1}\rangle$. This lifts to a δ-ring $P_0 = \mathbf{Z}_p\langle T_1^{\pm1}, \dots, T_n^{\pm1}\rangle$ determined by $\delta(T_i) = 0$. Let $\widetilde{f} \in P_0$ be any lift of f, and extend P_0 to a δ-ring $P_1 = P_0\langle t\rangle$ by setting $\delta(t) = 0$. Since $E(u)$ is an Eisenstein polynomial, $d := E(t\widetilde{f})$ is a distinguished element of P_1, and its (p,d)-adic completion $\underline{P} = (P := (P_1)^\wedge_{(p,d)}, (d))$ forms a bounded prism with quotient $P/(d) \simeq B$. Let B_∞ be the Hodge–Tate quotient of the perfection of $\underline{P}$. Then B_∞ is a perfectoid ring and the natural map $B \to B_\infty$ is a quasisyntomic cover as it is a p-completed base-change of

$$\pi_\infty \colon \mathbf{Z}_p\langle T_1^{\pm1}, \dots, T_n^{\pm1}, t\rangle \to \mathbf{Z}_p\langle T_1^{\pm1/p^\infty}, \dots, T_n^{\pm1/p^\infty}, t^{1/p^\infty}\rangle.$$

The perfectoid cover $B \to B_\infty$ comes naturally equipped with a continuous action of the profinite group $\Gamma := \mathbf{Z}_p(1)^{n+1} \simeq \varprojlim \mu_{p^m}^{n+1}$. Explicitly, an element $(\zeta_{1,m}, \dots, \zeta_{n,m}, \xi_m)_m \in \Gamma$ acts on B_∞ via continuous B-algebra automorphisms by scaling the chosen p^m-th roots: $T_i^{1/p^m} \mapsto \zeta_{i,m} T_i^{1/p^m}$ and $t^{1/p^m} \mapsto \xi_m t^{1/p^m}$. This yields a pro-étale Galois cover when p is inverted. Indeed, the variables T_i are units in B by definition and t becomes a unit when p is inverted because $tf = \pi$. Therefore, at each finite level m, the $B[1/p]$-subalgebra generated by T_i^{1/p^m} and t^{1/p^m} is a finite étale Kummer extension of $B[1/p]$ with Galois group $\mu_{p^m}^{n+1}$. Passing to the inverse limit over m, $B[1/p] \to B_\infty[1/p]$ becomes a Γ-torsor in pro-étale topology.

Let $\mathcal{A}$ be a 1-truncated $(\mathcal{G},\mu)$-aperture over B. We know how to solve the problem for the base change to the perfectoid cover $\mathcal{A}_{B_\infty}$ and obtain a (unique) reduction of structure to $\mathcal{P}_\mu^+$. To show that this descends to B, we have to show the induced reductions on the intersection $B_\infty \otimes_B B_\infty$ coincide. By Lemma 4.1.14, it suffices to show that the reductions induced on the associated $\mathcal{G}(\mathbf{Z}/p\mathbf{Z})$-local systems agree. But this is true by pro-étale descent because the Γ-action on $\mathcal{A}_{B_\infty}$ must preserve the reduction due to the uniqueness assertion. □

4.2. **Overconvergent canonical quasi-isogeny.** For abelian schemes and p-divisible groups, quotienting by the (overconvergent) canonical subgroup yields an isogeny which we call the (overconvergent) canonical isogeny. This isogeny has the property that it lifts the relative Frobenius isogeny modulo a suitable power of p depending on the valuation of the Hasse invariant. In this section, we construct the overconvergent canonical (quasi-)isogeny for $(\mathcal{G},\mu)$-apertures. As before, we assume that μ is defined over $\mathbf{Z}_p$.

Setup 4.2.1. Let R be a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, and let $\mathcal{A} \in \mathrm{BT}_\infty^{\mathcal{G},\mu}(R)$ be a $(\mathcal{G},\mu)$-aperture over R. This corresponds to a $\mathcal{G}$-torsor $\mathcal{E}$ over R^{syn}. Let $\mathcal{A}_1 \in \mathrm{BT}_1^{\mathcal{G},\mu}(R)$ denote the 1-truncated aperture over R obtained by reducing $\mathcal{A}$ modulo p, corresponding to the $\mathcal{G}$-torsor $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ over $R^{\mathrm{syn}} \otimes \mathbf{F}_p$. We assume that the Hasse invariant of $\mathcal{A}_1$ satisfies $\frac{p}{p+1} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{R/pR}))$. Under this assumption, Theorem 4.1.3 provides a unique reduction of structure $\mathcal{E}_1^- \subset \mathcal{E}_1$ to $\mathcal{P}_\mu^+$ that agrees with $\mathcal{Q}^+$ modulo $p^{1-\varepsilon}$.

Remark 4.2.2 (Parahoric subgroup)**.** Let $\mathcal{H}_\mu$ be the parahoric group scheme over $\mathbf{Z}_p$ defined as the dilatation of $\mathcal{G}$ along the closed subgroup scheme $\mathcal{P}_{\mu,\mathbf{F}_p}^+ \subset \mathcal{G}_{\mathbf{F}_p}$. For any p-torsion free $\mathbf{Z}_p$-algebra B, its B-points are given by the Cartesian square:

$$\begin{array}{ccc} \mathcal{H}_\mu(B) & \longrightarrow & \mathcal{G}(B) \\ \downarrow & & \downarrow \\ \mathcal{P}_\mu^+(B/pB) & \longrightarrow & \mathcal{G}(B/pB) \end{array}$$

Conjugation by $\mu(p)$ (defined as $x \mapsto \mu(p)x\mu(p)^{-1}$) on the generic fiber extends to a homomorphism of group schemes

$$\mathrm{inn}_{\mu(p)} \colon \mathcal{H}_\mu \to \mathcal{G}.$$

Construction 4.2.3 (The modified aperture)**.** The $\mathcal{P}_\mu^+$-reduction $\mathcal{E}_1^-$ over the special fiber tautologically lifts to a reduction of structure $\mathcal{E}^- \subset \mathcal{E}$ to $\mathcal{H}_\mu$. We define a new $\mathcal{G}$-torsor over R^{syn} by pushing forward $\mathcal{E}^-$ along $\mathrm{inn}_{\mu(p)}$:

$$\mathcal{E}' := \mathcal{E}^- \times^{\mathcal{H}_\mu, \mathrm{inn}_{\mu(p)}} \mathcal{G}.$$

As in Remark 2.1.21, we get a canonical modification $\mathcal{E} \dashrightarrow \mathcal{E}'$ along $p = 0$ bounded by μ.

Proposition 4.2.4. *The $\mathcal{G}$-torsor $\mathcal{E}'$ defines a $(\mathcal{G}, \mu)$-aperture over R, which we denote by $\mathcal{A}'$.*

Proof. We verify the condition on a geometric point $R \to \kappa$ of characteristic p by evaluating on the prism $(W(\kappa), (p))$. The $\mathcal{G}$-torsor $\mathcal{E}$ evaluates to a $\mathcal{G}$-torsor $\mathcal{E}_{W(\kappa)}$ over $W(\kappa)$ equipped with a Frobenius modification $m \colon \varphi^*\mathcal{E}_{W(\kappa)}[1/p] \xrightarrow{\sim} \mathcal{E}_{W(\kappa)}[1/p]$ of type μ. Locally trivializing the $\mathcal{H}_\mu$-reduction $\mathcal{E}^-_{W(\kappa)}$, the parahoric pushforward via $\mathrm{inn}_{\mu(p)}$ yields the Frobenius modification $m' = \mu(p)m\varphi(\mu(p))^{-1} = \mu(p)m\mu(p)^{-1}$ for $\mathcal{E}'_{W(\kappa)}$ over $W(\kappa)[1/p]$ since μ is defined over $\mathbf{Z}_p$. Since $\mathcal{E}^-_{W(\kappa)}$ lifts the conjugate reduction to $\mathcal{P}_\mu^+$ modulo p (which is preserved by m), m acts on $\mathcal{E}^-_{W(\kappa)}$, which means $m \in \mathcal{H}_\mu(W(\kappa)[1/p])$. Because $\mathcal{H}_\mu$ is the dilatation along $\mathcal{P}_\mu^+$ and μ is minuscule, conjugation by $\mu(p)$ maps $\mathcal{H}_\mu$ into $\mathcal{G}$, so $m' \in \mathcal{G}(W(\kappa))$. As m' is conjugate to m over $W(\kappa)[1/p]$, their invariant factors agree. Since μ is minuscule, any integral matrix bounded by μ has type exactly μ. Thus m' has type μ, making $\mathcal{E}'$ an aperture. $\square$

Definition 4.2.5 (Overconvergent canonical quasi-isogeny)**.** The quasi-isogeny $\mathrm{can}_{\mathcal{A}} \colon \mathcal{A} \dashrightarrow \mathcal{A}'$ bounded by μ corresponding to the modification $\mathcal{E} \dashrightarrow \mathcal{E}'$ defined in Construction 4.2.3 is called the **overconvergent canonical quasi-isogeny of level** 1.

Theorem 4.2.6 (Frobenius lifts)**.** *The overconvergent canonical quasi-isogeny* $\mathrm{can}_{\mathcal{A}}$ *reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$.*

Proof. Recall from the setup that $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ is the $\mathcal{G}$-torsor over $R^{\mathrm{syn}} \otimes \mathbf{F}_p$ corresponding to the mod p fiber $\mathcal{A}_1$. The conjugate realization of $\mathcal{E}_1$, denoted $T_{\mathrm{conj}}(\mathcal{E}_1)$, naturally provides a canonical reduction of structure of $\mathcal{E}_1$ to $\mathcal{P}_\mu^+$. By Theorem 3.5.7, the relative Frobenius quasi-isogeny on $\mathcal{A}_1$ corresponds precisely to the parahoric modification induced by this conjugate reduction $T_{\mathrm{conj}}(\mathcal{E}_1)$. On the other hand, the overconvergent canonical quasi-isogeny $\mathrm{can}_{\mathcal{A}}$ is defined as the parahoric modification associated to the reduction $\mathcal{E}^-$, which modulo p yields $\mathcal{E}_1^- \subset \mathcal{E}_1$. By Theorem 4.1.3, the reduction $\mathcal{E}_1^-$ agrees with the conjugate realization $T_{\mathrm{conj}}(\mathcal{E}_1)$ modulo $p^{1-\varepsilon}$. This congruence implies that the induced modifications agree modulo $p^{1-\varepsilon}$. Thus, $\mathrm{can}_{\mathcal{A}}$ reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$. $\square$

Remark 4.2.7 (Analytic fiber of overconvergent canonical quasi-isogeny)**.** Denote by $\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\leqslant\epsilon}$ the pullback of $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\leqslant\epsilon}$ under the 1-truncation map $\mathrm{BT}^{\mathcal{G},\mu}_{\infty} \otimes \mathbf{Z}_p^{\mathrm{cycl}} \to \mathrm{BT}^{\mathcal{G},\mu}_{1} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. For any p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, we have the following commuting diagram:

$$\begin{array}{ccc}
\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\geqslant p^{-1}\epsilon}(R) & \xrightarrow{T_{\text{ét}}} & \mathrm{Loc}_{\mathcal{H}_\mu(\mathbf{Z}_p)}(R[1/p]) \\
\downarrow & & \downarrow{\scriptstyle \mathrm{inn}_{\mu(p)}} \\
\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\geqslant\epsilon}(R) & \xrightarrow[T_{\text{ét}}]{} & \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(R[1/p])
\end{array}$$

We emphasize that our work on the overconvergent canonical quasi-isogeny implies that even though the top horizontal arrow *a priori* maps to $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$, it actually admits a canonical factorization via $\mathrm{Loc}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$ by virtue of the functor in Lemma 4.1.14. Here, the right vertical arrow is induced by the homomorphism $\mathcal{H}_\mu(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}_p)$ given by the adjoint action of $\mu(p)$. Indeed, this commutativity is essentially immediate from the construction of $\mathrm{can}_{\mathcal{A}}$. The left vertical arrow has the property that it reduces to the relative Frobenius modulo $p^{1-\varepsilon}$.

Theorem 4.2.8 (Level-n Frobenius lifts)**.** *Let $n \geqslant 1$ be an integer. Suppose that the Hasse invariant of $\mathcal{A}_1$ satisfies the stricter bound*

$$\frac{p}{p^{n-1}(p+1)} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{R/pR})).$$

Then one can iteratively apply the overconvergent canonical quasi-isogeny n times to obtain a composition

$$\mathcal{A} = \mathcal{A}^{(0)} \dashrightarrow \mathcal{A}^{(1)} \dashrightarrow \cdots \dashrightarrow \mathcal{A}^{(n)}$$

denoted $\mathrm{can}_{\mathcal{A}}^{(n)} \colon \mathcal{A} \dashrightarrow \mathcal{A}^{(n)}$ and called the ***overconvergent canonical quasi-isogeny of level*** *n. Furthermore, $\mathrm{can}_{\mathcal{A}}^{(n)}$ reduces to the n-th relative Frobenius quasi-isogeny modulo $p^{1-p^{n-1}\varepsilon}$.*

Proof. We proceed by induction on n, the base case $n = 1$ being Theorem 4.2.6. For $n \geqslant 2$, the level-1 modification $\mathrm{can}_{\mathcal{A}}^{(1)} \colon \mathcal{A} \dashrightarrow \mathcal{A}^{(1)}$ reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$. Consequently, the Hasse invariant of the modified aperture satisfies $\mathrm{Ha}(\mathcal{A}^{(1)}|_{R/pR}) \equiv \mathrm{Ha}(\mathcal{A}|_{R/pR})^p \pmod{p^{1-\varepsilon}}$. Since $p\varepsilon \leqslant p^{n-1}\varepsilon < \frac{p}{p+1} < 1-\varepsilon$, this congruence forces the valuation to be exactly $\nu_p(\mathrm{Ha}(\mathcal{A}^{(1)}|_{R/pR})) = p\varepsilon$. Applying the inductive hypothesis to $\mathcal{A}^{(1)}$ at level $n-1$ produces a sequence of modifications up to $\mathcal{A}^{(n)}$ which reduces to the $(n-1)$-th relative Frobenius modulo $p^{1-p^{n-2}(p\varepsilon)} = p^{1-p^{n-1}\varepsilon}$. Composing this with $\mathrm{can}_{\mathcal{A}}^{(1)}$ yields the result, noting that the initial congruence modulo $p^{1-\varepsilon}$ is strictly finer since $1-\varepsilon > 1-p^{n-1}\varepsilon$. □

4.3. **Comparison with existing results.** Canonical subgroups for abelian varieties and Barsotti–Tate groups have been extensively investigated, notably by Abbes–Mokrane [AM04], Andreatta–Gasbarri [AG06], Tian [Tia10], Rabinoff [Rab12], and Fargues [Far11]. Scholze in [Sch15, §3.2.1] also develops a theory of canonical subgroups *integrally* by using Illusie's work on the cotangent complex Although Rabinoff obtains larger overconvergence radii, his notion of canonical subgroups differs a bit from the one considered here, as it (a priori) does not involve lifting the kernel of the relative Frobenius. For this reason, and because the work of Fargues encompasses and refines the earlier results of Abbes–Mokrane, Andreatta–Gasbarri, and Tian, we shall restrict our comparison to [Far11]. Furthermore, the overconvergence bounds established here improve upon those obtained by Scholze in [Sch15]. A unifying characteristic of these classical approaches is their restriction to complete rank-1 valuation rings of mixed characteristic. In contrast, the linear algebraic approach taken using prismatic Dieudonné theory in this paper yields an integral, group-theoretic construction that works uniformly over arbitrary flat p-adic formal bases while recovering and strengthening existing results.

Setting $\mathcal{G} = \mathrm{GL}_h$ and $\mu(z) = \mathrm{diag}(z, \ldots, z, 1, \ldots, 1)$ with d occurrences of z and $h-d$ occurrences of 1, we obtain:

Theorem 4.3.1 (Canonical subgroup for 1-truncated Barsotti–Tate groups)**.** *Let G be a 1-truncated Barsotti–Tate group of height h and dimension d over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, with $G_0 := G|_{R/pR}$ its reduction modulo p. Assume that the Hasse invariant of G_0 has p-adic valuation $\varepsilon := \nu_p(\mathrm{Ha}\, G_0) < \frac{p}{p+1}$. Then G admits a unique finite locally free subgroup scheme $C_1 \subset G$ of rank p^d which agrees with the kernel of the relative Frobenius $\mathrm{Ker}(G_0 \to G_0^{(p)})$ modulo $p^{1-\varepsilon}$.*

Proof. This is just the specialization of Theorem 4.1.3 to $\mathcal{G} = \mathrm{GL}_h$. The assertion about rank can be proven exactly as in [Sch15, Proposition 3.2.8 (iv)]. □

Remark 4.3.2 (General bases)**.** The assumption requiring the base ring R to be a $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra is not essential for the validity of Theorems 4.3.1 and 4.3.3. The results carry over to an arbitrary p-complete flat $\mathbf{Z}_p$-algebra R, provided that references to the congruence condition "modulo $p^{1-\varepsilon}$" are systematically replaced by congruence modulo the ideal generated by all elements of R having p-adic valuation at least $1-\varepsilon$.

Although the overconvergence bound on the Hasse invariant established above is strictly stronger than that of [Far11, Théorème 4], a direct comparison warrants caution. Fargues characterizes the canonical subgroup via the ramification filtrations of Abbes–Saito, whereas our construction relies on lifting the kernel of the relative

Frobenius morphism. As we do not establish a comparison with the Abbes–Saito filtration, it is not right to directly compare our results with that of Fargues.

We also have a higher-level analogue where we recover the fundamental properties delineated in [Far11, Théorème 6] and [Sch15, Proposition 3.2.8] under these stronger bounds:

Theorem 4.3.3 (Level-n canonical subgroup for p-divisible groups)**.** *Let $n \geqslant 1$ be an integer, and let G be a p-divisible group of height h and dimension d over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, with reduction $G_0 := G|_{R/pR}$ modulo p. Assume that the Hasse invariant of G_0 satisfies the valuation bound*

$$\varepsilon := \nu_p(\operatorname{Ha} G_0) < \frac{1}{p^{n-2}(p+1)}.$$

*Then the p^n-torsion subgroup $G[p^n]$ admits a unique finite locally free subgroup scheme $C_n \subset G[p^n]$ of rank p^{nd}, called the **canonical subgroup of level** n, satisfying the following properties:*

*(1) **Congruence with Frobenius kernel:** The subgroup scheme C_n agrees with the kernel of the n-th iterated relative Frobenius isogeny $\operatorname{Ker}(G_0 \to G_0^{(p^n)})$ modulo $p^{1-p^{n-1}\varepsilon}$.*

*(2) **Truncation compatibility:** For any integer $1 \leqslant k < n$, the p^k-torsion subgroup scheme $C_n[p^k]$ is precisely the level-k canonical subgroup $C_k \subset G[p^k]$ of rank p^{kd}. Furthermore, for any characteristic 0 geometric point $R \to \overline{\kappa}$,*

$$C_n(\overline{\kappa}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^d.$$

*(3) **Quotient compatibility and Hasse invariant scaling:** For any integer $1 \leqslant k < n$, let $G^{(k)} := G/C_k$ denote the quotient by the level-k canonical subgroup. Then the Hasse invariant of $G^{(k)}$ has valuation $\nu_p(\operatorname{Ha}(G^{(k)}|_{R/pR})) = p^k\varepsilon$, and under the canonical identification $(G/C_k)[p^{n-k}] \simeq G[p^n]/C_k$, the level-$(n-k)$ canonical subgroup of $G^{(k)}$ coincides with the quotient C_n/C_k.*

*(4) **Cartier duality:** The Cartier dual p-divisible group G^D of height h and dimension $h-d$ has Hasse invariant valuation $\varepsilon < \frac{1}{p^{n-2}(p+1)}$, and its canonical subgroup of level n, denoted $C_n^D \subset G^D[p^n]$, has rank $p^{n(h-d)}$ and coincides with the orthogonal complement of C_n under the perfect Weil pairing:*

$$C_n^D = C_n^{\perp} \subset G^D[p^n].$$

Proof. (1), (2), (3) are clear as this is a direct specialization of Theorem 4.2.8 to $\mathcal{G} = \mathrm{GL}_h$ combined with the reasoning in [Sch15, Proposition 3.2.8 (iv)]. Under Cartier duality, the general linear group GL_h and cocharacter μ are replaced by the dual representation, which transposes dimensions: the dual p-divisible group G^D has height h and dimension $h-d$. Because the Hasse invariant is invariant under Cartier duality ($\operatorname{Ha}(G^D) = \operatorname{Ha}(G)$), G^D satisfies the same overconvergence bound $\varepsilon < \frac{1}{p^{n-2}(p+1)}$ and therefore admits a level-n canonical subgroup $C_n^D \subset G^D[p^n]$ of rank $p^{n(h-d)}$. In our group-theoretic framework, Cartier duality corresponds to applying a Chevalley involution $C\colon \mathcal{G} \to \mathcal{G}$ (cf. Remark 3.5.9), under which the overconvergent canonical quasi-isogeny and the properties established in Theorem 4.2.8 are canonically preserved. Consequently, the dual canonical subgroup C_n^D coincides precisely with the orthogonal complement of C_n under the perfect Weil pairing: $C_n^D = C_n^{\perp}$. □

5. Perfectoid Shimura varieties

We prove Theorem A by following Scholze's method [Sch15].

Setup 5.1. Maintain the notations of Section 3.1. For brevity, we rename $-\mu_v$ to $-\mu$.

Assumption 5.2. The Shimura datum has reflex field $\mathbf{Q}_p$, i.e., $-\mu$ is defined over $\mathbf{Z}_p$. We also assume that the boundary of minimal compactification has codimension at least 2 (cf. Assumption 3.6.6).

5.1. **The overconvergent anticanonical tower.** We use the theory developed in Section 3.7 to define the anticanonical tower. Let us first make some preliminary definitions:

Construction 5.1.1. Recall that we have a formally étale syntomic realization $\varpi\colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$ by Theorem 3.1.5.

- (Ordinary loci) Let $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ be the adic generic fiber of $\widehat{\mathscr{S}}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ from Definition 3.3.6, an open adic subspace of $\S_{\mathcal{G}(\mathbf{Z}_p)}$. In the notations of Remark 3.1.4, let $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)_h}$ be the preimage of $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ along the analytification of the level map s_h. Let $\mathcal{S}^{*,\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ be the adic generic fiber of the unique open formal subscheme of $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$ whose special fiber is the extended ordinary locus supplied by Corollary 3.6.9. Define $\mathcal{S}^{*,\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)_h}$ analogously as the preimage along analytification of level maps.
- (ε-neighbourhoods) For $0 \leqslant \varepsilon < 1$, we define $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ and $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ as the formal schemes over $\mathrm{Spf}\, \mathbf{Z}_p^{\mathrm{cycl}}$ representing the functors which assign to any p-adically complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R the set of pairs (f,u) where $f\colon \mathrm{Spf}\, R \to \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ (resp. $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$) and $u \in \mathrm{H}^0(\mathrm{Spf}\, R, f^*\omega^{-\chi})$ is a section such that $u\,\mathrm{Ha}(f) = p^\varepsilon$ modulo p (up to natural equivalence). By Proposition 3.6.8, these are admissible blow-ups and their adic generic fibers, denoted $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ and $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$, are exactly the open adic subspaces of $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}$ and $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ where $|\,\mathrm{Ha}\,| \geqslant |p|^\varepsilon$. For any deeper level K_p, we define $\mathcal{S}_{K_p}(\varepsilon)$ and $\mathcal{S}^*_{K_p}(\varepsilon)$ as their preimages under the level maps. The adic spaces involved in this item are all analytic adic space over $\mathrm{Spa}(\mathbf{Q}_p^{\mathrm{cycl}}, \mathbf{Z}_p^{\mathrm{cycl}})$.
- (ε-neighbourhoods in $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$) We define the formal stack $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(\varepsilon)$ as the base change of the open formal substack $\mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ along the natural 1-truncation map $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v} \otimes \mathbf{Z}_p^{\mathrm{cycl}} \to \mathrm{BT}_1^{\mathcal{G}^c,-\mu_v} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. Here $\mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ is the locus where the Hasse invariant satisfies $|\,\mathrm{Ha}\,| \geqslant |p|^\varepsilon$ (see Construction 5.1.1).

Theorem 5.1.2 (Canonical Frobenius lifts modulo $p^{1-\varepsilon}$)**.** *Let $0 \leqslant \varepsilon < \frac{p}{p+1}$. There exists a unique Cartesian square of p-adic formal stacks over* $\mathrm{Spf}\, \mathbf{Z}_p^{\mathrm{cycl}}$

$$\begin{tikzcd}
\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[d, "\varpi"] \\
\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(p^{-1}\varepsilon) \arrow[r] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(\varepsilon)
\end{tikzcd}$$

such that the vertical arrows are induced by syntomic realizations, the horizontal arrows are unique lifts of the relative Frobenii modulo $p^{1-\varepsilon}$, and the bottom horizontal arrow is induced by the overconvergent canonical quasi-isogeny of level 1 *(cf. Theorem 4.2.6).*

Proof. Set $h = \mu(p)^{-1}$. In what follows, $(-)^{\mathrm{an}}$ denotes the analytification functor. Recall that we have a Hecke correspondence diagram:

$$\begin{tikzcd}
 & \mathrm{Sh}_{\Gamma_0(p)} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\
\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} & & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} .
\end{tikzcd}$$

Over $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon)$, the p-adic étale realization map factors through $\mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$ (cf. Remark 4.2.7). The description in Remark 3.1.4 therefore shows that $t_h^{\mathrm{an}}|_{\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon)}$ is canonically isomorphic to the base-change of

$$[\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}} \times_{\mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \simeq [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}$$

along $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$. In other words, we have the following Cartesian squares:

$$\begin{tikzcd}
\mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] \arrow[d, "t_h^{\mathrm{an}}"'] & {[\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)} / \underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}} \arrow[r] \arrow[d] & {[\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)} / \underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}}} \arrow[d, "\alpha_h^{\mathrm{an}}"] \\
\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}
\end{tikzcd}$$

We have a conjugation map $\mathrm{inn}_{\mu(p)} \colon \mathcal{H}_\mu(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}_p)$. This gives a section:

$$\mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \xrightarrow{\simeq} [h\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}} \to [\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}.$$

This implies that we have a section

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon).$$

Composing this section with the map $s_h^{\mathrm{an}} \colon \mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \to \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}$ gives a map $\Phi' \colon \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}$ such that the diagram

$$\begin{tikzcd}
\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] \arrow[d, "\Phi'"'] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \arrow[d, "\mathrm{inn}_{\mu(p)}"] \\
\mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}
\end{tikzcd}$$

commutes. By Remark 4.2.7, the composition of the top horizontal arrow with the right vertical arrow factors as

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \xrightarrow{\varpi^{\mathrm{ad}}} \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v}(p^{-1}\varepsilon)^{\mathrm{ad}} \xrightarrow{\text{Theorem 4.2.6}} \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v}(\varepsilon)^{\mathrm{ad}} \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}.$$

The argument from [MY26, Proposition 4.3.1] now shows that this is the adic generic fiber of a morphism $\widetilde{F} \colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ and considerations with the Hasse invariant tells us that it actually factors through $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$. The reduction property modulo $p^{1-\varepsilon}$ now follows from Corollary 3.2.6. □

The following is the analogue of [Sch15, Theorem 3.2.15]:

Theorem 5.1.3. *Let $0 \leqslant \varepsilon < \frac{p}{p+1}$.*

(a) There is a unique commutative diagram of formal schemes over $\mathrm{Spf}\, \mathbf{Z}_p^{\mathrm{cycl}}$*:*

$$\begin{tikzcd}
\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] \arrow[d, hook] & \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[d, hook] \\
\widehat{\mathscr{S}}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] & \widehat{\mathscr{S}}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)
\end{tikzcd}$$

where the horizontal arrows modulo $p^{1-\varepsilon}$ agree with the relative Frobenii.

(b) For any integer $m \geqslant 0$, the inverse of the overconvergent canonical quasi-isogeny of level m (cf. Theorem 4.2.8) induces a morphism on the adic generic fibers

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \to \mathcal{S}_{\Gamma_0(p^m)}$$

which extends uniquely to a morphism of adic spaces $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \to \mathcal{S}^*_{\Gamma_0(p^m)}$. These morphisms are open embeddings. Denote their images by $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ inside $\mathcal{S}_{\Gamma_0(p^m)}$ and $\mathcal{S}^*_{\Gamma_0(p^m)}$, respectively. Moreover, for $m \geqslant 1$, the diagram

$$\begin{tikzcd}
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m-1}\varepsilon) \arrow[r, hook] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^*_{\Gamma_0(p^{m+1})} \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \arrow[r, hook] & \mathcal{S}^*_{\Gamma_0(p^m)}
\end{tikzcd}$$

is commutative and Cartesian, where the right vertical map is the level map.

(c) The images $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ are open and closed adic subspaces of the ε-neighbourhoods $\mathcal{S}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon)$, respectively.

Proof. Let us first consider the open Shimura variety loci. For part (a), Theorem 5.1.2 provides the unique lift $\widetilde{F}\colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ of the relative Frobenius modulo $p^{1-\varepsilon}$.

For part (b), the inverse of the level-m overconvergent canonical quasi-isogeny $(\mathrm{can}^{(m)}_{\mathcal{A}})^{-1}\colon \mathcal{A}^{(m)} \dashrightarrow \mathcal{A}$ is a p-isogeny bounded by $m(-\nu)(p)$. By Theorem 3.5.8, the space of p-isogenies bounded by $m(-\nu)(p)$ defines the adic space $\mathcal{S}_{\Gamma_0(p^m)}$. Hence, we obtain a classifying morphism $f\colon \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \to \mathcal{S}_{\Gamma_0(p^m)}$. Consider the composition $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \xrightarrow{f} \mathcal{S}_{\Gamma_0(p^m)} \to \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}$, where the second map is the natural projection tracking the target aperture of the isogeny (i.e., the Hecke-translated map). By definition, this composition maps $\mathcal{A}$ exactly to the target of $(\mathrm{can}^{(m)}_{\mathcal{A}})^{-1}$, which is $\mathcal{A}$ itself. Therefore, the composite map is exactly the natural open immersion $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \hookrightarrow \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}$. Since the target projection $\mathcal{S}_{\Gamma_0(p^m)} \to \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}$ is étale, it follows that f must itself be an open immersion. The commutativity of the diagram for $m \geqslant 1$ follows directly from the uniqueness property of overconvergent canonical quasi-isogeny. To deduce that the diagram is Cartesian, we compute the degrees of both vertical maps. The left vertical map $(\widetilde{F})^{\mathrm{ad}}$ lifts the relative Frobenius of the special fiber (which is a smooth scheme of relative dimension $\dim \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$), so its degree is exactly $p^{\dim \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}}$ (moreover, it is finite étale on the adic generic fiber because its differential is multiplication by p). The right vertical map is the natural level projection, whose degree is the group index $[\Gamma_0(p^m) : \Gamma_0(p^{m+1})]$. Because $m \geqslant 1$, this index measures the size of the $\mathbf{F}_p$-points of the unipotent radical of the parabolic subgroup associated to μ, which is exactly $p^{\dim \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}}$. Since both vertical maps are finite étale of the same degree and the horizontal maps are open immersions, the commutative square must be Cartesian.

For part (c), we need to show that the image is open and closed. It is open because the map extending to the minimal compactification is an open immersion by part (b). To see that it is closed, observe that the composition

$$\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \xrightarrow{\sim} \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \subset \mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon) \to \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$$

is exactly the m-th iterate of the Frobenius lift $(\widetilde{F})^{\mathrm{ad}}$, which is finite. Since the level map $\mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon) \to \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ is also finite, the inclusion of the image $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ into $\mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon)$ must be finite, and hence the image is closed. The corresponding statement for the open Shimura variety follows by taking preimages.

Since the boundary of the minimal compactification $\widehat{\mathscr{S}}^*_{\mathcal{G}(\mathbf{Z}_p)}$ has codimension $\geqslant 2$ by assumption, a Hartog's principle principle argument (cf. [Sch15, Lemma 3.2.10]) works to uniquely extend everything to minimal compactifications just like in the proof of [Sch15, Theorem 3.2.15]. □

Remark 5.1.4 (The overconvergent anticanonical tower)**.** The **overconvergent anticanonical tower** is the inverse system on the right side of the following commutative diagram of Cartesian squares:

$$\begin{tikzcd}
\vdots \arrow[d] & \vdots \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-(n+1)}\varepsilon) \arrow[r, "\sim"] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{n+1})}(\varepsilon) \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-n}\varepsilon) \arrow[r, "\sim"] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{n})}(\varepsilon) \arrow[d] \\
\vdots \arrow[d] & \vdots \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[r, equal] & \mathcal{S}^{*,\mathrm{anti}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)
\end{tikzcd}$$

A completely analogous statement holds for the open Shimura variety $\mathcal{S}$. The upshot is that the tower on the right has an integral model given by the obvious integral model of the tower on the left and this integral model has the crucial property that it reduces to a tower of relative Frobenii modulo $p^{1-\varepsilon}$.

5.2. **Perfectoid anticanonical neighbourhood.** This subsection essentially mirrors [Sch15, §3.2.3]. We show that the inverse limit of the overconvergent anticanonical tower constructed in Remark 5.1.4 is a perfectoid space.

Lemma 5.2.1 (Anticanonical neighbourhoods are affinoid). *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. For m sufficiently large, the adic space $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ is affinoid.*

Proof. By Theorem 5.1.3(b), $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \simeq \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$. Since ω^χ is ample on $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$, for $m \gg 0$, the higher cohomology of $(\omega^\chi)^{\otimes p^m}$ vanishes, allowing us to lift Ha^{p^m} to a global section $\widetilde{\mathrm{Ha}}_m$ over $\widehat{\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}}$. The locus $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$ where $|\,\mathrm{Ha}\,| \geqslant |p|^{p^{-m}\varepsilon}$ is equivalently defined by $|\widetilde{\mathrm{Ha}}_m| \geqslant |p|^\varepsilon$. As $\widetilde{\mathrm{Ha}}_m$ is a global section of an ample line bundle, this defines an affinoid open subspace. $\square$

Fix $t \in (\mathbf{Z}_p^{\mathrm{cycl}})^\flat$ with $|t| = |p|$, chosen to admit compatible p-power roots, yielding $(\mathbf{Z}_p^{\mathrm{cycl}})^\flat \simeq \mathbf{F}_p[\![t^{1/p^\infty}]\!]$. Let $\mathcal{S}'^*$ be the t-adic completion of $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes_{\mathbf{Z}_{(p)}} \mathbf{F}_p[\![t^{1/p^\infty}]\!]$ over $\mathbf{F}_p[\![t^{1/p^\infty}]\!]$, and $\mathcal{S}'^*$ its generic fiber. As in characteristic 0, let $\mathcal{S}'^*(\varepsilon)$ be the locus where $|\,\mathrm{Ha}\,| \geqslant |t|^\varepsilon$. Taking the inverse limit along relative Frobenius yields a perfect formal scheme $\mathcal{S}'^*(\varepsilon)_{\mathrm{perf}} = \varprojlim_\Phi \mathcal{S}'^*(\varepsilon)$ with perfectoid generic fiber $\mathcal{S}'^*_{\mathrm{perf}}(\varepsilon)$.

Theorem 5.2.2 (Perfectoid anticanonical neighborhood). *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. There are unique perfectoid spaces $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ over $\mathbf{Q}_p^{\mathrm{cycl}}$ such that*

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \quad \textit{and} \quad \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon).$$

Moreover, the tilt $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)^\flat$ identifies naturally with the open subset $\mathcal{S}'^*_{\mathrm{perf}}(\varepsilon) \subset \mathcal{S}'^*_{\mathrm{perf}}$ where $|\,\mathrm{Ha}\,| \geqslant |t|^\varepsilon$. In particular, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is affinoid perfectoid, and the boundary

$$\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon) \subset \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$$

is strongly Zariski closed.

Proof. We focus on the minimal compactification; the open Shimura variety follows analogously. By Theorem 5.1.3(b), $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ admits the integral model $\widehat{\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}}(p^{-m}\varepsilon)$. Crucially, the transition maps in this tower reduce modulo $p^{1-\varepsilon}$ to the relative Frobenius map. Let $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon) := \varprojlim_m \widehat{\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}}(p^{-m}\varepsilon)$. Working affine locally with $\mathrm{Spf}\, R_m \subset \widehat{\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}}(p^{-m}\varepsilon)$, the inverse limit yields a flat p-adically complete $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra $R_\infty = (\varinjlim_m R_m)^\wedge_p$. Since the transition maps reduce to Frobenius modulo $p^{1-\varepsilon}$, absolute Frobenius induces an isomorphism

$$R_\infty/p^{(1-\varepsilon)/p} \xrightarrow{\sim} R_\infty/p^{1-\varepsilon}.$$

This implies that R_∞ is a perfectoid algebra. Thus, its generic fiber is an affinoid perfectoid space over $\mathbf{Q}_p^{\mathrm{cycl}}$, which guarantees the existence and uniqueness of $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ (cf. [SW13, Definition 2.4.1, Proposition 2.4.2, Proposition 2.4.5]).

The canonical identification $\mathcal{S}'^*(p^{-m}\varepsilon)/t^{1-\varepsilon} = \widehat{\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}}(p^{-m}\varepsilon)/p^{1-\varepsilon}$ from Theorem 5.1.3(a) passes to the limit, yielding an isomorphism of coordinate rings modulo $t^{1-\varepsilon}$ and $p^{1-\varepsilon}$. This implies that the generic fibers are tilts of each other. Since the tilt $\mathcal{S}'^*_{\mathrm{perf}}(\varepsilon)$ is affinoid with a Zariski closed boundary, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is affinoid perfectoid with a strongly Zariski closed boundary. $\square$

5.3. **Passage to full level.** We now pass to the full level tower $\Gamma(p^m)$ at p. For any $m \geqslant 0$, let

$$\pi_m : \mathcal{S}_{\Gamma(p^m)} \to \mathcal{S}_{\Gamma_0(p^m)} \quad \text{and} \quad \pi_m^* : \mathcal{S}^*_{\Gamma(p^m)} \to \mathcal{S}^*_{\Gamma_0(p^m)}$$

be the natural level projection maps. Since these projection maps are finite, we can define the anticanonical loci at full level by pulling back the corresponding loci of parahoric level.

Definition 5.3.1 (Anticanonical loci at level $\Gamma(p^m)$)**.** Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. For any integer $m \geqslant 0$, we define the **anticanonical locus of level** $\Gamma(p^m)$ **and radius** ε on the open Shimura variety and its minimal compactification to be the preimages:

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) := \pi_m^{-1}\Big(\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)\Big)$$

and

$$\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) := (\pi_m^*)^{-1}\Big(\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)\Big)$$

respectively. Since the anticanonical loci of level $\Gamma_0(p^m)$ are open and closed in the respective ε-neighbourhoods, the same holds for $\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$.

To study the limit of these spaces as $m \to \infty$, we will use the following general result on inverse limits of adic spaces along finite transition maps:

Lemma 5.3.2 ([HJ23, Lemma 5.9])**.** *Let $(X_m)_{m\in I} \to (Y_m)_{m\in I}$ be a morphism of cofiltered inverse systems of locally Noetherian adic spaces. Assume that the maps $f_m : X_m \to Y_m$ and the transition maps in both inverse systems are finite, and that $Y_\infty := \varprojlim_m Y_m^\diamondsuit$ is perfectoid. Then $X_\infty := \varprojlim_m X_m^\diamondsuit$ is perfectoid, and the morphism $f_\infty : X_\infty \to Y_\infty$ is quasicompact. Moreover, if $U \subset Y_\infty$ is an open affinoid perfectoid subset which arises as the preimage of an open affinoid $U_m \subset Y_m$ for some m, then $f_\infty^{-1}(U) \subset X_\infty$ is also affinoid perfectoid. In particular, f_∞ is affinoid in the sense of [HJ23, Theorem 5.8].*

Applying this lemma, we obtain the analogue of Theorem 5.2.2 at full level:

Theorem 5.3.3 (Perfectoid anticanonical neighborhood at full level)**.** *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. There are unique perfectoid spaces $\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ over $\mathbf{Q}_p^{\mathrm{cycl}}$ such that*

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) \quad \text{and} \quad \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon).$$

Moreover, the natural projection

$$f_\infty : \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$$

is an affinoid morphism of perfectoid spaces (in the sense of [HJ23, Theorem 5.8]). In particular, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is affinoid perfectoid, and the boundary

$$\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \subset \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$$

is strongly Zariski closed.

Proof. We focus on the minimal compactification; the statement for the open Shimura variety follows by taking the open part. For each $m \geqslant 0$, the level projection $\pi_m^* : \mathcal{S}^*_{\Gamma(p^m)} \to \mathcal{S}^*_{\Gamma_0(p^m)}$ is a finite morphism of locally Noetherian adic spaces. Restricting this map to the anticanonical loci, we obtain finite maps of locally Noetherian adic spaces:

$$f_m : \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon).$$

Furthermore, the transition maps in both towers are induced by the natural level projections, which are finite. Therefore, the collection of maps $(f_m)_m$ defines a morphism of cofiltered inverse systems of locally Noetherian adic spaces with finite maps and finite transition maps. By Theorem 5.2.2, the limit

$$Y_\infty := \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \simeq \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$$

is a perfectoid space, which is moreover affinoid perfectoid. Applying Lemma 5.3.2, we deduce that the limit

$$\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) := \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$$

is a perfectoid space, and the projection map $f_\infty\colon \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is an affinoid morphism. Since $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is affinoid perfectoid and f_∞ is affinoid, we conclude that $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is affinoid perfectoid.

Finally, the boundary $\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is the preimage of the boundary $\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ under f_∞. Since $\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is strongly Zariski closed in $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ (by Theorem 5.2.2) and f_∞ is an affinoid morphism, it follows that $\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is strongly Zariski closed in $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$. □

5.4. **The Hodge–Tate period map.** We explain how to construct the Hodge–Tate period map in diamonds from toroidal and minimal compactifications. The contents of this subsection is well-known to experts.

Setup 5.4.1 (Flag varieties)**.** In this subsection, we do not enforce any assumptions on the Shimura data, that is, the reflex field is not assumed to be $\mathbf{Q}_p$ and there is no assumption on boundary codimensions of the minimal compactification. Let $\mathrm{Fl}_{\mathcal{G},\mu}$ be the Grassmannian scheme over $\mathcal{O}$ associated with μ. It can be presented as the fppf quotient $\mathcal{G}_{\mathcal{O}}/P^-_\mu$ and it only depends on the conjugacy class of μ. Let $\mathcal{F}l^\diamond_{\mathcal{G},\mu}$ and $\mathcal{F}l_{\mathcal{G},\mu}$ be the associated diamond and rigid generic fiber. In what follows, all diamonds are spatial.

Remark 5.4.2 (Construction of the Hodge–Tate filtration)**.** Consider a choice of log-smooth toroidal compactification $\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}$ of the Shimura variety. For any algebraic representation W of $\mathcal{G}^c$, one associates a local system $\mathcal{W}_p$ on the pro-Kummer-étale site of $\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}$, alongside a filtered vector bundle $\mathcal{W}_{\mathrm{dR}}$ equipped with an integrable log-connection. The work of Diao–Lan–Liu–Zhu [Dia+23, Theorem 5.3.1] establishes that these two objects are canonically associated via the relative period sheaves. Specifically, there is an isomorphism

$$\mathcal{W}_p \otimes_{\mathbf{Q}_p} \mathcal{O}\mathbf{B}_{\mathrm{dR},\log} \simeq \mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \mathcal{O}\mathbf{B}_{\mathrm{dR},\log}$$

which respects the filtrations, the connections, and the Hecke action.

Leveraging the canonical isomorphism, we can construct the Hodge–Tate filtration on the pro-étale vector bundle $\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}$. We introduce two $\mathbf{B}^+_{\mathrm{dR}}$-local systems defined by

$$\mathbf{M} := \mathcal{W}_p \otimes_{\mathbf{Q}_p} \mathbf{B}^+_{\mathrm{dR},\log} \quad \text{and} \quad \mathbf{M}_0 := (\mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \mathcal{O}\mathbf{B}^+_{\mathrm{dR},\log})^{\nabla=0}.$$

These serve as lattices inside the common $\mathbf{B}_{\mathrm{dR}}$-local system $\mathbf{M} \otimes_{\mathbf{B}^+_{\mathrm{dR}}} \mathbf{B}_{\mathrm{dR}} \simeq \mathbf{M}_0 \otimes_{\mathbf{B}^+_{\mathrm{dR}}} \mathbf{B}_{\mathrm{dR}}$. Because $\mathbf{M}$ and $\mathbf{M}_0$ both carry filtrations, they jointly define an ascending filtration via

$$\mathrm{Fil}_{-j}\big(\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}\big) := (\mathbf{M} \cap \mathrm{Fil}^j \mathbf{M}_0)\big/(\mathrm{Fil}^1\mathbf{M} \cap \mathrm{Fil}^j \mathbf{M}_0).$$

This ascending filtration satisfies the grading relation $\mathrm{Gr}_j\big(\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}\big)(j) \simeq \mathrm{Gr}^j\mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}$.

Due to the functoriality of the aforementioned construction with respect to Hecke actions we can deduce analogous structures on principal bundles by Tannakian formalism. Over $\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}$, we consider the pro-Kummer-étale $\mathcal{G}^c(\mathbf{Q}_p)$-torsor $\mathcal{G}_{\text{p-ét},p}$ and the $\mathcal{G}^c$-torsor $\mathcal{G}_{\mathrm{dR}}$ on the étale site. The filtration on $\mathcal{W}_{\mathrm{dR}}$ induces a reduction of structure group of $\mathcal{G}_{\mathrm{dR}}$ to a $\mathcal{P}^{c,-}_\mu$-torsor, which we denote by $\mathcal{P}_{\mathrm{dR}}$. This defines the space

$$\mathcal{M}_{\mathrm{dR}} := \mathcal{P}_{\mathrm{dR}} \times^{\mathcal{P}^{c,-}_\mu} \mathcal{M}^c_\mu.$$

Similarly, the Hodge–Tate filtration on $\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}$ reduces the structure group of $\mathcal{G}_{\text{p-ét},p} \times^{\mathcal{G}^c(\mathbf{Q}_p)} \mathcal{G}^c$ to a $\mathcal{P}^{c,-}_\mu$-torsor $\mathcal{P}_{\mathrm{HT}}$, giving rise to

$$\mathcal{M}_{\mathrm{HT}} := \mathcal{P}_{\mathrm{HT}} \times^{\mathcal{P}^{c,-}_\mu} \mathcal{M}^c_\mu.$$

Theorem 5.4.3 (The Hodge–Tate period map in diamonds)**.** *By taking the inverse limit over p-power levels $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, we arrive at the diamond $\mathcal{S}^{\mathrm{tor},\diamond}_{K^p,\Sigma} := \varprojlim_{K_p} \mathcal{S}^{\mathrm{tor},\diamond}_{K^pK_p,\Sigma}$. Over this space, the pro-Kummer-étale torsor*

$\mathcal{G}_{p\text{-ét},p}$ becomes trivial and the Hodge–Tate filtration induces a $\mathcal{G}(\mathbf{Q}_p)$-equivariant morphism

$$\pi_{\mathrm{HT}}^{\mathrm{tor}} \colon \mathcal{S}_{K^p,\Sigma}^{\mathrm{tor},\diamond} \to \mathcal{F}l_{\mathcal{G},\mu}^{\diamond}.$$

This Hodge–Tate period map factors $\mathcal{G}(\mathbf{Q}_p)$-equivariantly through a map

$$\pi_{\mathrm{HT}}^{*} \colon \mathcal{S}_{K^p}^{*,\diamond} \to \mathcal{F}l_{\mathcal{G},\mu}^{\diamond},$$

where $\mathcal{S}_{K^p}^{*,\diamond}$ is the minimal compactification inverse limit $\mathcal{S}_{K^p}^{*,\diamond} := \varprojlim_{K_p} \mathcal{S}_{K^pK_p}^{*,\diamond}$ as diamonds.

Proof. The existence of $\pi_{\mathrm{HT}}^{\mathrm{tor}}$ is clear and $\mathcal{G}(\mathbf{Q}_p)$-equivariance is true because the discussion in Remark 5.4.2 is compatible with Hecke action by [Dia+23, Theorem 5.3.1]. For factorization through the minimal compactification, choose a very ample character $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu_v}^c)$ so that it induces a very ample line bundle $\mathcal{L}$ on $\mathcal{F}l_{\mathcal{G},\mu}$.

The Hodge–Tate period map $\pi_{\mathrm{HT}}^{\mathrm{tor}} \colon \mathcal{S}_{K^p,\Sigma}^{\mathrm{tor},\diamond} \to \mathcal{F}l_{\mathcal{G},\mu}^{\diamond}$ pulls this ample line bundle back to the automorphic line bundle ω^χ on the diamond infinite-level toroidal compactification $\mathcal{S}_{K^p,\Sigma}^{\mathrm{tor},\diamond}$. To show that $\pi_{\mathrm{HT}}^{\mathrm{tor}}$ descends to the minimal compactification, it suffices to prove that it is constant on the fibers of the projection $q \colon \mathcal{S}_{K^p,\Sigma}^{\mathrm{tor},\diamond} \to \mathcal{S}_{K^p}^{*,\diamond}$. We analyze these fibers by descending to finite levels. For any finite level $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, let $q_{K_p} \colon \mathcal{S}_{K^pK_p,\Sigma}^{\mathrm{tor}} \to \mathcal{S}_{K^pK_p}^{*}$ denote the natural proper projection between the rigid-analytic toroidal and Baily–Borel minimal compactifications. By the algebraic theory of canonical extensions and rigid-analytic GAGA, the automorphic line bundle $\omega_{\mathrm{tor},K_p}^\chi$ on $\mathcal{S}_{K^pK_p,\Sigma}^{\mathrm{tor}}$ is the pullback of a locally free sheaf (in fact, an ample line bundle) $\omega_{\min,K_p}^\chi$ on $\mathcal{S}_{K^pK_p}^{*}$. Consequently, the restriction of $\omega_{\mathrm{tor},K_p}^\chi$ to any fiber of q_{K_p} is trivial. Taking the inverse limit to infinite level, any fiber F of q is the inverse limit of the proper, connected fibers of q_{K_p}. On F, the pullback $(\pi_{\mathrm{HT}}^{\mathrm{tor}})^*\mathcal{L} \simeq \omega^\chi|_F$ is therefore trivial. However, $\mathcal{L}$ is an ample line bundle on the flag variety $\mathcal{F}l_{\mathcal{G},\mu}^{\diamond}$. Any morphism from a proper connected diamond F to $\mathcal{F}l_{\mathcal{G},\mu}^{\diamond}$ along which an ample line bundle pulls back to the trivial bundle must be constant. This implies $\pi_{\mathrm{HT}}^{\mathrm{tor}}$ is constant on all fibers of q. Because q is a proper surjective morphism (and thus a v-cover), the period map factors uniquely as a composition

$$\mathcal{S}_{K^p,\Sigma}^{\mathrm{tor},\diamond} \xrightarrow{q} \mathcal{S}_{K^p}^{*,\diamond} \xrightarrow{\pi_{\mathrm{HT}}^{*}} \mathcal{F}l_{\mathcal{G},\mu}^{\diamond}. \qquad \square$$

5.5. **Global perfectoidness.** Using Hecke-equivariance of the Hodge–Tate period map, we spread perfectoidness of the anticanonical locus ε-neighbourhood to the whole infinite-level minimal compactification. Maintain Assumption 5.2.

Lemma 5.5.1. *The preimage of $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p) \subset |\mathcal{F}l_{\mathcal{G},\mu}^{\diamond}|$ under $|\pi_{\mathrm{HT}}^{*}|$ is given by the closure of $\mathcal{S}_{\Gamma(p^\infty)}^{*}(0)$.*

Proof. The fact that the μ-ordinary locus can be seen from the universal isocrystal with $\mathcal{G}$-structure is explained in Remark 3.7.2. Now the desired result follows from [CS17, §3]. $\square$

Lemma 5.5.2. *Fix $0 < \varepsilon < 1$. There is an open subset $U \subset \mathcal{F}l_{\mathcal{G},\mu}^{\diamond}$ containing $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ such that*

$$|\pi_{\mathrm{HT}}^{*}|^{-1}(U) \subset |\mathcal{S}_{\Gamma(p^\infty)}^{*}(\varepsilon)|.$$

Proof. The period map $\pi_{\mathrm{HT}}^{*} \colon \mathcal{S}_{K^p}^{*,\diamond} \to \mathcal{F}l_{\mathcal{G},\mu}^{\diamond}$ is a map of spatial diamonds. By Lemma 5.5.1, the preimage $(\pi_{\mathrm{HT}}^{*})^{-1}(\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p))$ is exactly the extended μ-ordinary locus. In particular, it is contained in the locus $|\mathcal{S}_{\Gamma(p^\infty)}^{*}(\varepsilon)|$. Since $|\mathcal{S}_{K^p}^{*,\diamond}|$ is a spectral space, the closed subset $|\mathcal{S}_{K^p}^{*,\diamond}| \setminus |\mathcal{S}_{\Gamma(p^\infty)}^{*}(\varepsilon)|$ is quasi-compact for the constructible topology. As the intersection of the closed subsets $(\pi_{\mathrm{HT}}^{*})^{-1}(U)$ over all open neighborhoods $U \supset \mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ is exactly $(\pi_{\mathrm{HT}}^{*})^{-1}(\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p))$, which is disjoint from this complement, compactness implies that there exists some open $U \supset \mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ satisfying $|\pi_{\mathrm{HT}}^{*}|^{-1}(U) \subset |\mathcal{S}_{\Gamma(p^\infty)}^{*}(\varepsilon)|$. $\square$

Lemma 5.5.3. *For any open subset $U \subset \mathcal{F}l_{\mathcal{G},\mu}$ containing a $\mathbf{Q}_p$-rational point, we have $\mathcal{G}(\mathbf{Q}_p) \cdot U = \mathcal{F}l_{\mathcal{G},\mu}$.*

Proof. Since $\mathcal{G}(\mathbf{Q}_p)$ acts transitively on $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$, we may assume without loss of generality that the $\mathbf{Q}_p$-rational point in U is the base point $x = eP_\mu^- \in \mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$. Let P_μ be the opposite parabolic subgroup to P_μ^-, and let N_μ be its unipotent radical. The orbit of the base point under N_μ forms the big cell $\mathcal{C} = N_\mu \cdot x$, which is an open subspace of $\mathcal{F}l_{\mathcal{G},\mu}$ isomorphic to the adic affine space $(\mathbf{A}^d_{\mathbf{Q}_p})^{\mathrm{ad}}$, where $d = \dim(\mathcal{G}/P_\mu)$. Because U is an open neighborhood of x, it contains an affinoid polydisk $D \subset \mathcal{C}$ centered at x.

Let p be the uniformizer of $\mathbf{Z}_p$, and define the element $\gamma = \mu(p) \in P^-_\mu(\mathbf{Q}_p)$. We consider the adjoint action of γ on the Lie algebra $\mathfrak{n}_\mu$. By definition of P_μ, the roots α appearing in $\mathfrak{n}_\mu$ are precisely those that are strictly positive with respect to μ, meaning $\langle \mu, \alpha \rangle > 0$. For any root space $\mathfrak{g}_\alpha \subset \mathfrak{n}_\mu$ and $X_\alpha \in \mathfrak{g}_\alpha$, we have

$$\gamma \exp(X_\alpha)\gamma^{-1} = \exp(\mathrm{Ad}(\mu(p))X_\alpha) = \exp(p^{\langle \mu,\alpha\rangle} X_\alpha).$$

Since μ is minuscule, the only possible strictly positive pairing with a root is 1, so $\langle \mu, \alpha \rangle = 1$ for all roots in $\mathfrak{n}_\mu$. Thus, conjugation by γ acts as uniform scalar multiplication by p on the canonical coordinates of the big cell $\mathcal{C}$.

Consequently, γ acts as a global contraction mapping on the rigid analytic space $\mathcal{C}$ with x as the unique attracting fixed point. Let $W \subset \mathcal{C}$ be any quasicompact open subset. Because γ is a strict contraction, there exists an integer $n \gg 0$ such that $\gamma^n \cdot W \subset D \subset U$. Applying γ^{-n}, we find $W \subset \gamma^{-n} \cdot U \subset \mathcal{G}(\mathbf{Q}_p) \cdot U$. Since the big cell $\mathcal{C}$ is an ascending union of quasicompact subsets, it follows that $\mathcal{C} \subset \mathcal{G}(\mathbf{Q}_p) \cdot U$.

Finally, since $\mathcal{F}l_{\mathcal{G},\mu}$ is proper, it is covered by the $W(\mathcal{G}, T)$-translates of the big cell $\mathcal{C}$. Because the normalizer of the torus $N_{\mathcal{G}}(T)(\mathbf{Q}_p)$ surjects onto the Weyl group, there exist finitely many elements $w_1, \ldots, w_k \in \mathcal{G}(\mathbf{Q}_p)$ such that $\mathcal{F}l_{\mathcal{G},\mu} = \bigcup_{i=1}^k w_i \cdot \mathcal{C}$. Since $\mathcal{C} \subset \mathcal{G}(\mathbf{Q}_p) \cdot U$ and $\mathcal{G}(\mathbf{Q}_p) \cdot U$ is stable under left translation by $\mathcal{G}(\mathbf{Q}_p)$, we conclude that $\mathcal{G}(\mathbf{Q}_p) \cdot U = \mathcal{F}l_{\mathcal{G},\mu}$, as desired. □

Theorem 5.5.4. *The infinite-level minimal compactification $\mathcal{S}^*_{\Gamma(p^\infty)}$ is a perfectoid space.*

Proof. Fix a sufficiently small radius $0 < \varepsilon < 1/2$. By Lemma 5.5.2, we may find a quasicompact open subset $U \subset \mathcal{F}l^\diamond_{\mathcal{G},\mu}$ containing the $\mathbf{Q}_p$-rational points $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ such that its preimage under the Hodge–Tate period map satisfies

$$|\pi^*_{\mathrm{HT}}|^{-1}(U) \subset |\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon)|.$$

By Lemma 5.5.3, the $\mathcal{G}(\mathbf{Q}_p)$-translates of the underlying spatial diamond of U cover the entirety of $\mathcal{F}l^\diamond_{\mathcal{G},\mu}$. Because $\mathcal{F}l^\diamond_{\mathcal{G},\mu}$ is quasicompact, we can extract a finite subcover. That is, there exist finitely many elements $\gamma_1, \ldots, \gamma_k \in \mathcal{G}(\mathbf{Q}_p)$ such that

$$\mathcal{F}l^\diamond_{\mathcal{G},\mu} = \bigcup_{i=1}^k \gamma_i \cdot U.$$

Since the Hodge–Tate period map π^*_{HT} is equivariant for the action of $\mathcal{G}(\mathbf{Q}_p)$, we derive

$$|\mathcal{S}^*_{\Gamma(p^\infty)}| = \bigcup_{i=1}^k \gamma_i \cdot |\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon)|.$$

Since $\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon) = \mathcal{G}(\mathbf{Z}_p) \cdot \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$, the desired result follows from Theorem 5.2.2. □

5.6. **Conclusion.** Finally, we conclude by reducing the problem of infinite level perfectoidness of general Shimura varieties to that of Shimura varieties with reflex field $\mathbf{Q}$.

Setup 5.6.1. Suppose (G, X) is a general Shimura data. We do not enforce Assumption 5.2.

We borrow the following piece of terminology from [HJ23]:

Definition 5.6.2 (Property P)**.** We say that a Shimura datum (G, X) satisfies Property P if for every open compact subgroup $K^p \subset G(\mathbf{A}^p_f)$, the diamond of the minimal compactification at infinite level $\mathrm{Sh}^*_{K^p}(G, X) := \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathrm{Sh}^*_{K^pK_p}(G, X)^\diamondsuit$ is a perfectoid space. Similarly, a connected Shimura datum (G, X^+) satisfies Property P if for every arithmetic subgroup $\Gamma \subset G(\mathbf{Q})^+$, the diamond $\mathrm{Sh}^*_{\Gamma,\infty}(G, X^+) := \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathrm{Sh}^*_{\Gamma \cap K_p}(G, X^+)^\diamondsuit$ is a perfectoid space.

Proposition 5.6.3. *A Shimura datum (G, X) satisfies Property P if and only if its adjoint connected Shimura datum (G^{ad}, X^+) does.*

Proof. See [HJ23, Proposition 5.18]. □

Remark 5.6.4 (Reduction to absolutely simple Shimura data)**.** A standard dévissage allows one to reduce the question of Property P for Shimura varieties to the case of absolutely simple Shimura data. The adjoint group G^{ad} decomposes into a direct product $\prod_{i=1}^m G_i$ of $\mathbf{Q}$-simple adjoint groups. Correspondingly, the connected Shimura datum (G^{ad}, X^+) decomposes as a product $\prod_{i=1}^m (G_i, X_i^+)$, where each factor (G_i, X_i^+) is an absolutely simple connected Shimura datum. Because fiber products of perfectoid spaces are perfectoid, Property P is preserved under finite products. Consequently, by Proposition 5.6.3, to establish Property P for Shimura varieties of a certain type, it suffices to prove it for absolutely simple connected Shimura data.

Theorem 5.6.5. *(G, X) satisfies Property P for all sufficiently large primes p.*

Proof. By the preceding remark, we may assume (G, X) is a $\mathbf{Q}$-simple adjoint Shimura datum. If (G, X) is of pre-abelian type, we know that integral canonical models exist for all primes and hence a standard Weil restriction argument allows us to reduce to a Shimura data with reflex field $\mathbf{Q}$ because a closed adic subspace of a perfectoid space is automatically perfectoid. So this is solved by Theorem 5.5.4. Therefore, we only consider $\mathbf{Q}$-simple adjoint exceptional Shimura data. If it is of type E_7 then we are done again because $\mathbf{Q}$ is the only possible reflex field. Otherwise, the embedding of Theorem 3.8.1 extends to a closed embedding of integral canonical models for all sufficiently large primes p. Since a closed adic subspace of a perfectoid space is automatically perfectoid, the desired result follows. □

The Hodge–Tate period map a priori exists as a map of diamonds. However, since the source $\mathcal{S}^*_{K^p}$ is now the diamond of a perfectoid space, this map upgrades to a morphism of adic spaces:

Theorem 5.6.6. *Let (G, X) be a Shimura datum with reflex field E. For all sufficiently large primes p, the following holds. Fix a complete algebraically closed field $C/\mathbf{Q}_p$ and an embedding $E \hookrightarrow C$. Fix any open compact subgroup $K^p \subset G(\mathbf{A}_f^p)$. For any open compact subgroup $K_p \subset G(\mathbf{Q}_p)$, let $\mathcal{S}^*_{K^pK_p}$ denote the adic space over $\operatorname{Spa} C$ associated with the base change of the minimal compactification $\mathrm{Sh}^*_{K^pK_p}(G, X)$ along $E \hookrightarrow C$. Then there is a perfectoid space $\mathcal{S}^*_{K^p}$ such that*

$$\mathcal{S}^*_{K^p} \sim \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathcal{S}^*_{K^pK_p}$$

as diamonds over $\operatorname{Spd} C$. Moreover, there is a canonical $G(\mathbf{Q}_p)$-equivariant Hodge–Tate period map

$$\pi_{\mathrm{HT}} \colon \mathcal{S}^*_{K^p} \to \mathscr{F}\ell_{G,\mu}$$

of adic spaces over $\operatorname{Spa} C$ which is functorial in the tame level. Finally, the boundary of $\mathcal{S}^*_{K^p}$ is Zariski-closed.

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Massachusetts Institute of Technology, Cambridge, MA, USA

Email address: ayannath@mit.edu